

The global expansion of science has not made breakthrough ideas easier to find

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Over the past half century, knowledge and the number of countries producing it have both expanded. Theories of scientific progress hold that such expansion should accelerate discovery. Yet disruptive work has grown rarer and breakthroughs remain concentrated in a few countries, even as many more countries now produce highly cited work. Expansion may have widened participation without widening where new ideas are born. Here we track 2.3 million new ideas, dated by the first co-occurrence of two units of knowledge, across 119 million papers, and find them born on increasingly crowded ground: since the early 1970s the share involving the most crowded decile has risen by half. The most crowded new ideas are three times as likely to become highly cited and less than half as likely to be highly disruptive, a penalty explained not by the exhaustion of easy opportunities or the number of attempts that follow, but by narrower search. Countries that entered science during this period went furthest, their new ideas moving from less crowded than those of incumbent countries to more crowded and their relative disruption intensity falling from above the incumbents' to below, while the incumbents held at the world average. Two incentives created by expansion drove this movement: rapid growth tilted national funding towards crowded work, the faster a country expanded the more so, and researchers followed because crowded ground brings their citation returns closest to incumbents' while their disruption falls furthest behind. The impact so earned then became the record on which the next round of funding was allocated. Eleven Central and Eastern European countries, however, expanded comparably without following this path. Neither proximity and tradition nor distance from the frontier explains the difference, because their own funding favoured crowded work as strongly as the entrants' and their non-European funding was no larger. The difference lies instead in institutional access to external funding frameworks, opened by political treaties timed by diplomacy rather than research performance, after which their European-funded share rose from near parity with the entrants' to almost

four times theirs, leaving their funded work less crowded than even the incumbents'. Foreign funding lowers crowding in every country we examine, incumbents included. Entrants began with wider access than incumbents and lost it as they expanded. Scientists who returned from incumbent countries to entrant ones were drawn onto more crowded ground by the incentives at home, even those whose standing or continuing ties abroad kept them closest to the frontier. These results suggest that both the incentives and the arrangements built to steer rapid expansion towards crowded ground tend to outlast the expansion that created them, so that the institutions which made scientific catch-up succeed lock latecomers into positions where breakthroughs are least likely. Expansion determines how many take part in science, but the institutions that fund and judge research determine where its new ideas are born.

Introduction

Over the past half century, science has expanded on two fronts at once. The body of knowledge has grown, with more topics studied and more combinations among them, and the number of countries producing it has multiplied. Recombinant growth and collective-brain theories hold that such expansion should accelerate discovery, because more participants drawing on more diverse knowledge create more opportunities for novel combinations^{1,2}. Part of this expectation has been borne out. Participation has widened, and countries that entered science late now account for a growing share of its most highly cited work. But two things have not followed. Across science, technology and the wider economy, creative destruction has slowed. Papers and patents have become less disruptive on average³, the research effort needed to sustain a given rate of advance keeps rising⁴, and new firms displace incumbents less often than they once did⁵. At the same time, the breakthroughs that do occur have remained concentrated in a small number of countries. More than 80% of the discoveries behind science Nobel prizes were made in just five countries, and this concentration has persisted across more than a century of scientific expansion⁶. A similar asymmetry has long been observed beyond science, where catch-up economies have brought substantial incremental improvement but frontier innovation has remained concentrated in a few countries^{7,8}.

Existing accounts of the slowdown point to the growing burden of knowledge and the exhaustion of easy opportunities^{4,9}, to properties of the disruption index itself^{10,11}, or to a slowdown in the diffusion of knowledge from frontier to follower⁵. These explanations ask how hard new ideas have become to find, but not where they are born. An expansion can widen participation without widening the ground on which work is done. If new participants and new work gather on directions that already command broad attention, the expansion is a crowded one¹². Crowded ground has real attractions. Its paradigm is mature, peers are available to judge the work¹³, and its methods are known to succeed, so work done there is credible and readily cited^{14,15}. But these attractions may come at a cost. On crowded ground, researchers tend to continue along their established lines rather than pivot away from them^{16,17}, and to draw on nearby

rather than distant material^{18,19}, narrowing the very search from which breakthroughs arise. A crowded expansion would then deliver impact while withholding disruption. Whether the expansion of science has changed where new ideas are born, or has mainly multiplied work on ground that was already occupied, has not been examined.

Alongside the growth of knowledge, new countries have entered science. Because the slowdown in creative destruction is shared by all countries, the question for latecomers is not whether disruption has declined but whether they have narrowed the gap with the world in breakthroughs as they have in impact, or, in the language of development economics, whether catch-up in output has become catch-up in opening frontiers. Entry has brought real gains. To join the world research system, latecomers invested on a large scale, and their output of highly cited work rose accordingly²⁰. But large-scale investment may also decide where the work goes. The faster a country expands, the more its funding is drawn towards directions where global attention has already converged, because such directions offer the credibility, visibility and recognition that make proposals easier to evaluate¹⁴. For individual researchers, crowded ground offers the same credibility, visibility and recognition, whereas on sparse ground a contribution's reception depends more on the standing of whoever proposes it, which is precisely what latecomers lack. Yet, as noted above, crowded ground narrows the search from which breakthroughs arise. The faster the expansion, the stronger these pressures become. The impact earned on crowded ground becomes the record on which the next round of funding is allocated, and institutions that serve catch-up well may fail at the frontier²¹. Catch-up in impact and failure to catch up in breakthroughs would then be two outcomes of one process. Prior accounts attribute latecomers' gap to their distance from the frontier, whether in knowledge not yet diffused or in the capacity to absorb it^{5,22,23}. Whether the gap arises instead from where entry places their work, and from the incentives and institutions that place it there, has not been measured.

Here we start from where new ideas are born and ask what places them there. We treat keywords as units of knowledge and date the birth of a new idea by the first co-occurrence of two units anywhere in the literature. The crowding of a birthplace

measures how widely countries had already worked on the two units, and how much work they carried, in the five years before they met. Across 119 million papers, we identify 2.3 million new ideas born between 1971 and 2017. The expansion of science has not diversified where new ideas are born. Since the early 1970s, the share of new ideas with at least one unit in the most crowded decile has risen by half. Crowding cuts both ways. The most crowded new ideas are roughly three times as likely as the least crowded to produce highly cited papers, but less than half as likely to rank among the most disruptive of their year. The penalty comes from narrower search, as researchers on crowded knowledge pivot less far from their own prior work, lean more on elite institutions and draw on nearer material. The contraction of pivots alone accounts for about 70% of it, and most of it survives controls for exhausted opportunities and too many attempts. Latecomers lived out the same trade-off. Over half a century, the relative disruption intensity of 13 incumbent countries, the share of their highly cited papers that are highly disruptive relative to the world share, held at the world average, while that of 21 entrant countries, which began above the incumbents', fell below both. Their papers grew more crowded than the incumbents', and synthetic control places both shifts after entry. Two incentives created by expansion drive this movement. Entrants' national funding came to favour crowded work more strongly than the incumbents' did, and for entrant researchers crowded work narrows the citation gap with incumbents even as their disruption falls further behind. As predicted, the faster a country expanded, the more crowded its funding and its papers became and the further its disruption fell. Together these incentives form a loop, as the impact researchers earn on crowded ground becomes the record for the next round of funding. Foreign funding pulls the other way in every country we examine, so how far a country resists the loop depends on its institutional access, the share of its funded work carried out inside another country's funding framework and so reviewed by that framework's standards. The eleven Central and Eastern European countries that signed agreements with the European Communities between 1991 and 2001 (Europe Agreement countries) show what can break the loop. Although they expanded on a comparable scale, crowding fell

and relative disruption rose in most of them. The agreements were exogenous to research, political treaties timed by diplomacy rather than research performance that opened European funding programmes to these countries. The obvious alternative is that their closeness to the frontier, in geography, culture and scholarly tradition, did the work, by widening their access without the agreements, steering their own funding away from crowding or letting them draw more from each unit of foreign funding. None of this happened. Their access did not precede the agreements, since before signature their European share matched the entrants'. Nor did closeness raise it, since their funding from outside Europe, which the agreements did not govern, stayed at the entrants' level before and after accession. Their own funding favoured crowded work as strongly as the entrants', and each unit of European funding pulled their work back no further. What grew was the breadth of their access, not its depth. After accession their share of funded work in European programmes rose to nearly four times the entrants', leaving their funded work less crowded than even the incumbents'. The entrants, by contrast, lost the access they had. In 1991–2003, the first half of the period, 30% of their funded papers drew on foreign funding against 19% for incumbents, and foreign funding pulled their work further from crowding than it pulled the incumbents'; by 2004–2017 their share had fallen to the incumbents' 26% and that advantage had gone, as domestic funding outgrew foreign funding. Returning scientists, the classic carriers of tacit knowledge, test the account within careers. The 36,887 who came home from incumbent countries had begun narrowing their pivots and leaning on elite institutions while still abroad, conforming in advance to the system they were about to enter. After return, crowding rose in every subgroup, even among those with high pre-return prestige or continuing ties abroad, who should have resisted it most if distance from the frontier were the constraint. On return they lost their connection to elite institutions and their highly cited output fell, while the knowledge they brought widened their pivots and raised average disruption, though not its top tail, and crowding barely moved. By the second funding cycle that connection had not returned. Highly cited output recovered only in part as pivots contracted and crowding rose, and highly disruptive work declined. What decides where knowledge goes is not what returnees

carry home but the incentives to which they return.

Our study makes three contributions. For the science of science, we show that where a new idea is born shapes whether it becomes a breakthrough, and that the global expansion of science has multiplied participants without diversifying where new ideas are born. For development and institutional economics, we provide evidence from science that latecomers' shortfall in breakthroughs lies less in their distance from the frontier than in the institutions through which they catch up. For policy, the findings point to deeper cooperation in research funding. Because foreign funding lowers crowding everywhere, incumbents included, opening funding frameworks to one another is less aid to latecomers than a correction from which every participant gains. For entrant countries, it also calls for evaluation that rewards more than bibliometric impact and for openness that keeps pace with expansion. Whether expansion makes breakthroughs easier to find depends not on how many join but on where institutions steer them.

Measurement of global crowding

To quantify the amount of global attention and effort into which a new idea is born, we treat keywords extracted from papers as units of knowledge, define the first link between two units as the emergence of a new idea, and measure its global crowding as the mean crowding of the two units it connects (Fig. 1a,b and Global crowding in the Methods).

Crowding is computed for each unit of knowledge as follows. For every five-year window, we count the papers of each country that use a given keyword and divide by that country's total output over the same window, yielding a distribution of country-level attention that is free of country-size effects. Crowding is the product of the normalized entropy of this distribution and a volume factor derived from the log-scaled percentile rank of the keyword's publication count. Crowding C varies on the $[0, 1]$ interval. It approaches 1 when a unit of knowledge draws attention from countries worldwide and carries a large volume of work, and approaches 0 when attention is concentrated in a few countries or the volume of work is negligible. Both components are required: entropy alone would misclassify an obscure direction shared evenly among a handful of countries as crowded, whereas volume alone would misclassify a large programme run by a single country as globally invested.

Fig. 1c, d shows the distribution and evolution of crowding. Among the 2,343,892 new ideas that emerged between 1971 and 2017, crowding ranges across most of the $[0, 1]$ interval with a median of 0.71 (Fig. 1d), indicating that new ideas are born on ground of widely varying density. Relative to the 1971–1975 baseline, the mean crowding of all units of knowledge rose by 51% and the Gini coefficient of crowding across units fell by 54%, while the share of new ideas with at least one unit in the top decile of that year's crowding distribution rose by 51% (Fig. 1c): as global effort has deepened and spread, new ideas have increasingly emerged on already crowded ground. This rise proceeds through several distinct phases of acceleration, possibly reflecting major shifts in how information has been connected over recent decades.

The two ends of the distribution can be anchored by notable new ideas. Near the

crowded end is beta-amyloid peptide–cognitive impairment ($C = 0.92$, 2005), a direction that has guided Alzheimer research for three decades and whose supporting 2006 paper was retracted in 2024; near the sparse end are artificial neural network–backpropagation ($C = 0.37$, 1988) and vanilloid–VR1 ($C = 0.22$, 1998), the former associated with the 2018 Turing Award and the 2024 Nobel Prize in Physics, the latter with the 2021 Nobel Prize in Physiology or Medicine.

A similar tendency appears in the aggregate data. The 187,556 new ideas appearing in an NSF or NIH grant (8.0% of all new ideas) have a median crowding of 0.76, the 63rd percentile of the overall distribution, and those in the top decile are roughly 4.4 times as likely to be funded as those in the bottom (Fig. 1e). By contrast, the 138 new ideas underlying Nobel-prize-winning work have a median crowding of 0.67, the 41st percentile, and the least-crowded quintile is enriched roughly 1.73-fold relative to the most-crowded (Fig. 1f; because Nobel recognition typically lags by decades, this sample is restricted to 1971–2003). Across these two samples, funding is directed more towards where global effort has converged, whereas prize-winning work leans towards the sparser end.

Impact dividend and breakthrough penalty of crowded combinations

Fig. 2 shows that new ideas born on crowded ground enjoy a substantial impact dividend but pay a breakthrough penalty. For impact, the share of a new idea's papers ranking in the top 5% of citations (Hit 5pct) rises monotonically with crowding, from 0.046 in the least-crowded bin to 0.136 in the most crowded, a 2.96-fold difference (Spearman $\rho = 1.00$; Fig. 2a). The relationship holds after adjusting for year and field (0.059 to 0.135). Comparing the two ends of the distribution directly reveals the same contrast: new ideas whose two units of knowledge both fall in the bottom decile of that year's crowding distribution have a mean Hit 5pct of 0.037, against 0.146 for those in the top decile, 291.9% higher ($n = 25,886$ and $20,043$; two-sided Welch's t -test, $P < 0.001$).

For disruptiveness, the relationship reverses (Fig. 2b). The probability that the mean CD5 of a new idea's papers ranks in the top 5% of new ideas emerging in the same year falls monotonically with crowding, from 7.13% in the least-crowded bin to 2.95% in the most crowded, a decline of 58.6% (Spearman $\rho = -1.00$). The decline persists after adjusting for year and field (6.29% to 3.69%). Were the most disruptive work distributed equally across levels of crowding, this share would remain at 5%; the observed curve falls below that benchmark at a crowding of roughly 0.75 and continues downwards. The contrast between the two ends is equally pronounced: 9.09% of new ideas in the bottom decile reach the top 5%, against 2.53% in the top decile, a 3.60-fold difference ($P < 0.001$). In this analysis, then, crowded ground yields new ideas that are more widely cited but less likely to make the most disruptive contributions. Both relationships have strengthened over time: across three cohorts defined by year of birth, the association between crowding and impact becomes increasingly positive and that with disruptiveness increasingly negative, with significant crowding $\times$ decade interactions in both cases (impact $\beta = +0.187$; disruptiveness $\beta = -0.021$; both $P < 0.001$; Fig. S3).

These relationships prove robust to a range of alternative measures, estimation strategies, and sample restrictions (Figs. S1 and S2). Results are robust to multiple

alternative definitions of scientific impact, including Hit at varying thresholds (top 1%, 5%, 10%) across different field levels (LV0 and LV1), citation counts within fixed windows, and mean citations three and five years after publication (Fig. S1a–g). For disruptiveness, the association with crowding persists when using alternative variants of the disruption index (the within-year percentile of CD5 and the share of papers with $CD5 > 0$) and under minimum reference count restrictions of 5, 15, 20, 30 and 50 references (Fig. S2a–g). Both relationships also hold when computed separately within each of the four macro-fields (Figs. S1h and S2h).

The relationship is robust in measurement, but its source admits several possibilities. Lower disruptiveness on crowded ground may arise because these units of knowledge have been worked on longer and accumulated more, leaving only incremental work for those who follow; it may be a by-product of scale, as crowded ground attracts more attempts and returns diminish with their number; or it may be of little consequence, if the disruptiveness found on sparse ground goes unattended. We examined all three in regressions of top-tier disruption with $\text{year} \times \text{field}$ fixed effects (Table S1). Adding two measures of prior accumulation — the cumulative papers on the two units of knowledge before the edge formed, and keyword age — did not attenuate the coefficient on crowding but raised it slightly, from 74% to 86% of the baseline effect once scale and citations, examined below, are held fixed; both measures were themselves significantly positive ($P < 0.001$), so that more mature, more heavily accumulated units of knowledge were in fact more likely to yield top-tier disruption, prior accumulation here resembling a ladder for further progress rather than a mine worked out. Turning to scale, the number of papers on a new idea over the following five years was significantly negatively associated with top-tier disruption ($P < 0.001$), confirming diminishing returns; yet controlling for it attenuated crowding by only about 8%, retaining 92%. Controlling in addition for mean citations over the following five years — where impact is higher, top-tier disruption may be harder to come by, itself a direct expression of the opposing directions of impact and disruptiveness — crowding still retained 74%, indicating that the disruptiveness of sparse ground is not a disruptiveness no one attends

to. In sum, the breakthrough penalty appears not as a matter of exhausted fruit, of too many attempts, or of work going unnoticed: each accounts for part of the relationship, but most of it remains. Where, then, does it come from?

What drives these two opposing differences in performance? A natural starting point is who works on crowded ground. The ‘Matthew effect’ describes the tendency of reputation and resources to converge further on where they have already converged¹⁴, and this convergence translates into visibility along two paths: widely recognized knowledge bases both harbour and attract established scholars, who bring visibility of their own, and work built on such bases gains credibility while sitting within a larger pool of potential readers. These considerations indicate that the impact dividend of crowded ground may rest on visibility itself — on who works there and on where they draw their material. To test this, we successively added two characteristics to regressions of Hit 5pct with year $\times$ field fixed effects (Table S2). As crowding rises, team prestige increases by 34.3% and elite-institution reliance by 30.0%; adding team prestige explains 24.2% of the crowding effect, adding elite-institution reliance 61.1%, and together 62.0% (all $P < 0.001$). Most of the impact dividend is thus carried by the convergence of reputation and resources, consistent with the Matthew effect. Yet this makes the breakthrough penalty more puzzling: if crowded ground concentrates more prestigious teams and more widely recognized sources of knowledge, it should produce more disruptive work, not less. The answer appears to lie not in who arrives but in how ideas are generated.

Turning to frameworks of idea generation, a canonical perspective emphasizes an ‘explore versus exploit’ trade-off in creative search, where exploitation involves lower-risk, lower-return search around the edges of one’s current focus and exploration involves higher-risk, higher-return forays into more distant areas¹⁶. Accordingly, the amount of exploration that had already occurred on the two knowledge bases before an edge formed should be positively associated with its subsequent disruptiveness. We measure this by pivot size — how far the researchers working on these knowledge bases had departed from their own prior work in the five years before the new idea emerged.

As crowding rises, pivot size falls by 22.1%: on more crowded ground, researchers are more inclined to continue along their own established direction rather than move away from it. Adding pivot size to the disruption regression explains 69.6% of the crowding effect, and it is significantly positively associated with top-tier disruption ($P < 0.001$; Table S3). Notably, the breadth of a team's knowledge does not by itself yield breakthroughs: team knowledge distance is significantly positively associated with top-tier disruption when entered alone, but its main effect attenuates sharply and loses significance once pivot size is controlled for, while its interaction with pivot size is significantly positive ($P < 0.001$; Table S3). Broad knowledge is thus a latent capacity, realized as a breakthrough only when it is put to use through an actual pivot.

A further set of frameworks emphasizes the breadth of search itself, linking creative advantage not only to departure from established paths but to how widely material is drawn^{18,19}. This perspective stands in tension with our findings on the impact side: concentrating on a few authoritative sources for the sake of visibility may be precisely what constrains broader search. The data support this. Elite-institution reliance is significantly negatively associated with top-tier disruption, and remains so after controlling for pivot size ($P < 0.001$; Table S3), with an explained share of 85.6% in this specification. The two amplify one another: when researchers both depart from their own established direction and draw material widely from beyond authoritative sources, the likelihood of a breakthrough exceeds what either contributes alone ($P < 0.001$). The same variable is positive on the impact side and negative on the disruption side — reliance on authoritative sources raises citations while suppressing breakthroughs. Finally, new work can be seen as a new combination of existing material²⁴, and the distance a link spans bears directly on its novelty. We measure this by neighborhood overlap — the share of neighbours shared by an edge's two units of knowledge — where lower overlap means the link joins two more distant regions of knowledge. Neighborhood overlap is significantly negatively associated with top-tier disruption ($P < 0.001$; Table S3): links spanning greater distances are more likely to yield breakthroughs. It too amplifies pivoting: the greater the distance a link spans, the

higher the likelihood that a pivot yields a breakthrough ($P < 0.001$). The explained share in this specification is 91.0%.

These findings indicate that crowded ground does not lack resources so much as room to depart from established paths. Pivoting is the route to breakthrough, while the breadth of a team's knowledge, the breadth of its search, and the distance of the link together determine how much a given pivot can deliver; crowding depresses all four. Nor does this narrowing stop at the moment a new idea is born: over the five years that follow, the same four characteristics shift with crowding in the same direction (Fig. 2c) — the work taken up on crowded ground continues to draw its material, depart from established paths, and connect across distances more narrowly. The convergence of reputation and resources yields returns in impact, while the contraction of search comes at the cost of disruption — the two occur together, and together they constitute the impact dividend and the breakthrough penalty of crowded combinations.

More global expansion, more crowding, fewer breakthrough ideas

The expansion of the knowledge network has been matched by the expansion of science in geographic space. Over the past half century the world research system has drawn in an increasing number of countries, and among the most highly cited work the combined share of the five largest producers fell from 77.9% to 51.4% while the effective number of contributing countries rose from 7.8 to 24.7 (Fig. S5a), so that high-impact output has indeed become more dispersed rather than more concentrated. A range of existing theories further anticipates that such widening should make breakthrough ideas more likely to emerge^{1,2}, yet over the same period the share of top-5%-cited papers (Hit 5pct) that are disruptive fell from 2.02% to 0.33% (Fig. S5b). A declining share of highly disruptive work has been documented across the whole of science³, and beyond science in the slowing of creative destruction across the economy⁵, while the causes of the disruption decline remain contested. Some studies locate it in the construction of the CD index itself, pointing to its sensitivity to the completeness of reference lists, to database coverage and to the length of the citation window^{10,25,26}, whereas others advance substantive accounts grounded in how knowledge diffuses between frontier and follower firms⁵. Whatever its source, this background decline is common to all countries in the same period, and once the contemporaneous world rate is removed as a benchmark what remains is each country's position relative to the world, which leads to the question we take up here, whether the geographic expansion of science over the same period made breakthrough ideas easier to find.

Countries are classified as entrants or incumbents by a vote across 55 classification variants, giving 21 entrants and 13 incumbents, and what entrant countries share is a large-scale expansion of research output within the observation period (Methods). We restrict the denominator throughout to Hit 5pct papers with a computable CD5 index and end the observation window in 2015, since disruption measures are unstable for sparsely cited work and citation accumulation typically saturates within a decade of publication^{27,28} while our data end in 2025. Measured against the contemporaneous world rate, the relative disruption intensity of entrant countries declined across the

period, standing at 1.24 in 1980, rising to a peak of 1.35 in the mid-1980s, then falling steadily, crossing the world average in the mid-2000s and ending at 0.89, while incumbent countries held at the global average throughout and barely moved (Fig. 3a). The entrant decline holds under all 55 classification variants, and dropping any single entrant or any single incumbent leaves both trajectories unchanged (Fig. S6), as it does under alternative specifications and when computed separately within fields (Figs. S7 and S9). The crowding of entrant papers rose over the same period, with both the knowledge base they built on and the new ideas they introduced growing steadily more crowded through the 1990s and overtaking incumbent papers in the mid-2000s, since when entrant work has been the more crowded of the two (Fig. 3d), a rise that likewise does not depend on the particular specification or field assignment (Figs. S8 and S9). The two crossings fall in the same years, and entrants lost their disruptive advantage exactly as their work moved onto crowded ground. To separate these movements from global trends, we use the synthetic control method, fitting a counterfactual trajectory for the entrant group from 25 untreated donor countries, which allows a comparison between where entrant countries actually went after scaling up and where they would have gone without it, with treatment dated at 1994, the median entry year of the group (Methods). The results match Fig. 3a and 3d, in that entrant countries' relative disruption intensity runs 0.27 below this path ($P = 0.050$; Fig. 3b), while the crowding of their new ideas runs 0.038 above it and that of their knowledge base 0.048 above it (both $P = 0.008$; Fig. 3e). Both gaps open late, with entrants and their counterfactual nearly identical before 1994 and separating only afterwards, disruption intensity falling below the world average and crowding climbing above it by 2010. All three estimates hold under in-space placebo tests, under leave-one-out removal of any entrant or donor country, when each entrant is dated from its own entry year, and across treatment years from 1988 to 2002 (Fig. S10).

One possible account of these movements lies in the nature of large-scale investment itself. To expand research output within a short period, a country must rapidly train and absorb a large number of researchers, and those entering research need to produce

results quickly. Under such pressure, moving towards directions that already command wide attention is the safer choice, since these directions come with an established knowledge base and a paradigm to work within, and they more readily attract recognition from peers, the impact dividend documented in Fig. 2 being precisely the return to that choice. These are also the most crowded directions. If this account holds, countries that invested more heavily and expanded faster should show both a steeper decline in relative disruption intensity and a larger rise in crowding, and the data are consistent with that implication, as annualised growth in output is negatively associated with the change in relative disruption intensity ($r = -0.30$) and positively with the change in crowding ($r = +0.70$; Fig. 3c,f), with both relationships keeping their sign under alternative samples, weights and thresholds (Fig. S11). The Europe Agreement countries form a notable exception. Eleven countries signed agreements with the European Communities between 1991 and 2001, committing to institutional and regulatory convergence, and their entry date is fixed by treaty rather than inferred from publication data. They too underwent considerable expansion in output, yet at comparable expansion multiples their crowding rose markedly less than that of entrant countries and their relative disruption intensity fell less steeply (Fig. 3c,f). The scale of expansion therefore does not by itself determine where a country's work comes to sit. What set the Europe Agreement countries on a different trajectory, and whether the incentives this account presumes are in fact at work in the allocation of funding and in researchers' individual choices, are the questions we take up below.

The incentives behind global crowding

The preceding analysis showed that the faster a country expanded its research output, the more crowded the positions its papers came to occupy and the more its relative disruption intensity fell—but also that the Europe Agreement countries did not follow this pattern. Expansion alone does not determine where a country's work finally sits. To understand why, we turn to the first incentive a researcher faces: what kind of work attracts funding. For each group of countries and each ten-year window we compute a crowding gap—the mean within-year crowding percentile of the group's funded papers minus that of its own unfunded papers. Subtracting a country's own unfunded work removes everything that acts on both kinds of paper in the same country and year: field composition, language, and the secular thickening of the knowledge space. The gap itself, however, moves with the period. Funding coverage has widened, communication has quickened, and acknowledgement practice has changed, so the distance between funded and unfunded work contracts for reasons unrelated to any country's allocation choices. We therefore divide each group's gap by the incumbents' national-money gap in the same window. The incumbents are a fixed and stable reference; dividing by them leaves how far a country's own money pushes its work towards crowded ground, measured against a common yardstick that moves with the times. National money favours crowded positions in all three groups. Over 1991–2017 the national-money crowding gap is positive everywhere—+0.0347 percentile points for the incumbents, +0.0550 for the entrants and +0.0512 for the Europe Agreement countries (Table S4, New ideas)—and the two expanding groups stand roughly half again above the incumbent benchmark, at $\times 1.59$ and $\times 1.48$ (Fig. 4a). The pattern is consistent with the impact dividend documented in Fig. 2: work on crowded ground is more likely to reach the highly cited tier, and funding agencies select on visible returns, since a direction that already commands global attention comes with an established knowledge base, precedents to cite, and reviewers able to judge whether a proposal is credible²⁹. What appears on the allocation side is money converging further on where reputation and resources have already converged^{14,30}. If selection pressure is what produces this bias,

the bias should be strongest where the pressure is greatest. A country that multiplies its research base within a short period confronts its funding agencies with a sharply rising volume of proposals and a sharply expanding population of researchers whose track records are thin. The data follow this expectation. Taking each country as one observation and correlating its annualised growth in output with its own national-money crowding gap over 2011–2015, across forty countries spanning all four groups, the faster a country expanded the larger that gap: the knowledge-base layer, which covers 99.9% of papers, gives $r = +0.41$ unweighted and $+0.56$ weighted ($P < 0.01$); the new-ideas layer agrees in sign but more weakly ($+0.19$ and $+0.67$), covering only 13–15% of papers and therefore carrying more measurement noise in the dependent variable—as the sample threshold rises and that variable is measured more precisely the correlation strengthens (from $+0.26$ to $+0.61$ on the knowledge base), the direction attenuation predicts (Fig. S13b,c). The relationship holds on a single regression line pooling all four groups, and no group breaks it. Fig. 3f recorded the downstream end of this process; here we record the upstream end, where the money went first.

The entrant–incumbent comparison nonetheless left an exception unresolved. The Europe Agreement countries underwent comparable expansion, yet their crowding rose far less and their relative disruption intensity fell less steeply (Fig. 3c,f). If national money pushes towards crowded ground everywhere, and pushes harder the faster a country expands, this exception calls for an explanation. The natural one is proximity and culture. The Europe Agreement countries lie close to Western Europe and stand nearer the frontier not only geographically but in language, disciplinary tradition and long-standing scholarly contact, and several carry a tradition of scientific achievement of their own reaching back well before the period we observe³¹. Knowledge reaches them more readily, and their researchers may simply have been better placed—by training and tradition—to attempt what others had not. Were that so, their own money should push their work towards less crowded positions than the entrants’ does. It does not. On national money the two groups stand equally far above the incumbents, almost level over 2004–2017, the late half of the period falling entirely after entry ($\times 1.59$

against $\times 1.61$), and with the Europe Agreement countries higher at the last ten-year window, 2008–2017 ($\times 1.85$, 95% CI 1.70–1.99, against $\times 1.71$, 1.68–1.75); their solid lines in Fig. 4a essentially coincide, and the remaining difference in funded position between the two groups falls almost entirely on the other side of the ledger (Table S4). Neither geographic proximity nor cultural affinity leaves any trace on where their own money goes.

A second difference between the two groups lies not in circumstance but in institutions. Between 1991 and 2001, eleven Europe Agreement countries signed agreements on institutional and regulatory convergence with the European Communities and their successors, and with them the right to participate in European research programmes. The entrants had no comparable instrument over the same period. The distinction may lie not in what these countries are but in what they can enter. We define, for each group, w : the share of its funded papers acknowledging at least one foreign or EU funder. Writing n_f for a group's funded papers, n_{EU} for those naming a supranational funder—the European Union's framework programmes, the European Research Council and a small number of comparable international bodies, denoted EU throughout—and n_x for those naming a non-EU foreign funder but no supranational one, $w = (n_{EU} + n_x) / n_f$, and it splits into an EU share n_{EU} / n_f and a non-EU foreign share n_x / n_f —papers naming at least one foreign national funder but no supranational one, whether or not a domestic funder is also named—that sum to it. w contains no crowding information; it records only how large a share of a country's researchers work inside another country's funding framework, which is not the same as collaborating across borders: a foreign co-author who attaches no grant does not raise it (Methods). We call it institutional access. Over 1991–2017 on the new ideas layer, w stands at 0.459 for the Europe Agreement countries against 0.263 for the entrants and 0.246 for the incumbents—close to twice the other two groups, with the entrants and incumbents at comparable levels. What matters more is where that advantage comes from. Their EU share is 0.278, three fifths of their total access, against 0.087 for the incumbents

and 0.077 for the entrants. On the non-EU foreign share the three groups are close, and the entrants and the Europe Agreement countries are all but identical—0.187 against 0.182, with the entrants in fact the higher of the two—while the incumbents sit at 0.159 (Table S4). These are not, in any general sense, more internationally connected than other countries. Outside what the agreements cover they are like everyone else; inside it they are three times everyone else. What pulls them back towards sparser ground is not proximity to Western Europe but the agreements themselves.

The agreements were above all political—research participation was one clause among many in instruments designed for regulatory convergence and eventual accession—but it might still be asked whether the Communities chose to contract with countries whose researchers were already well connected, in which case access would precede the agreements rather than follow them. Before signature the two groups are nearly indistinguishable in the channel the agreements govern: over 1981–1990 the EU share of the Europe Agreement countries stands at 1.16 times the entrants’, and over the five years immediately preceding the first signatures, 1986–1990, at 1.04. It rises to 1.35 over 1991–2003 and 3.91 over 2004–2017. And the divergence is dated not by signature but by accession and full participation in the framework programmes a decade later: on ten-year rolling windows the ratio between the two groups is flat from 1984 to 2004 and rises only afterwards. It might also be asked whether these countries were simply closer to the frontier—in geography, language and scholarly tradition—so that their access would have been wider with or without the agreements. Were that so, the non-EU foreign channel, which the agreements do not govern, should be elevated alongside the one they do. It is not: over the same three windows the non-EU foreign share of the Europe Agreement countries stands at 1.05, 1.03 and 0.95 times the entrants’, flat throughout and slightly below by the last. At the country level, the EU share rises in 11 of 11 countries after their own accession year while the non-EU foreign share falls in 11 of 11. The same endowment was tested twice and left no trace either time: not on where their own money goes, and not on the channel the agreements do not govern. What separates the two groups is therefore not how near they stand to the frontier but

whether an arrangement existed that changed how their work was reviewed and selected. What the agreements changed is the channel they opened, and nothing else.

Access translates into position because external money sits in sparser places. The pull-back can be written $w \times \Delta$, where Δ is how much more crowded purely domestically funded papers are than papers touching foreign money. Δ is positive in 44 of 44 countries (Fig. S14): foreign money pulls work towards sparser ground everywhere, not only in the Europe Agreement countries. A regularity that holds without exception across the data is itself something policy can rely on. The entrants and incumbents are comparable on both factors (w 0.263 against 0.246, Δ 0.0422 against 0.0374), while the Europe Agreement countries stand roughly twice as high on each (0.459 and 0.0780); the product gives them a pull-back more than three times the entrants' (+0.0358 against +0.0111). That both factors differ admits a second reading. The Europe Agreement countries might draw more from each unit of external money—because they participate more deeply within a project, or because proximity makes what is transmitted easier to absorb—rather than simply drawing from more of it. Δ does differ (0.083 against 0.043 over 2004–2017), but the pull-back decomposes exactly into its two channels, $w \times \Delta = (n_{EU} / n_f) \cdot \Delta_{EU} + (n_x / n_f) \cdot \Delta_x$ where Δ_{EU} is the crowding gap between domestically funded papers and those naming an EU funder, and Δ_x the same gap against papers naming a non-EU foreign funder, and the decomposition shows that the apparent difference in Δ is a consequence of the difference in w , not a separate force. Over 2004–2017 the two groups are indistinguishable in how far each unit of EU money pulls ($\Delta_{EU} = 0.107$ against 0.101) and differ almost fourfold in how much of their work enters that channel (EU share 0.299 against 0.076). The non-EU foreign channel tells the opposite story: over the same window its per-unit pull is higher for the Europe Agreement countries ($\Delta_x = 0.041$ against 0.020), but because the two groups enter it at the same rate (non-EU foreign share 0.175 against 0.184), it contributes only 13% of the total pull-back difference—the remaining 87% arises in the EU channel. The overall Δ is higher for the Europe Agreement countries not because external money works

harder there but because a larger share of their external money flows through the EU channel, where per-unit pull is stronger for everyone. On the knowledge-base layer the same conclusion holds more simply: Δ falls within the narrow band 0.023–0.027 across all three groups while w ranges from 0.216 to 0.374, so the difference in pull-back can only come from access. Multiplying the two factors yields the position of the dashed lines in Fig. 4a: at the last window the Europe Agreement countries' funded work as a whole falls to $\times 0.56$ (0.45–0.67), below the incumbents themselves, while the entrants fall only to $\times 1.36$ (1.33–1.40). The two groups start from almost the same point in 2000 ($\times 0.75$ and $\times 0.65$) and their national-money lines end close together; the difference lies entirely in what happens once foreign money is added. The same countries travel the same course over time. Splitting at 2004 into periods of roughly fourteen years each, the entrants' total access in the earlier period exceeds the incumbents' ($w = 0.299$ against 0.191) and their foreign money pulls them back significantly further (extra pull-back +0.0040, $Z = 2.9$), whereas in the later period their w falls to the incumbents' own 0.261 and the advantage goes with it (-0.0001 , $Z = -0.4$). The Europe Agreement countries' w rises over the same interval from 0.331 to 0.474, their extra pull-back from +0.0084 to +0.0278 (Table S4). The entrants entered the world research system better connected to external frameworks than the incumbents were, and lost that position in the course of expanding.

Access is itself not fixed but moves with expansion, and it moves in opposite directions across the three groups. Correlating each country's annualised growth in output with its own w over 2011–2015, the coefficient is positive for the incumbents and the Europe Agreement countries (+0.32 to +0.77 across weighted and unweighted fits and both layers) and negative for the entrants (-0.40 to -0.61): the faster an incumbent or a Europe Agreement country expanded the wider its access, and the faster an entrant expanded the narrower its own. All four panels agree in sign, weighted and unweighted estimates agree, and the pattern survives both definitions of the entry year (Fig. S16); the pull-back $w \times \Delta$ splits the same way (Fig. S15). The entrants' foreign-funded output grew in absolute terms over this period; what fell was its share, because domestic

funding grew faster still. This may be the other face of the same force. The larger a country's own funding becomes, the more its researchers are both able and required to organise their work around a domestic agenda, and the narrower the place external frameworks occupy within it. Once established, such an arrangement tends to perpetuate itself, since each round of countable output becomes the record on which the next allocation rests. The structures built to expand at scale — in which domestic funding comes to dominate and external frameworks to sit at the margin — are also likely to outlast the expansion, the faster it went the more firmly³². The force that pushes work towards crowded ground also squeezes the one channel capable of pulling it back. Other channels are possible, among them a widening distance in question selection between domestic and external funding as expansion proceeds, and the fact that a national agency abroad does not enlarge its programmes because a partner country has expanded, though we do not test them here.

The contrast between the two forces is now clear. What expansion does to national money is universal—all four groups fall on a single regression line running in the same direction (Fig. S13b,c). What it does to institutional access is not—the three groups run in opposite directions (Fig. S16). The push is available to everyone; the pull is not. Yet the pull is itself universally available: the incumbents' own national money pushes their work towards crowded ground, and once foreign money is counted their funded work as a whole falls from the benchmark to $\times 0.73$ over 1991–2017, a pull-back of +0.0092 percentile points on a national-money gap of +0.0347 (Table S4). Foreign money pulls on the incumbents too. Institutional access is therefore not aid from developed countries to latecomers but a correction that works for everyone who takes part—a point we return to in the Discussion. The results of this section hold on both layers; the knowledge-base counterparts are given in Fig. S13a and Table S4. The answer is institutional. Money chooses before people do, and what decides where it goes is the speed of expansion and the breadth of access.

Money went to crowded ground, but crowding is the sum of individual choices, and whether moving there is rational, and rewarded, for a researcher in an entrant country

is a separate question. We divide all papers into five equal groups by the crowding of the new ideas they introduce and compare entrants with incumbents within each group (Fig. 4b). The two returns move in opposite directions. The entrants' probability of producing a top-5% cited paper, relative to the incumbents', rises from 0.63 in the least crowded group to 0.79 in the most crowded (15.83% to 19.37% against 25.10% to 24.50%), whereas the share of such papers reaching $CD5 > 0.25$ falls from 1.15 times the incumbents' to 0.75 times (0.642% to 0.178% against 0.560% to 0.236%): entrants are 15% more disruptive than incumbents where the ground is sparsest and 25% less where it is most crowded. These directions hold in grouped binomial regressions with country and year fixed effects and standard errors clustered by country-year and by country, and the disruption coefficients grow as the definition tightens (Fig. S12). Crowded ground narrows this gap since it offers credibility, because its paradigm is settled; recognisability, because there are peers able to judge the work and disposed to cite it; and viability, because its methods are known and its risk of failure is low. There, a piece of work is legible against a standard that everyone shares. On sparse ground no such standard yet exists, and how a contribution is received rests more on the standing of whoever proposes it¹⁴—the dimension on which entrants lag—so the gap widens. The same position therefore offers entrants a higher private return and a lower social one. This is not a system that fails to recognise latecomers but one that recognises them in the wrong place, and it closes a loop. Funders favour work on crowded ground (Fig. 1e and above); in the Global South, where most entrants lie, evaluation leans most heavily on quantitative metrics³³; researchers go where their visible return is highest; and the impact they earn there becomes the record on which the next round of money is allocated. Each step is rational where it is taken, and together they hold the system on the crowded side.

The most forceful alternative to this account is that crowding reflects distance from the frontier rather than incentives. In the economics of innovation, a slowdown in the diffusion of knowledge from frontier to laggard firms has been identified as the principal force behind the decline of business dynamism and creative destruction⁵;

applied here, latecomers would cluster on ground that already commands global attention because it is the ground within their reach, lacking the knowledge to venture further. The previous section found no trace of this at the level of countries: proximity to Western Europe did not steer the Europe Agreement countries' own money towards sparser ground. Its sharpest test, however, lies with individuals. Mobile scientists are the classic carriers of tacit knowledge across borders^{34,35} and a principal route by which latecomers build the capacity to absorb what is produced elsewhere²²; if distance were what held latecomers back, scientists who trained and worked in an incumbent country and then returned home should move towards sparser ground. We test this on 36,887 scientists who moved from an entrant to an incumbent country and later returned to the same entrant country, comparing each with two scientists who stayed abroad, matched on origin country, first publication year, return year and pre-return research trajectory and position, in a weighted event study with scientist and calendar-year fixed effects (Methods).

The crowding of the new ideas returnees introduce rises after return, reaching +0.0086 above the counterfactual over $\tau \in [5,10]$ ($P < 0.05$; 65,791 authors, 577,379 author-years; Fig. 4c,d); on the knowledge-base layer, which covers 95.7% of author-years against 23.9%, the rise is +0.0146 ($P < 0.001$; Fig. S17). The drift is more pronounced among those who return to countries that expanded faster (Fig. S18)—precisely where the incentive of Fig. 4b does its work. A system that multiplies its research base within a short period places a large cohort of newcomers in competition for recognition under a common standard; the room for work that may fail narrows; and the higher private return on crowded ground becomes the rational choice. No subgroup is exempt. Measured as the change between the early window $\tau \in [0,2]$ and the late window $\tau \in [5,10]$ —which lies entirely after return and in which the reference terms cancel—crowding rises in every subgroup on both layers (+0.0089 to +0.0134 on new ideas, all $P < 0.05$; +0.0065 to +0.0168 on the knowledge base, all $P < 0.001$; Table S6). Those who returned with a strong record—above the median in cumulative top-10% cited

papers before return—drift like the rest, and so do those who went on working with co-authors in incumbent countries (Figs 4c,d and S17; Table S6). The knowledge-diffusion account predicts that precisely these two groups, the best equipped and the best connected, should resist the pull. They do not. Co-authorship is not the same thing as institutional access (Methods).

The adjustment begins before the move. On both crowding layers, returnees and their matched stayers move in parallel within the placebo window $\tau \in [-5, -4]$ (Table S8), but by the three years before return they have already diverged: relative to the stayers, returnees' crowding has risen (+0.0042 on new ideas, a layer that never entered the matching), their pivots have narrowed (−0.0049), their reliance on top institutions has increased (+0.0038), their shares of top-5% and top-1% cited papers have edged up (+0.0021 and +0.0012; $P = 0.096$ and 0.086) and their share of highly disruptive papers has edged down (−0.0006; $P = 0.059$) (Table S8 and Fig. S20). This is the shape of the evaluation standard they are about to enter—one that rewards impact rather than disruption—and returnees conform to it before they arrive. Return is a process, not a date.

The move itself severs one channel and opens another. Reliance on top institutions drops 0.0321 below the counterfactual over $\tau \in [0, 2]$, and highly cited output falls with it (top-5% share −0.0322, top-1% −0.0133; Fig. S19). At the same time pivot size jumps 0.0272 above the counterfactual as returnees carry knowledge acquired abroad into a new research setting and recombine it with what they find there (Table S9). This brings a measure of novelty: mean CD5 and the share of papers with $CD5 > 0$ both rise in the early window. Crowding, however, barely moves. That of new ideas dips by an amount indistinguishable from zero (−0.0031 over $\tau \in [0, 2]$), and that of the knowledge base edges up (+0.0024; Table S6). With no real move onto sparser ground, the upper tail does not move either: the share with $CD5 > 0.25$ is indistinguishable from the counterfactual (+0.0004; Table S7).

The later window, roughly a second funding cycle after return, shows environment and

incentives at work. Access to leading institutions is not regained but lost further, reliance lying 0.0379 below the counterfactual over $\tau \in [5,10]$. With that channel closed and a home system that rewards visible output, returnees are pushed onto crowded ground: between the two windows crowding rises by 0.0116 on new ideas and 0.0122 on the knowledge base and pivot size contracts by 0.0092 (Tables S6 and S9), so that the slight dip at return does not outlast the first cycle. Highly cited output recovers in part—38% of the initial shortfall in the top-5% share, but only 19% in the top 1%—and the cost falls on disruption, visible separately in the average and in the tail. At return, when the pivot jumps and the crowding of new ideas has not yet risen, the average moves and the tail does not; thereafter, in the late window, as the pivot contracts, mean CD5 and the share above zero rise no further, and as crowding rises, the tail falls: the share with $CD5 > 0.25$ declines by 0.0010 between the two windows ($P < 0.05$; Table S7), much as the share of highly disruptive papers edged down in the years before return, when crowding was also rising and pivots narrowing. In the cross-section, lower reliance on top institutions goes with more disruptive work (Table S3); here it does not, because what accompanies it is a contracting pivot and rising crowding. What changes between the two windows is not what the returnees carried home but the setting in which they use it. Along a single career they trade discovery for citation on exactly the terms that Fig. 4b records across countries.

Taken together, these results describe a loop. National money flows towards crowded ground, and flows there more the faster a country expands; evaluation rewards the countable output that crowded ground yields most readily; researchers, returnees among them, move to where that return is highest; the work they produce there is widely cited and rarely disruptive; and the impact it earns becomes the record on which the next round is allocated. Every party along the loop is rewarded, which is why it holds. Expansion tightens every link, multiplying the newcomers who compete under the same standard while shrinking the share of work that enters external frameworks—the one channel capable of pulling it back. Mobility does not break the loop: it brings knowledge home, but the setting to which it returns draws that knowledge onto crowded

ground. The only force we find that pulls a country back is entry into other countries' funding frameworks, and for the entrants that might be precisely what expansion eroded.

Discussion

The global expansion of science has not made breakthrough ideas easier to find, either for science as a whole or for the countries that joined it late. Participants multiplied, but the ground on which new ideas are born did not diversify. New work gathered where global attention already converged — ground that rewards citations but rarely yields work that breaks from the past — and latecomers gathered there most, so that their share of highly cited work rose while their breakthroughs did not keep pace. The problem is not that frontier knowledge failed to reach them, nor that they went unrecognized. They were recognized most readily on crowded ground, where breakthroughs are least likely. Our evidence instead points to the incentives built into the way they expanded as the force that placed their work there, and the Europe Agreement countries show that the resulting lock-in can be broken. The predicament of latecomer science is not that it cannot reach the frontier, but that the institutions which made its catch-up succeed have locked it into the wrong position.

None of this refutes the expectation that a larger and more diverse scientific community should discover more^{1,2}; the premise behind it, that newcomers search in new places, was simply not met. Nor is it enough to say that the low-hanging fruit has been picked and that ideas have become harder to find^{4,9}. Most of the penalty of crowded ground survives controls for accumulated work and for the number of attempts that followed, and heavily accumulated knowledge turns out to be more, not less, likely to yield top-tier disruption. What the expansion produced was a decoupling of impact from disruption^{3,25}, and because the loss comes from where incentives placed that effort, it can in principle be recovered. The decoupling is rational for each researcher and costly for science as a whole. Crowded directions repay their settled paradigms, ready judges and proven methods in citations¹⁴, but they narrow the search that breakthroughs require. What crowded ground lacks is not resources but room to depart from established paths, and this cost falls hardest on latecomers. On crowded ground their work is judged against a standard everyone shares, so their citations gradually close in on the incumbents'. On sparse ground its reception rests on a reputation they do not yet have.

The recognition they earn therefore pulls them deeper into the place where breakthroughs are least likely.

Distance from the frontier is the prevailing explanation of latecomers' lag^{22,23,36}. If it were the binding constraint, two groups should have resisted the pull towards crowded ground: countries close to the frontier by geography, culture and scholarly tradition, and scientists who brought frontier knowledge home with them. The Europe Agreement countries stood closer to the frontier than the entrants, yet before the agreements their European funding was no greater than the entrants', their funding from outside Europe stayed at the entrants' level throughout, their own money pushed work towards crowded ground just as hard, and each unit of European money pulled them back no further. Rather than resting on a single estimate, each of these comparisons rules out one way in which closeness to the frontier could have mattered — by widening access without the agreements, by easing their own funding's tilt, or by making each unit of foreign money go further — and only the channel the agreements opened remains. This is consistent with the view that differences between countries stem from their institutions rather than from geography or culture^{37,38}, carried here into the production of ideas. Returning scientists tell the same story at the individual level. They drifted onto crowded ground in every subgroup we examined, including those with strong pre-return records and those who kept collaborating abroad, and what changed between their first and second funding cycles after return was not what they knew but what the system around them rewarded.

Institutions built for catching up are not those needed at the frontier. Far from the frontier, growth comes from investment and incremental improvement; near it, from selecting for novelty and tolerating the failures that come with it. Institutions suited to the first stage can hold a country back when the task shifts, the non-convergence trap that Acemoglu, Aghion and Zilibotti describe, building on Gerschenkron's insight that latecomers need institutions of their own^{21,39}. Our evidence shows this sequence playing out inside science. Large-scale investment raised latecomers' output and impact quickly, but the faster a country expanded, the more both its funding and its papers tilted towards

crowded ground and the further its disruption fell. Funding and attention alike moved along the Matthew effect towards directions the world already attends to, the more so the faster the expansion, consistent with the selection pressure that a surge of applications creates. It is a system well built for impact and poorly built for breakthroughs. Returnees to faster-expanding countries, where more newcomers compete under a common standard, drifted more markedly. Such institutions also persist, since each round of countable output becomes the record on which the next allocation rests^{32,37}. The trap can also be read through economic complexity, in which our crowding measure corresponds to ubiquity^{40,41}. Crowding describes how common the ground where new ideas are born is, not how broad a country's research portfolio is. Economies that concentrate on goods everyone can make tend to stall, yet in science the same convergence on common ground has usually been read as joining the global research community. It may be the same trap in a different setting, and a harder one to recognize, because every step of the way it is rewarded.

Expansion also eroded the one channel capable of pulling entrants back. Entrants joined the world research system better connected to external frameworks than incumbents were, but domestic funding outgrew foreign funding, so their share of work inside external frameworks fell to the incumbents' level, while incumbents and Europe Agreement countries widened theirs as they grew. Several channels could account for this, among them a widening distance between the questions domestic and external funders select, and foreign agencies that do not enlarge their programmes because a partner has expanded. Crowding out is one likely cause among them. What declined was the share of externally funded work rather than its volume, and it declined most where expansion was fastest. This is the path dependence described above seen from the other side. As a domestic system grows, it absorbs more of its researchers' attention and organises more of their work around its own agenda, leaving external frameworks a smaller place within it. The force that pushes work towards crowded ground appears also to squeeze the channel capable of pulling it back.

Because the problem lies in position rather than in capacity, policy can address it, and

the Europe Agreement countries demonstrate this directly. The question is pressing now, as many countries expand research at unprecedented speed while the terms of cross-border research funding are being redrawn. Once European funding programmes opened to their researchers, their funded work moved from ground as crowded as the entrants' to ground less crowded than the incumbents'. The aim is not to avoid crowded ground, which consolidates knowledge and is where latecomers first earned recognition, but to loosen the lock-in that keeps them there. Four levers apply to any research system. Evaluation could rely less on bibliometric indicators, citation counts among them, and make more room for work on unfashionable questions. Selection could incorporate a partial lottery among fundable proposals of comparable quality⁴², as funders in Switzerland, Austria, Sweden and New Zealand already do^{43,44}, preserving reviewers' role in guarding quality while blunting the advantage of proposals that are most legible against a shared standard²⁹; its effect on research direction has not been assessed, and crowding offers a way to do so. Funding portfolios could reserve a deliberate share for sparse ground and offer longer, more failure-tolerant grants⁴⁵. Institutional access could be widened by opening funding frameworks to one another, and because foreign money lowers crowding in every country we examine, incumbents included — a regularity without exception on which policy can rely — such access is not aid to latecomers but a correction from which every participant stands to gain. For countries still building their research systems, expansion itself is the point of intervention. Access should be built in from the start, through binding arrangements and reciprocal co-funding, since what set the Europe Agreement countries apart was a treaty, not proximity, and it should keep pace with expansion. Returning scientists need their connection to external frameworks preserved, not only domestic funding and collaboration abroad. And evaluation reform is most pressing where evaluation leans hardest on countable metrics³³. On this evidence, the continued concentration of breakthroughs in North America and Europe tells us more about institutions than about distance.

These conclusions come with limits. We measure position rather than ability, with keyword-based knowledge units and a disruption index sensitive to reference

completeness, and the Europe Agreement evidence rests on eleven countries whose selection may have partly reflected institutional capacity. Two readings also remain open: that competition discourages frontier work far from the frontier^{46,47}, although what matters here appears to be the distance of the system rather than of the individual, a distance that institutional access may shorten; and that latecomers, facing a large competitive gap, may rationally shift towards variety-expanding and incremental innovation rather than creative destruction^{48,49}, which is consistent with what returnees showed on return, when their pivots widened and their average disruption rose while the upper tail did not follow. Yet the trade-off we document may extend beyond science to any system that steers catch-up by visible returns, and it may apply to the next expansion of science as well. That expansion will be driven by machines rather than by countries, and AI-driven research already concentrates on data-rich and popular problems⁵⁰. If it grows under the same incentive structure, it may crowd onto the same ground. Expansion determines how many take part in science. Institutions determine whether they can change it.

Methods

Data

Our analyses draw on three sources. Publication records, titles and abstracts, disambiguated author identifiers, institutional affiliations, fields of study, funder acknowledgements and citation links come from OpenAlex (release 26 June 2026), from which we retain 119,040,244 papers published between January 1966 and December 2021, with citations accumulated through 2025 so that a five-year citation window is complete for every paper in the observation period. Award titles and abstracts come from the National Science Foundation and the National Institutes of Health, downloaded from each agency's public award database. The discoveries underlying Nobel prizes in physics, chemistry and physiology or medicine, together with the papers reporting them, come from SciSciNet; because recognition typically lags by decades, the Nobel sample is restricted to 1971–2003.

Units of knowledge are keywords extracted from titles and abstracts with RAKE, which yields 1,505,343 candidates, not all of which name scientific concepts. To separate those that do, we drew a random sample of 5,000 candidates, presented each in place within the title and abstract of a paper using it, and had two annotators judge independently whether it names a scientific concept; the two agreed on 4,866 of them (97.3%), and these, of which 3,569 were positive, were split 8:1:1 and used to fine-tune SciBERT, which reaches an F1 of 0.920 on the held-out test set (precision 0.903, recall 0.938). Applying the classifier to up to three randomly drawn papers per candidate, we retain as units of knowledge the 1,137,039 candidates (75.5%) whose mean predicted probability is at least 0.5. Two units are linked at their first co-occurrence anywhere in the literature, so that every edge is dated by the moment its combination enters the literature (Fig. 1a); because crowding requires five years of prior history, 1971 is the earliest measurable year, giving 2,343,892 new ideas born between 1971 and 2017, of which 187,556 (8.0%) appear in an NSF or NIH award and 138 underlie Nobel-prize-winning work. Each new idea defines two five-year windows around its year of birth t . The earlier window, $t-5$ to $t-1$, is used to measure the crowding of the two units of

knowledge it connects, and the characteristics of the work done on them, before they met; the later window, t to $t+4$, is used to measure the impact and disruptiveness of the new idea itself.

Global crowding

For a new idea born in year t , crowding is computed over the preceding five years, $t-5$ to $t-1$. For each unit of knowledge k , let N_c be the total publication output of country c over the window and $N_{k,c}$ the number of that country's papers using k . The attention of country c to k , expressed relative to that country's own research volume so that the distribution is free of country-size effects, is

$$p_{k,c} = \frac{N_{k,c}}{N_c}, \quad c = 1, 2, \dots, n_k \quad (1)$$

where n_k is the number of countries with at least one paper on k . The evenness of this distribution is its entropy normalized by its maximum:

$$H_{norm}(k) = \frac{-\sum_{c=1}^{n_k} p_{k,c} \log(p_{k,c})}{\log(n_k)} \in [0, 1] \quad (2)$$

The volume of work a unit carries is a log-scaled transformation of $T_k \in (0, 100]$, the percentile rank of k 's publication count within the window:

$$f_k = \frac{\log(1 + T_k)}{\log(1 + T_{k \max})} \in [0, 1] \quad (3)$$

Global crowding is the product of the two:

$$GC_k = H_{norm}(k) \times f_k \in [0, 1] \quad (4)$$

and the crowding of a new idea is the mean of GC over the two units it connects. Both components are required: entropy alone would misclassify an obscure direction shared evenly among a handful of countries as crowded, whereas volume alone would misclassify a large programme run by a single country as globally invested. The measure approaches 1 when a unit draws attention from countries worldwide and carries a large volume of work, and 0 when attention is concentrated in a few countries

or the volume of work is negligible. A new idea enters the analysis only if its two units together carry at least 10 papers over t to $t + 4$.

Classification of entrant and incumbent countries

We restrict attention to countries with a sustained presence in high-impact research: a country enters the universe if its fractional share of world publications reaches 0.2% in either the base period (1971–1980) or the late period (2011–2015) and if it accumulates at least 3,000 papers in the top 10% of citations within their field and publication year over 1971–2015, which yields 50 countries. Shares are computed on fractional counts and smoothed with a five-year centred moving average. Because any single criterion for entry into the world research system has a known failure mode (Fig. S6), we classify countries by a vote across five independent strategies, each run over its own grid of thresholds: a level rule, based on a country's share in the base and late periods (18 variants); a growth rule, based on the ratio of its late-period to base-period share (18); a persistence rule, based on whether its share crosses a threshold during the window and stays above it (9); a diffusion rule, based on whether the inflection year of its share curve falls inside the window (8); and a policy rule, based on whether a documented turning point in national research policy falls inside the window (2). A country is assigned to a category if it receives at least three of the 55 votes for that category and more votes for it than for the other; countries meeting neither condition are left unclassified and enter only as part of the rest of the world. This yields 21 entrants and 13 incumbents, whereas individual variants identify between 7 and 33 entrants, and the decline in entrants' relative disruption intensity holds under all 55 (Fig. S6).

For each entrant, the entry year is the median of four signals: the year its share first reaches a set level, the years of maximum first and second differences in its share, and the documented policy turning point where one exists. Entry years range from 1982 to 2006, with a median of 1994, which serves as the treatment year in the synthetic control analyses; because entry is staggered, identification rests on in-space placebo tests rather than on a sharp break in time (Fig. S10). Eleven Central and Eastern European countries signed agreements on institutional and regulatory convergence with the European

Communities and their successors between 1991 and 2001; for these countries the entry year is the signature year, fixed by treaty rather than inferred from publication data. They are excluded from both the entrant group and the donor pool and serve instead as a comparison group. Romania, which signed in 1993 and also meets the publication-based entrant criteria in 1999, is assigned to the Europe Agreement group in the funding analysis and to the entrants elsewhere.

Institutional access

Funders named in a paper's acknowledgements are classified relative to the author country, and each paper is assigned to one of five mutually exclusive categories: domestic, where every named funder is a national body of the author country; supranational, where at least one named funder is supranational—the European Union's framework programmes, the European Research Council and a small number of international bodies classified by organisation type rather than by country of registration—denoted EU throughout, whatever else the paper names; foreign, where every named funder is a national body of another country; mixed, combining domestic and foreign national funders; and unfunded. Writing n_d , n_{EU} , n_x and n_n for the numbers of papers in the domestic, EU, foreign-plus-mixed and unfunded categories, and $n_f = n_d + n_{EU} + n_x$ for all funded papers, institutional access is the share of funded papers naming at least one foreign or supranational funder:

$$w = \frac{n_{EU} + n_x}{n_f} = w_{EU} + w_x, \quad w_{EU} = \frac{n_{EU}}{n_f}, \quad w_x = \frac{n_x}{n_f} \quad (5)$$

where w_{EU} is the EU share and w_x the non-EU foreign share, the latter covering papers that name at least one foreign national funder and no supranational one, whether or not a domestic funder is also named. w contains no crowding information.

Let $\bar{C}_d$, $\bar{C}_{EU}$, $\bar{C}_x$ and $\bar{C}_n$ be the mean within-year crowding percentiles of the papers in each category, $\bar{C}_{ext}$ that of the $n_{EU} + n_x$ externally funded papers and $\bar{C}_f$ that of all funded papers. The national-money gap and the all-funded gap measure how

far a group's nationally funded work, and its funded work as a whole, sit above its own unfunded work:

$$g_{dom} = \bar{C}_d - \bar{C}_n, \quad g_{all} = \bar{C}_f - \bar{C}_n \quad (6)$$

Each gap is then expressed as a multiple of the incumbents' national-money gap in the same window,

$$m = \frac{g}{g_{dom}^{inc}} \quad (7)$$

so that 1 marks the incumbent benchmark; the rationale for this double difference is given in the main text. The denominator is treated as a fixed scale rather than an estimate, since the 13 incumbents are a defined list and dropping any one of them moves it only between 0.0299 and 0.0350 at the last window.

Because every funded paper is either domestic or external, $\bar{C}_f$ is a weighted mean of $\bar{C}_d$ and $\bar{C}_{ext}$, and the all-funded gap decomposes exactly:

$$\bar{C}_f = (1-w)\bar{C}_d + w\bar{C}_{ext} \Rightarrow g_{all} = g_{dom} - w \cdot \Delta, \quad \Delta = \bar{C}_d - \bar{C}_{ext} \quad (8)$$

The three terms separate the two forces acting on funded work. g_{dom} is the push: how far a group's own money places its work on crowded ground. $w \cdot \Delta$ is the pull-back: how far external money draws that work back, the product of how much of the group's funded work enters external frameworks (w) and how much less crowded that work is than nationally funded work (Δ). g_{all} is the net position of the group's funded work. The identity has no residual, so any difference between groups in the position of their funded work must appear in g_{dom} , in w or in Δ .

The pull-back splits further into the two channels through which external money enters:

$$w \cdot \Delta = w_{EU} \cdot \Delta_{EU} + w_x \cdot \Delta_x, \quad \Delta_{EU} = \bar{C}_d - \bar{C}_{EU}, \quad \Delta_x = \bar{C}_d - \bar{C}_x \quad (9)$$

Each channel contributes the share of funded work entering it times its per-unit pull. This separates the breadth of access from its depth: a group may obtain more pull-back because more of its work enters a channel (w_{EU} or w_x), or because each unit of that

channel pulls further (Δ_{EU} or Δ_x). The extra pull-back of a group is its $w \cdot \Delta$ minus that of the incumbents in the same window.

The Europe Agreement countries allow us to ask whether access moves the position of funded work or merely accompanies other advantages. The agreements play the part an instrument for access would, and we assess the three conditions such a role requires rather than estimating a two-stage model. For relevance, the agreements must raise access: the EU share of the Europe Agreement countries relative to the entrants rises after accession, and it does so in each of the 11 countries after its own accession year. For exogeneity, access must neither precede the agreements nor share a common cause with them: the EU-share ratio is flat before signature, and the non-EU foreign share, which the agreements do not govern and which a common endowment of proximity would also have raised, stays at the entrants' level throughout. For exclusion, the agreements must act on the position of funded work only through access: by identity (8), any other route would have to pass through g_{dom} or through the per-unit pull, yet g_{dom} is indistinguishable between the two groups and so is Δ_{EU} , leaving w as the only term through which they differ. One concern cannot be tested with these data: the Communities may have selected partners partly on institutional capacity, which could independently raise participation in the framework programmes.

w records entry into another country's funding framework rather than co-authorship: a foreign collaborator listed as an author who attaches no grant does not raise it, whereas a domestic researcher working inside a foreign grant does. The two carry different content. A name on an author list transfers credit and visibility; entry into a funding framework transfers the criteria by which questions are judged worth pursuing, the review that enforces them and the working relationships through which such criteria are learned, knowledge that passes only through participation, and the terms of the arrangement—how much failure it tolerates, how long its cycle runs, how its review is pitched—change what researchers attempt independently of the sum involved. The effect is directional, but its direction is set by who is new to the framework rather than

by nationality. For a researcher from an entrant country, entering an external framework is exposure to a different standard of question selection. Between two incumbent countries, a cross-border collaboration in which each side brings its own grant is costly in coordination, distance and administration, and arises only when the expected return covers those costs, which for teams already at the frontier usually means a less obvious project. The first is exposure and the second selection; both move work towards sparser ground, which is why Δ is positive in every group while the mechanism differs.

All quantities are computed on both the new-ideas and the knowledge-base layers, in rolling ten-year windows and in three periods (1991–2017, 1991–2003 and 2004–2017). A country enters a window only if it has at least 200 funded and 200 unfunded papers there; on the new-ideas layer the rolling series begins with the window ending in 2000, since earlier windows leave fewer than two-thirds of the Europe Agreement countries above this threshold. Confidence intervals are computed at the paper level from $\text{Var}(\text{percentile}) = 1/12$, which is exact for a within-year percentile by construction; because they treat papers as independent, they are a lower bound. The Z statistic for an extra pull-back is its value divided by its standard error. For the country-level analyses (Figs S13–S16), the same quantities are computed for each country over 2011–2015 and related to its annualised growth in output since the five years before entry, with sample thresholds given in the figure captions.

Returning scientists and the matched comparison group

We identify returning scientists from annual publication records. A scientist's location in a given year is the country type contributing the largest number of that year's papers, with ties resolved in favour of the entrant country; years in which no institutional country information is available are left missing. Consecutive years of the same type are merged into spells, missing years do not break a spell, and spell length is the calendar span between a spell's first and last observed year. A scientist's origin country is the entrant country appearing most often in the earliest entrant spell.

A return requires a spell of at least four years in an incumbent country, followed within two years by a spell of at least five years back in the origin country; where a scientist

returns more than once, the first return is used. Stayers never satisfy this pattern, remain in an incumbent country in their final spell, and have published from an entrant country at some point in their career. Each stayer is assigned pseudo-return years placed at least four years into an incumbent spell, with at least five further years observed. Both groups are required to be at least five years past their first publication at the return or pseudo-return year, so that the same career-stage restriction applies on either side of the comparison. This yields 45,398 returnees and 2,761,323 stayer $\times$ pseudo-return-year candidates drawn from 185,602 stayers, of which 36,887 returnees have complete pre-return covariates.

Matching proceeds in two stages. Each returnee is first placed in an exact block defined by origin country, first publication year within one year, and return year within two years. Blocking on origin country holds constant the national research system to which the scientist returns, and blocking jointly on first publication year and return year fixes career age at return. Within a block, the two nearest stayers are selected on a standardised Euclidean distance over five covariates measured in $\tau \in [-3, -1]$, with each scale factor taken as the standard deviation of the candidate pool: the level and slope of crowding of the knowledge base, which fix the pre-return research position and the direction in which it was already moving; log annual output, which fixes the scale of production; and the shares of first- and last-authored papers, which fix authorship role and, with career age already held constant by the blocks, the stage of scientific independence. Candidates must have at least two years with a defined crowding value in that window. Matching is 1:2 with replacement, with returnees carrying weight 1.0 and each matched stayer 0.5, so that total control weight equals the number of returnees. Two matches rather than one are used because the median distance to the second neighbour exceeds that to the first by only 19% (0.507 against 0.399). The procedure yields 36,887 returnees and 65,392 distinct stayer $\times$ pseudo-return-year units drawn from 44,304 distinct stayers.

Crowding of new ideas, the primary outcome, is deliberately excluded from the matching distance. It is defined for only 23.9% of author-years, against 95.7% for

crowding of the knowledge base, so requiring it to be non-missing would restrict the comparison to returnees with unusually dense publication records and change the estimand. It enters instead as an out-of-sample check and is balanced in both the matching and the placebo window (Table S5). Two further pre-return quantities are excluded by design: the share of internationally co-authored papers and the cumulative count of top-10% cited papers are the pre-return counterparts of the variables used later to split returnees by post-return incumbent ties and by pre-return prestige, and matching on them would align the groups in advance and remove the contrast those splits are meant to test. Balance on all matched and unmatched covariates, in both windows, is reported in Table S5. Conclusions are unchanged when the block additionally requires the same transit incumbent country; when a sixth distance dimension is added measuring the Euclidean distance between the two scientists' field-frequency distributions over their entire pre-return publication record, with the two nearest neighbours selected on that augmented distance; under career-age thresholds of five, six, seven or eight years; under 1:1 and 1:3 rather than 1:2 matching; when China is excluded; and when returnees are split by return cohort.

Estimation and pre-return behaviour

Outcomes are measured at the paper level as within-year global percentiles, so that a paper's value depends only on which keywords or keyword combinations it uses and not on where it was written. An author-year value is the sum of percentiles across that year's papers divided by the number of papers. Series are smoothed over a ± 1 -year rolling window that pools papers rather than averaging annual means, is computed on a completed event-time grid so that missing years enter neither numerator nor denominator, and does not cross $\tau = 0$, so that $\tau = -1$ uses years $\{-2, -1\}$ and $\tau = 0$ uses $\{0, +1\}$. The rolling grid is keyed on author $\times$ return year rather than on author, since replacement allows one stayer to enter with several pseudo-return years; keying on author alone would pool observations from different calendar years into the same event-time cell. Observations require at least one paper in the smoothed window and are truncated at 2017. Crowding of the new ideas a scientist introduces is the primary

outcome throughout, consistent with the object of this study; crowding of the knowledge base, which covers 95.7% of author-years, is reported alongside it (Fig. S17).

We estimate

$$y_{it} = \sum_{\tau \neq -1} \beta_{\tau} \cdot \mathbf{1}\{\text{returnee}_i, \tau_{it} = \tau\} + \alpha_i + \delta_t + \varepsilon_{it} \quad (10)$$

where α_i is an author $\times$ return-year fixed effect and δ_t a calendar-year fixed effect.

The omitted category comprises $\tau = -1$ together with all control observations, which identify the calendar-year effects. The within transformation uses the same weights as the regression. No time-varying controls are included: output and authorship shares are themselves post-return outcomes, and career age is collinear with the two sets of fixed effects.

The event window is estimated over $\tau \in [-5, +12]$ and reported over $\tau \in [-5, +10]$. Estimation extends beyond the reporting window because under ± 1 smoothing the value at $\tau = +10$ pools years $\{9, 10, 11\}$, which a panel truncated at $\tau = +10$ could not supply. Reporting stops at $\tau = +10$ because returns are restricted to 2013 and earlier so that the return year itself is observed, crowding is measured to 2017, and only returns up to 2007 can be followed for a full decade. Returnees are on average 14.9 years past their first publication when they return, so $\tau = +10$ already corresponds to the twenty-fifth year of a career.

Coefficients are shifted by a constant so that three pre-return reference years, $\tau = -5$, -4 and -1 , average to zero. No single year is therefore fixed at zero, and the estimate at $\tau = -1$ equals minus that mean and carries its own sampling error. Shifting moves the whole path and leaves every difference between two points, and every test, unchanged. Confidence bands are computed for the shifted coefficients, so that the variance of each point is that of the same linear combination of β that the point represents rather than the variance relative to the omitted year. Summary contrasts are linear combinations of β : the early effect is the mean over $\tau \in [0, 2]$, the late effect the mean over $\tau \in [5, 10]$, and their difference is measured entirely within the post-return period, so that the reference terms cancel and it is unaffected by any pre-return trend. Standard errors for all

contrasts are computed from the same covariance matrix, are clustered by author, and use disjoint-sample standard errors for between-group differences, since no unit appears in both arms of any split.

Pre-return behaviour is tested in the placebo window $\tau \in [-5, -4]$, which is excluded from the matching distance. Placebo slopes are indistinguishable from zero in both crowding layers and in every subgroup. The matching window $\tau \in [-3, -1]$ shows systematic movement, and the evidence that this is anticipation rather than a failure of the design comes from outcomes that entered no matching dimension: pivot size falls and reliance on top institutions rises in that window, both at $P < 0.001$, and both also move within the placebo window itself, while the shares of top-5% and top-1% cited papers rise and the share of highly disruptive papers falls. Return is a process rather than a date, and scientists adjust before they move. Two readings of this adjustment are observationally identical here — the scientists have already decided to return and are adapting to the criteria of the receiving system, or receiving institutions recruit on those criteria so that those hired already display the pattern — and because unsuccessful applicants are not observed, they cannot be separated. This ambiguity bears on the interpretation of the pre-return window rather than on the post-return path, since the difference between the late and early effects is measured entirely after return. Full pre-return diagnostics are in Table S8.

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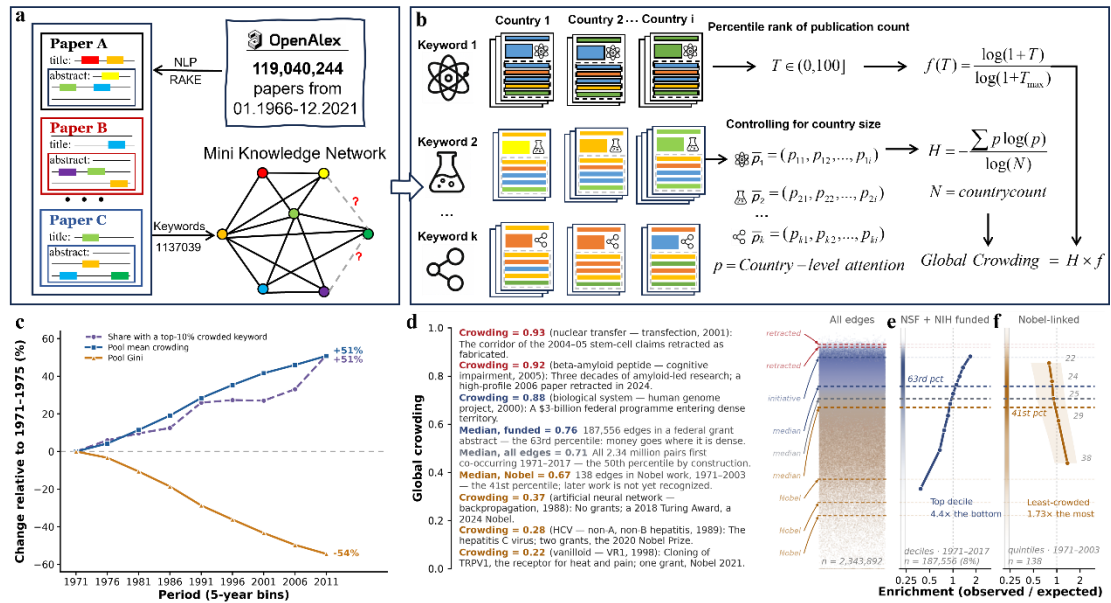


Fig. 1 Overview of global crowding. **a**, We illustrate the construction of the keyword network. RAKE is applied to the titles and abstracts of 119,040,244 OpenAlex papers published between 1966 and 2021, yielding 1,505,343 candidate keywords, of which 1,137,039 are retained as scientific concepts by a fine-tuned SciBERT classifier (Methods). Two keywords are linked when they co-occur in a paper, and we retain each pair's first co-occurrence anywhere in the literature (grey dashed edges), so that every edge is dated by the moment its combination enters the literature. **b**, We illustrate the calculation of global crowding, which quantifies how widely and how intensively the world attends to a keyword. For each keyword and each five-year window, papers using it are counted by country and divided by that country's total output, giving a country-level attention vector p that is free of country-size effects. Crowding is $H \times f$ and lies on the $[0, 1]$ interval, where H is the entropy of p normalized by $\log N$ (N , number of countries) and $f(T) = \log(1+T)/\log(1+T_{\max})$ rescales T , the percentile rank of the keyword's publication count. The crowding of an edge is the mean of its two keywords, measured over the five years preceding its first co-occurrence; because five years of history are required, 1971 is the earliest measurable year. **c**, The rise of crowding is shown relative to the 1971–1975 baseline in 5-year bins (final bin, 2011–2017). The mean crowding of all keywords rose by 51% and the Gini coefficient of crowding across keywords fell by 54%, while the share of new edges with at least one keyword in the top decile of that year's crowding distribution rose by 51%. Crowding has thus become both higher and more evenly shared, and new combinations increasingly form on ground that is already dense. **d**, The distribution of edge crowding is shown, with each dot representing a single first co-occurrence ($n = 2,343,892$). The vertical axis reflects the value of crowding, while the horizontal axis is jittered to minimize overlapping points; denser (darker) regions indicate more commonly observed values. Dashed lines mark the medians of edges appearing in an NSF or NIH grant title or abstract (0.76, 63rd percentile), of all edges (0.71, 50th percentile by construction) and of Nobel-linked edges (0.67, 41st percentile), together with six illustrative combinations spanning the range. **e**, We show the enrichment (observed/expected) of edges appearing in an NSF or NIH grant title or abstract, by decile of edge crowding ($n = 187,556$, 8.0% of all edges, 1971–2017), defined as the probability of appearing in a grant within a decile divided by the overall probability. Shading denotes 95% confidence intervals, and the ramp at left shows the crowding distribution of these edges. Edges in the top decile are

enriched $4.4\times$ relative to the bottom. **f**, As in **e**, for edges underlying Nobel-prize-winning work ($n = 138$, 1971–2003, in quintiles; italic numbers indicate edges per quintile). The least-crowded quintile is enriched $1.73\times$ relative to the most-crowded. We observe that funding concentrates where the frontier is dense, whereas prize-winning work does not.

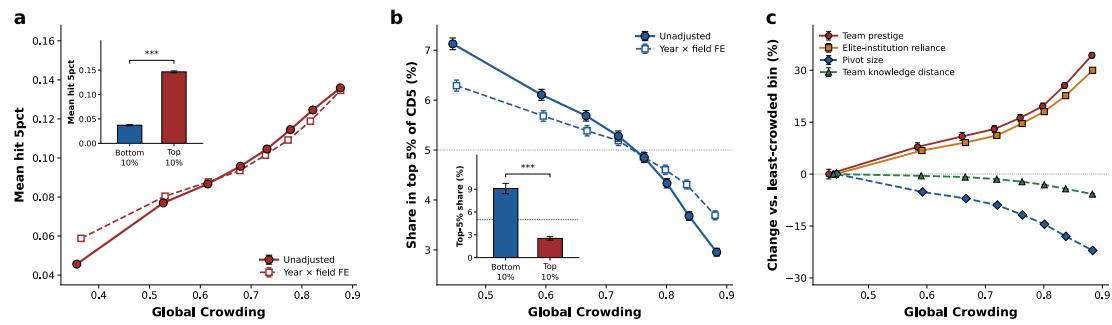


Fig. 2 Impact dividend and breakthrough penalty of crowded combinations. a, Relationship between the global crowding of a new combination (over the five years before its first co-occurrence) and its impact (over the five years after). $n = 2,343,892$ keyword pairs first co-occurring between 1971 and 2017. More crowded combinations exhibit higher mean values in impact (Hit 5pct, **a**). Solid curves are unadjusted; dashed curves are adjusted for year and field. Error bars denote 95% confidence intervals. Inset, combinations whose two keywords both fall in the bottom versus both in the top decile of that year's crowding distribution ($n = 25,886$ and $20,043$). Asterisks indicate two-sided Welch's t -test between the two group means. **b,** As in **a**, for disruptiveness (Top-5% share of CD5, $n = 2,343,892$). The relationship reverses: more crowded combinations are less likely to yield the most disruptive work. The horizontal grey line marks 5%, the share expected if such work were distributed equally across levels of crowding. **c,** Four characteristics in the later period (the five years after), expressed as the percentage change relative to the least-crowded bin. Points indicate group means; error bars denote 95% confidence intervals. Team prestige and elite-institution reliance rise with crowding, whereas pivot size and team knowledge distance fall. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

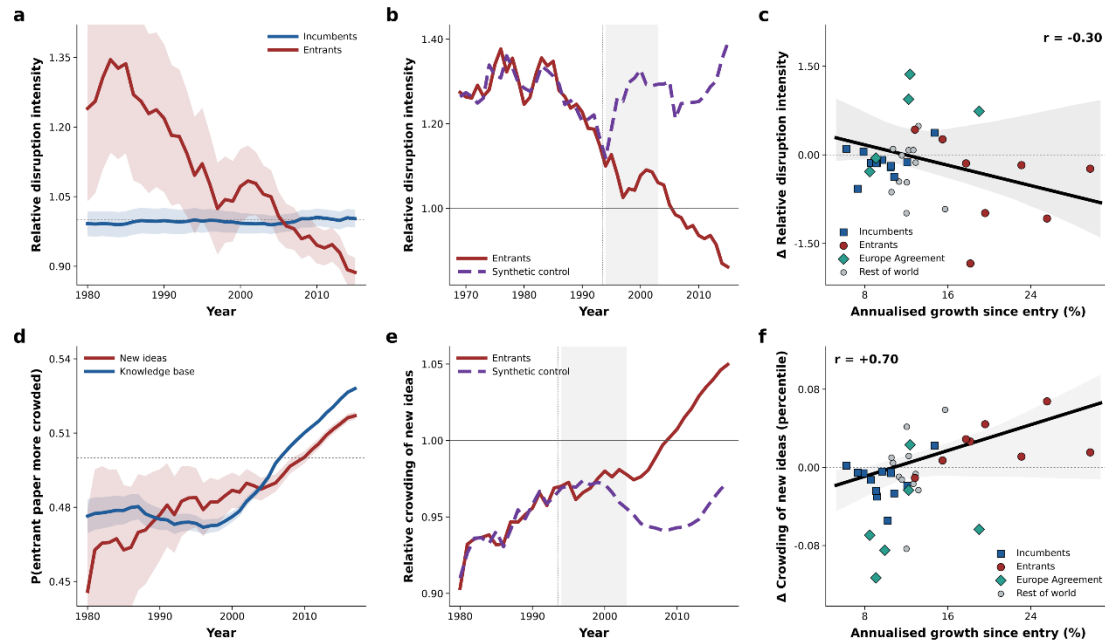


Fig. 3 Countries that expanded fastest moved onto the most crowded ground and lost their disruptive advantage. **a**, Entrant countries (21 countries that entered the world research system during the observation period) began well above the world average in disruption intensity but declined steadily and fell below it by the mid-2000s, while incumbent countries (13 established at the start of the period) held at the global mean throughout. Status is assigned by vote across 55 classification variants (Methods). Disruption intensity is the share of a group's top-5%-cited papers that are disruptive ($CD5 > 0.25$), divided by the contemporaneous world share, so that 1 denotes the global average; values are computed on 10-year trailing windows and plotted at the final year, with shading showing 95% confidence intervals. **b**, A counterfactual path for entrant countries is fitted by synthetic control from 25 untreated donor countries (countries classified as neither entrants nor Europe Agreement signatories, from which the counterfactual is composed), with treatment set at 1994 (the median entry year of the entrant group). Entrants diverge sharply downwards from this counterfactual, indicating that the decline does not track global trends. Because entry is staggered across 1982–2006, the divergence phases in rather than breaking sharply; the grey band marks windows still spanning pre-entry years. All 55 classification variants yield a decline, and 49 fall below the world benchmark (Fig. S6). **c**, The decline in disruptiveness was steepest where output grew fastest. Each point is a country; the horizontal axis is annualised growth in publication output between the five years preceding entry and 2011–2015, and the vertical axis the change in relative disruption intensity over the same interval, each measured against the contemporaneous world rate. Countries without an entry year are dated from 1994. Europe Agreement countries (11 countries that signed agreements with the European Communities between 1991 and 2001, committing to institutional convergence) provide an exogenous benchmark: most rose rather than fell in disruption intensity, appearing in the upper part of the panel. **d**, The difference in crowding between entrant and incumbent papers, measured by AUC, where 0.5 denotes no difference. Entrant papers once drew on less crowded ground but crossed over in the mid-2000s and have been the more crowded of the two since. Red, crowding of the new ideas a paper introduces; blue, crowding of the units of knowledge it builds on (Fig. 1a,b); shading shows 95% confidence intervals. **e**, Crowding of entrant papers against a counterfactual fitted by synthetic control under the same specification

as **b**. Entrants rose far above their counterfactual, meaning that their papers increasingly built on knowledge that had already drawn heavy global investment — the mirror image of the disruption result. **f**, Countries that expanded fastest moved furthest onto crowded ground. Axes and country dating follow **c**, with the change in mean crowding percentile on the vertical axis. Most Europe Agreement countries fell rather than rose in crowding, appearing in the lower part of the panel, consistent with their disruption result. Together with the breakthrough penalty of crowded combinations (Fig. 2b), these results indicate that the geographic expansion of science placed its newest participants on precisely the ground least likely to yield breakthroughs. Full definitions, country lists, donor pools, alternative treatment years and unweighted specifications are given in Methods and Figs. S6 and S10.

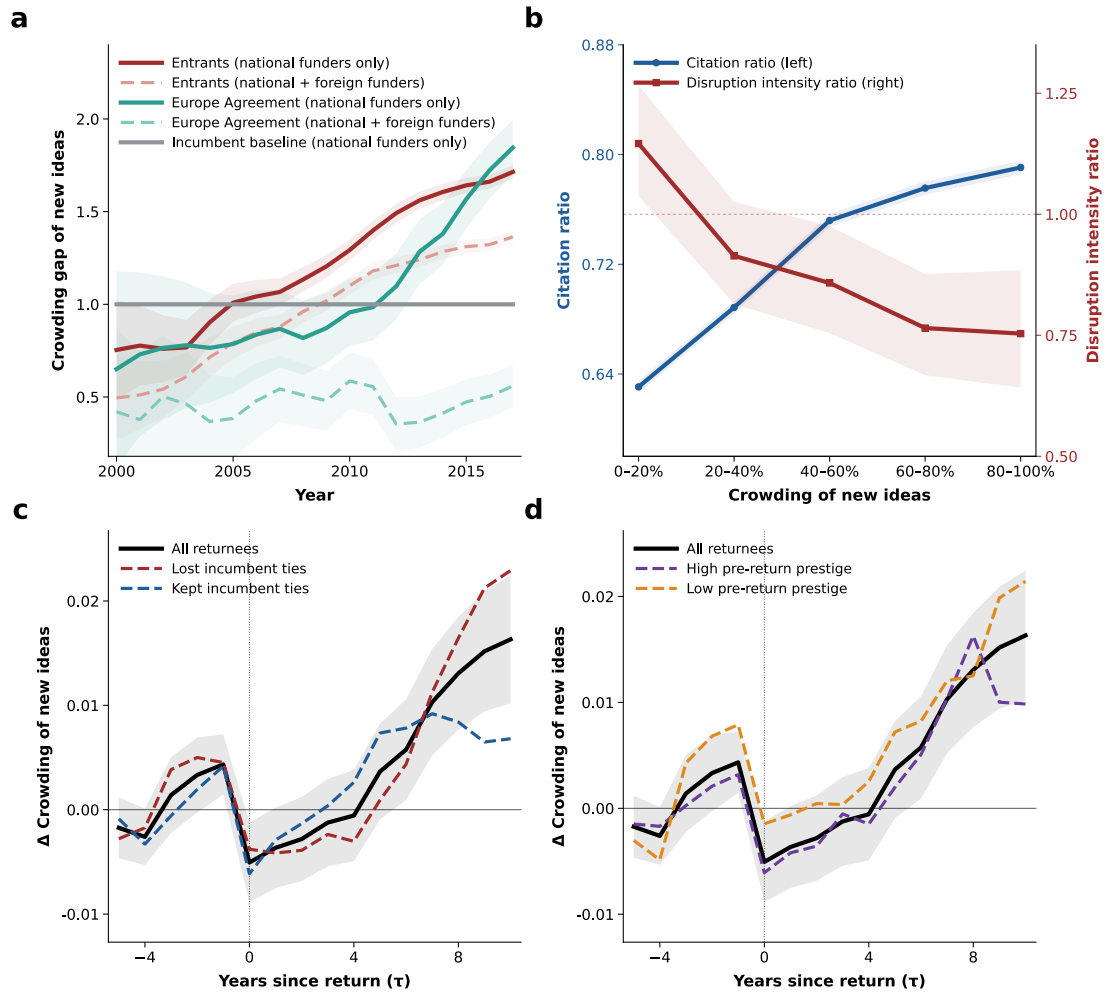


Fig. 4 The incentives behind global crowding. **a**, National money pushes every group towards crowded positions; foreign money pulls back, but only far enough to matter in the Europe Agreement countries. Rolling ten-year windows, plotted against the year in which the window ends, so the rightmost point covers 2008–2017. For each group and each window we compute the crowding gap: the mean within-year crowding percentile of the group’s funded papers minus that of its own unfunded papers, measured on New ideas. Solid lines count only papers whose funders are all domestic; dashed lines add every paper acknowledging at least one foreign or supranational funder. The series are divided by the incumbents’ national-money gap in the same window, so 1.0 is the grey reference line and the y-axis reads as multiples of the incumbent benchmark. Shaded bands are 95% confidence intervals. At the last window the entrants’ nationally funded work sits at $\times 1.71$ (95% CI 1.68–1.75) and their funded work as a whole at $\times 1.36$ (1.33–1.40); the Europe Agreement countries reach $\times 1.85$ (1.70–1.99) on national money alone but $\times 0.56$ (0.45–0.67) once foreign money is included. The two groups start from similar positions in 2000 ($\times 0.75$ and $\times 0.65$) and the Europe Agreement countries end above the entrants on national money alone ($\times 1.85$ against $\times 1.71$), so the divergence is confined entirely to the dashed lines. The construction is a double difference. Subtracting a group’s own unfunded papers removes whatever is common to that country and year — field mix, language, and the secular thickening of the knowledge space, which moves funded and unfunded papers alike. Dividing by the incumbents removes the global time trend in the funding premium itself. The normalising constant at the last window is +0.0327 percentile points; it is treated as a

fixed scale rather than an estimate, since the thirteen incumbents are a defined list and leave-one-out moves the constant only between 0.0299 and 0.0350, so the intervals shown come from the numerator alone. The series begins in 2000 because earlier windows leave fewer than two-thirds of the Europe Agreement countries above the sample threshold on New ideas. Group membership is fixed throughout: 13 incumbents, 20 entrants, 11 Europe Agreement countries. **b**, Private and social returns across crowded positions. All papers are divided into five equal groups by crowding of new ideas (horizontal axis gives the percentile among all papers with a value of the measure in that year), and entrant and incumbent countries are compared within each group. The blue line is the ratio of the probability of being a top-5% cited paper (entrants $\div$ incumbents, left axis); the red line is the ratio of the share of top-5% papers with CD5 > 0.25 (right axis). The dashed line at one marks parity. Shading gives 95% confidence intervals from a binomial approximation at the paper level. 1971–2021, 13 incumbent and 21 entrant countries, whole counting. The citation gap narrows as crowding rises: the entrant-to-incumbent ratio increases from 0.63 in the least crowded group to 0.79 in the most crowded (entrants 15.83% $\rightarrow$ 19.37%, incumbents 25.10% $\rightarrow$ 24.50%). The disruption gap moves the other way: the ratio falls from 1.15 to 0.75, so entrants are 15% more disruptive than incumbents at the least crowded positions and 25% less disruptive at the most crowded ones (entrants 0.642% $\rightarrow$ 0.178%, incumbents 0.560% $\rightarrow$ 0.236%). The two curves move in opposite directions: the same position offers entrants a higher citation return and a lower discovery return. Results on the knowledge-base layer are shown in Fig. S12. **c,d**, Crowding of new ideas after return. Event-study estimates from a matched design in which each returnee is compared with two stayers matched on origin country, first publication year, return year and pre-return research trajectory and position. The outcome is the within-year global percentile of the keyword combinations a paper uses, pooled over a three-year rolling window. Coefficients are normalised so that three pre-return reference years ($\tau = -5, -4$ and -1) average to zero, so no single year is fixed at zero. The shaded band is the 95% confidence interval for the all-returnees series only; subgroup series are shown without bands for legibility and the corresponding intervals are given in Table S6. Standard errors are clustered by author. **c**, Returnees split at the median of the share of their papers co-authored with a researcher in an incumbent country in the return year and the two years that follow. **d**, Returnees split at the median of cumulative top-10% cited papers before return. Both splits are shown against the pooled series in black; control observations enter both panels through the same matched design. Across all returnees crowding falls slightly at return and rises thereafter, reaching +0.0086 over $\tau \in [5, 10]$ ($P < 0.05$, $n = 65,791$ authors and 577,379 author-year observations). The rise is larger among returnees who lost incumbent ties (+0.0103, $P < 0.05$) than among those who kept them (+0.0061, $P = 0.21$), and larger among the low pre-return prestige (+0.0108, $P < 0.05$) than among the high pre-return prestige (+0.0071, $P = 0.12$), but none of the between-group differences reaches conventional significance in this layer ($P = 0.15$ to 0.91). Crowding of new ideas is observed for 23.9% of author-years against 95.7% for the knowledge base, so its standard errors are two to three times larger; the corresponding knowledge-base panels are shown in Fig. S17.

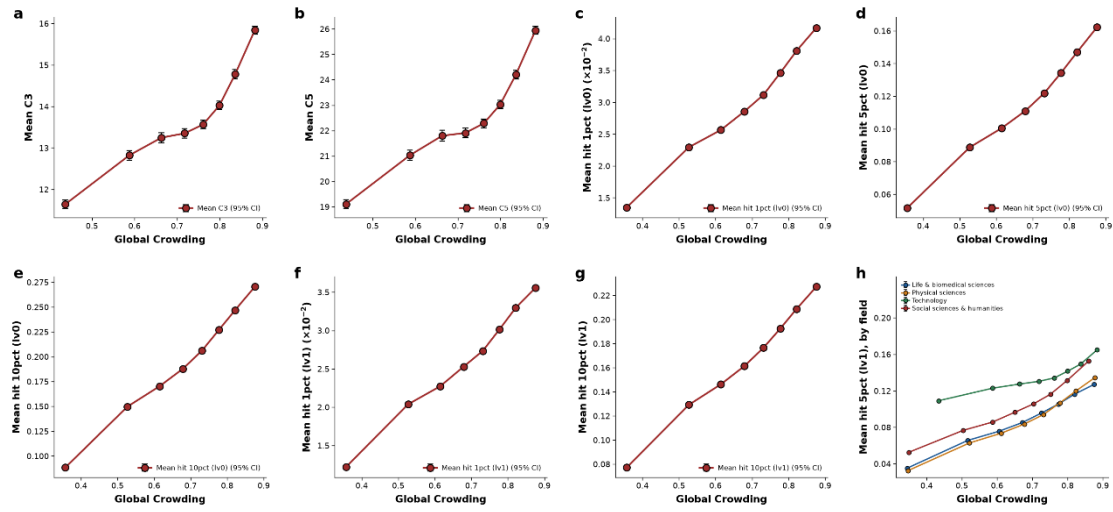


Fig. S1 Robustness of the impact relationship to alternative citation measures. **a**, Mean C₃. **b**, Mean C₅. **c–e**, Mean hit 1pct, 5pct and 10pct, normalized within year and level-0 concept. **f**, **g**, Mean hit 1pct and 10pct, normalized within year and level-1 concept. **h**, Mean hit 5pct (level-1) computed separately within each of the four macro-fields. All panels are as in Fig. 2a. Points are means of eight equal-count crowding bins; error bars, 95% confidence intervals. $n = 2,343,892$ keyword pairs first co-occurring between 1971 and 2017.

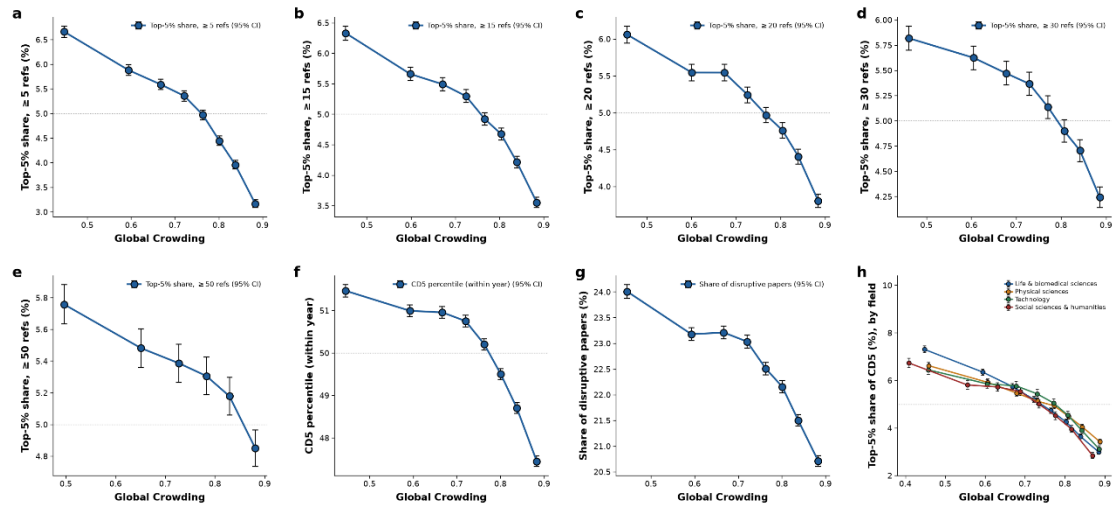


Fig. S2 Robustness of the disruption relationship to reference-count restrictions and alternative measures. **a–e**, As in Fig. 2b, but restricted to papers with at least 5, 15, 20, 30 and 50 references, respectively, addressing the sensitivity of CD5 to reference-list length. **f**, Mean within-year percentile of CD5. **g**, Share of disruptive papers (CD5 > 0). **h**, As in Fig. 2b, but computed separately within each of the four macro-fields. Points are means of eight equal-count crowding bins; error bars, 95% confidence intervals; horizontal grey lines mark the value expected in the absence of an effect. $n = 2,343,892$ keyword pairs.

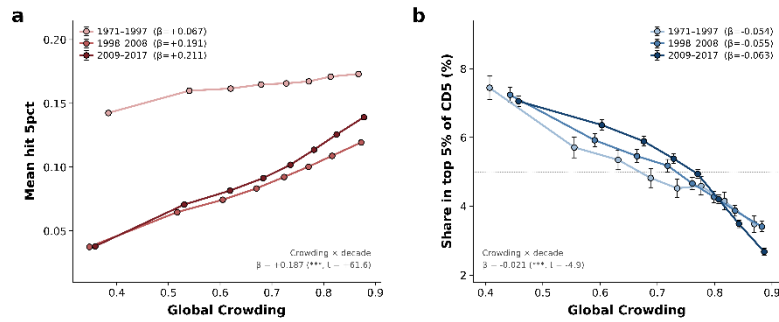


Fig. S3 Crowding effects strengthen over time. a, The association between crowding and high citation impact has become increasingly positive across successive cohorts. **b**, Over the same period, the association between crowding and top-5% disruption has become increasingly negative.

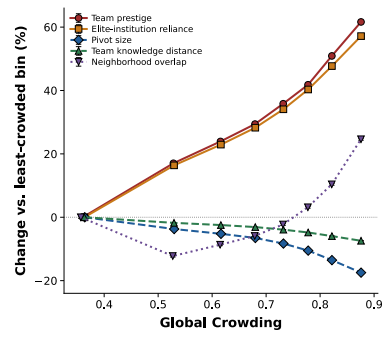


Fig. S4 Mechanisms across levels of global crowding. Five mechanisms in the earlier period (the five years before), expressed as the percentage change relative to the least-crowded bin. Points indicate group means; error bars denote 95% confidence intervals. Team prestige, elite-institution reliance and neighborhood overlap rise with crowding, whereas pivot size and team knowledge distance fall.

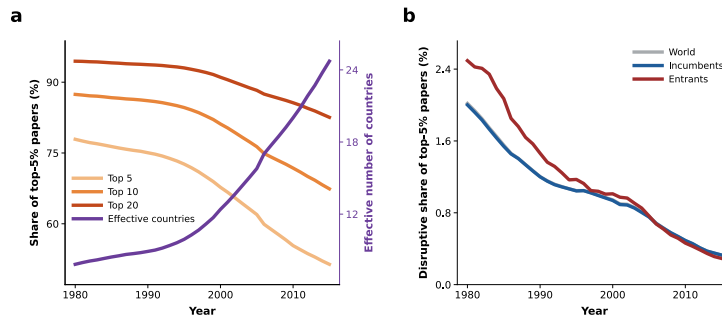


Fig. S5 Geographic spread of high-impact research and the world-wide decline in disruption. Both panels use ten-year backward rolling windows; the horizontal axis is the end year of the window, so the first point covers 1971–1980 and the last 2006–2015. **a**, Concentration of top-5% cited papers across countries. Papers are whole-counted, consistent with the main analysis, with each paper counted once for every country represented among its authors. National shares are calculated relative to the sum of these country-level counts. Orange lines show the combined shares of the five, ten and twenty largest producers; the purple line shows the effective number of countries, calculated as the exponential of the Shannon entropy of the national share distribution. The top-five share falls from 77.9% to 51.4%, and the top-twenty share from 94.4% to 82.6%, while the effective number of countries rises from 7.8 to 24.7. Under fractional counting, each paper is divided equally among its author countries. This approach yields a similar pattern, with the top-five share declining from 80.0% to 55.0% and the effective number of countries increasing from 6.9 to 21.1. **b**, Share of top-5% cited papers with a CD5 index above 0.25, for the world, the thirteen incumbent countries and the twenty-one entrant countries. The denominator is restricted to highly cited papers with a computable CD5 index to ensure the quality of works. The world share falls from 2.02% to 0.33%, a decline of 84%. Entrant countries begin at 2.49%, well above the world level, and converge to it by the mid-2000s, ending at 0.29%; incumbent countries track the world series throughout. The world and incumbent-country series nearly overlap, leaving the grey world line largely obscured by the blue incumbent-country line. Fig. 3a expresses the entrant and incumbent series as ratios to the world series.

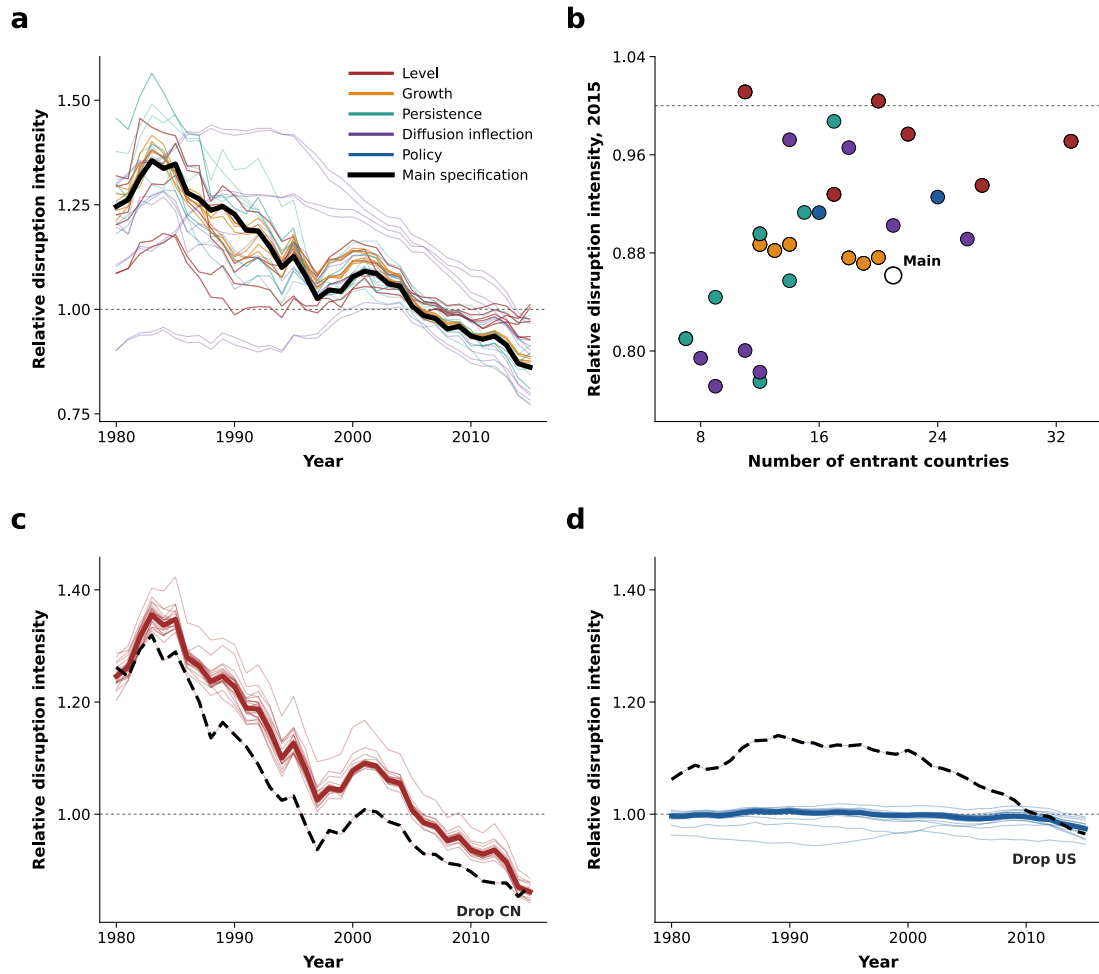


Fig. S6 Robustness of the country classification. **a** and **b** test the classification rule itself; **c** and **d** test individual members of the final lists. All panels use top-5% papers, $CD5 > 0.25$ and ten-year backward rolling windows. Entrant and incumbent countries are determined by a vote across five independent strategies: level (low share in the base period, high share at the end), growth (multiple by which the share rises), persistence (crossing a threshold and staying above it for N years), diffusion inflection (the inflection year of the share curve falling inside the window), and policy (a documented turning point in research policy). Each strategy carries its own grid of threshold parameters, with 18, 18, 9, 8 and 2 settings respectively, giving 55 variants in total. Using five strategies together is necessary because each has a known failure mode on its own: a pure level rule classifies countries whose base-period share is already substantial as incumbents, while a pure growth rule classifies very small countries with noisy growth as entrants. The main specification requires a country to receive at least three votes and to have a plurality. **a**, Relative disruption intensity of the entrant group identified by each of the 55 variants; colour indicates strategy and the black line is the main specification. All 55 variants decline, by between 0.101 and 0.561 (median 0.336; main specification 0.384). Starting values range from 0.902 to 1.457 and end values from 0.771 to 1.011, with 49 variants ending below the world benchmark. **b**, Number of entrant countries identified by each variant against its end value, with point colour matching the strategies in **a**. The count ranges from 7 to 33 (main specification 21, open

circle), so the different strategies and thresholds yield lists differing nearly fivefold in size, while all 55 series in a decline. **c**, Series recomputed after dropping one entrant country at a time (21 lines); the heavy line is the baseline using all 21 and the dashed black line drops China. End values range from 0.842 to 0.885 and all fall below the world benchmark; dropping China gives 0.872 against a baseline of 0.862, so China is not what sustains the result. **d**, Series recomputed after dropping one incumbent country at a time (13 lines), on the same vertical scale as **c**. The dashed black line drops the United States, which rises to 1.140 around 1989 and falls back to 0.965 after 2010, deviating from the baseline by at most 0.136; the remaining 12 countries end between 0.946 and 0.995. The dashed black lines in **c** and **d** drop the largest single member of each group in the late period: China accounts for 36.0% of the entrant denominator for relative disruption intensity in 2011–2015 and the United States for 40.1% of the incumbent denominator, in each case about three times the second-largest member.

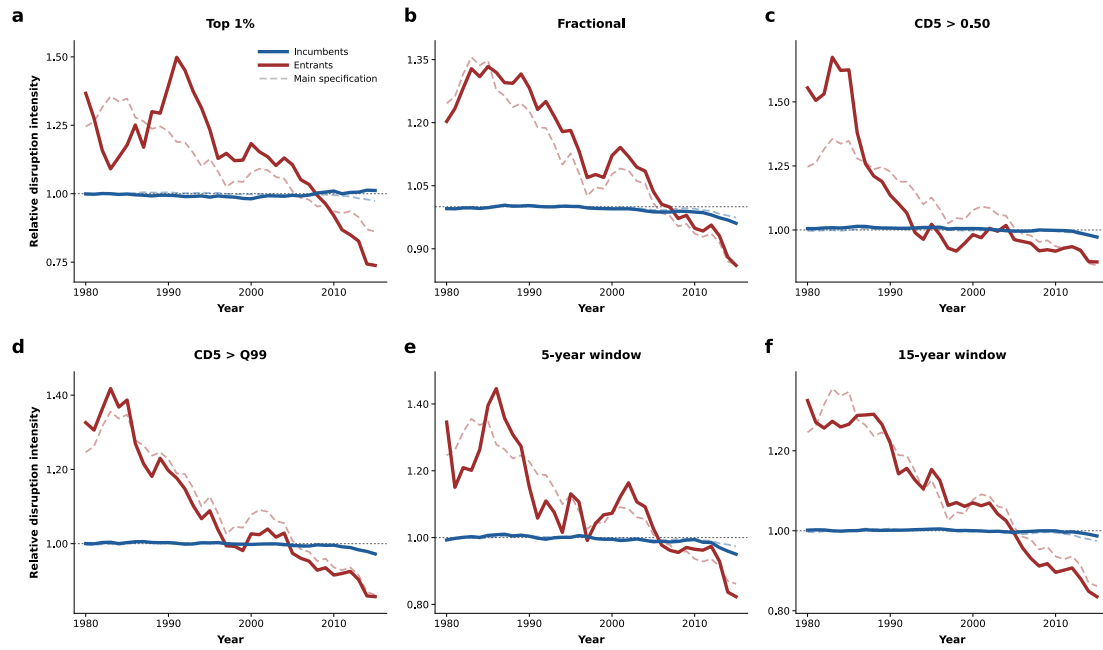


Fig. S7 Robustness of relative disruption intensity to alternative specifications. Each panel replaces one element of the main specification (top-5% papers, $CD5 > 0.25$, whole counting, ten-year backward rolling windows) with an alternative and holds the rest fixed. Solid lines give the alternative, dashed lines the main specification; red is entrants and blue incumbents. The horizontal axis is the end year of a ten-year window, so the first point covers 1971–1980 and the last covers 2006–2015. **a**, Citation threshold changed to the top 1%. **b**, Fractional counting, dividing each paper among its author countries, so that national shares sum to one. **c**, **d**, Disruption threshold changed to $CD5 > 0.50$ and to the 99th percentile of $CD5$ over the whole period. **e**, **f**, Rolling window changed to five and to fifteen years. Under all six alternatives the entrant group ends below the world benchmark (0.74 to 0.88) and the incumbent group remains close to it (0.96 to 1.01). The incumbent end value rises monotonically as the citation threshold tightens (0.963 at the top 10%, 0.974 at the top 5%, 1.012 at the top 1%): the stricter the threshold, the higher incumbents sit relative to the world.

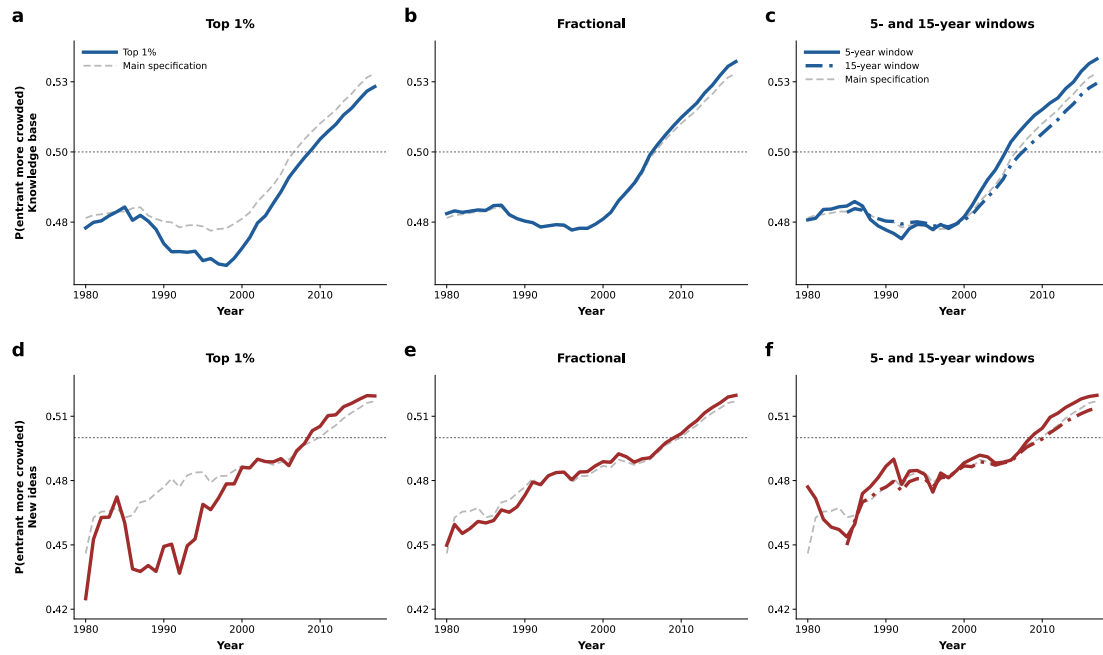


Fig. S8 Robustness of global crowding to alternative specifications. The vertical axis is the probability that a randomly drawn entrant paper occupies a more crowded position than a randomly drawn incumbent paper; the reference line at 0.5 indicates identical distributions. The upper row is crowding of the knowledge base and the lower row crowding of new ideas. Solid lines give the alternative specification and dashed lines the main one (top-5% papers, whole counting, ten-year backward rolling windows). **a, d**, Citation threshold changed to the top 1%. **b, e**, Fractional counting. **c, f**, Rolling window changed to five and to fifteen years. All ten alternatives end above 0.5: 0.523 to 0.533 for the knowledge base and 0.514 to 0.520 for new ideas. Tightening the citation threshold lowers the early values of both series without changing the end values, so the displacement is larger for the most highly cited work.

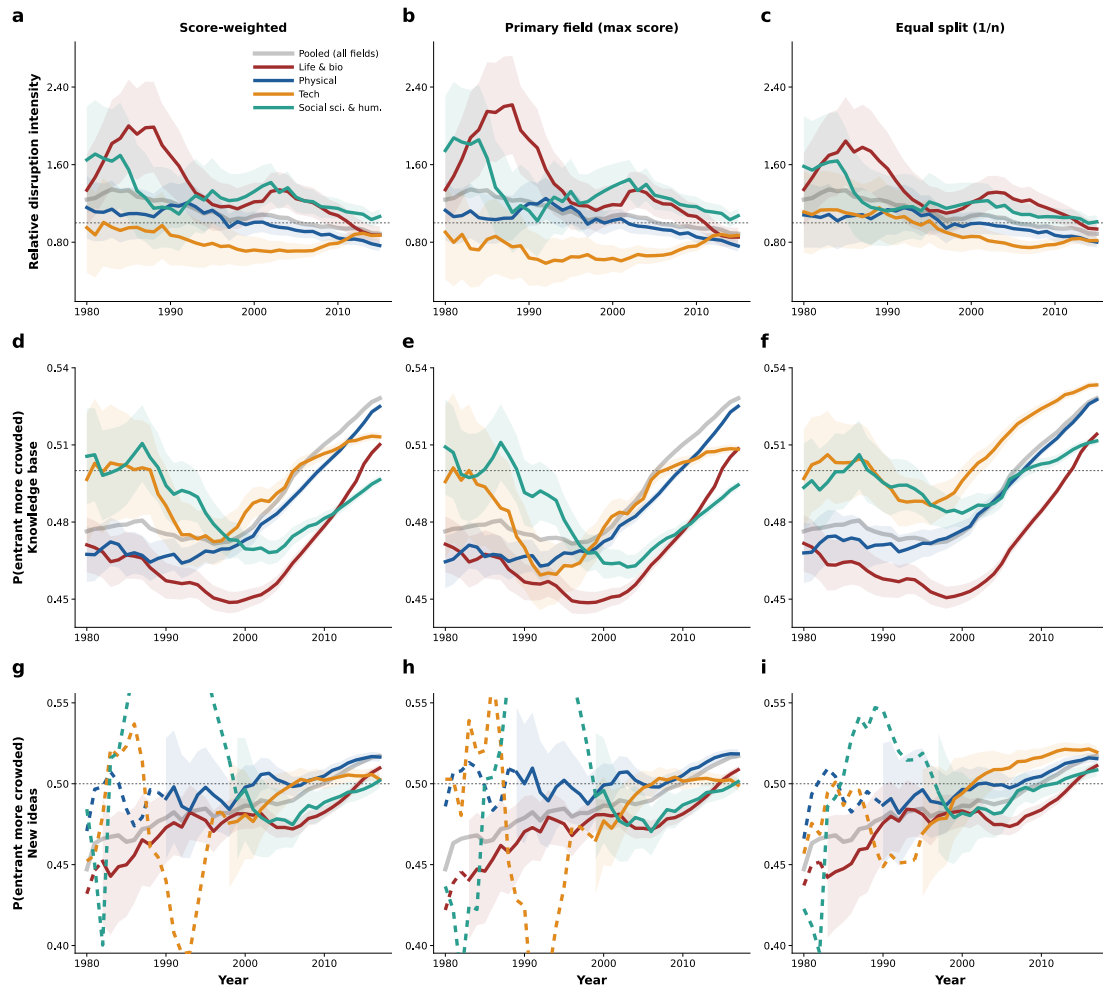


Fig. S9 Field robustness of the decline in relative disruption intensity and the rise in crowding among entrant countries. **a–c**, Relative disruption intensity of entrant countries, computed separately under three schemes for assigning papers to macro-fields: weighting by concept score (**a**), assigning each paper to its highest-scoring field (**b**), and splitting equally across all fields a paper touches (**c**). **d–f**, Knowledge-base crowding under the same three schemes; the vertical axis is the probability that a randomly drawn entrant paper occupies a more crowded position than a randomly drawn incumbent paper. **g–i**, New-idea crowding under the same three schemes. All panels use top-5% papers. Each point is a ten-year backward rolling window; the first covers 1971–1980 and the last covers 2006–2015 (**a–c**) or 2008–2017 (**d–i**). Thick grey, all fields pooled; coloured lines, the four macro-fields; shading, 95% confidence intervals. Segments in which the entrant group contributes fewer than 200 papers per window are dashed and carry no interval; the vertical range of each row is set by the solid segments. Relative disruption intensity declines in all twelve field-by-scheme combinations (**a–c**). Life and biological sciences, physical sciences and technology all end below the world benchmark under every scheme; social science and humanities ends above one (1.065, 1.073, 1.011) but shows the largest decline of the four fields, falling from 1.650, 1.745 and 1.582. The decline is smallest in technology (–0.03 to –0.30) and is close to flat on the vertical scale of this figure; at the top-1% threshold the same field declines by 0.28 to 0.87, suggesting that the top-5% cut may not be

selective enough to isolate frontier work in a field with large applied output. New-idea crowding rises in all twelve combinations and ends at or above 0.5 (**g–i**). Knowledge-base crowding rises and ends above 0.5 in every field except social science and humanities (**d–f**). The knowledge base of social science and humanities is the only series that does not end above 0.5, and it ends close to it (0.4965 under score weighting and 0.4944 under primary-field assignment, half-widths 0.0015; 0.5115 under equal splitting). It rises throughout the period following entry: under the first two schemes the series reaches its minimum in 2003, the first window falling entirely after the treatment year of 1994, and rises monotonically thereafter by 0.028 and 0.032. One possible explanation, which we do not test, is that knowledge in this field diffuses at lower cost than in the experimental sciences, so entrant countries were able to follow internationally common topics before large-scale investment and therefore started from widely diffused positions; that during the expansion they moved towards questions specific to their own national context, whose keywords are geographically concentrated and hence less crowded; and that once entry was largely complete, connecting to the global system drew their topics back towards more widely diffused ones.

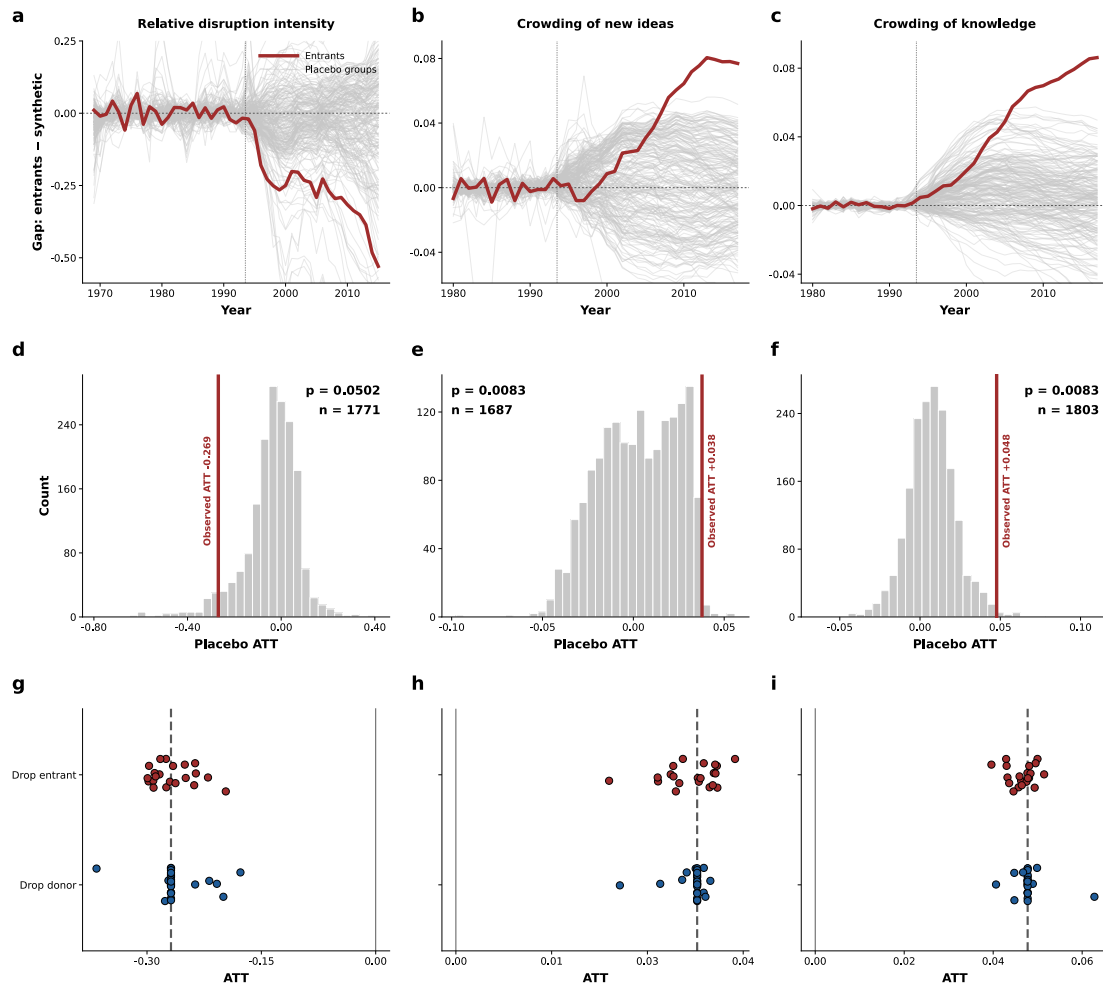


Fig. S10 Robustness for synthetic control. Columns correspond to the three synthetic controls: relative disruption intensity (main Fig. 3b; ATT -0.269 , pre-treatment fit ratio 0.50 , $P = 0.050$), crowding of new ideas (main Fig. 3e; $+0.038$, 0.29 , 0.008) and crowding of the knowledge base ($+0.048$, 0.33 , 0.008). The latter two columns give the weighted mean crowding percentile of entrant papers divided by the contemporaneous mean for non-entrant countries — incumbents and the rest of the world combined, entrants excluded — as in main Fig. 3e. Main Fig. 3d instead gives the pairwise probability that an entrant paper is more crowded than an incumbent paper, with no benchmark in the denominator and a reference line at 0.5 ; the comparison group, the statistic and the reference line all differ. Entrants are excluded from every benchmark so that their own displacement does not move the benchmark and dilute the measured effect. The 21 entrant and 13 incumbent countries are determined by a vote across five independent classification rules (see Methods). The treatment year is 1994, the median entry year across entrants; each country's entry year is the median of four signals — share level, slope, acceleration and policy turning point — and ranges from 1982 to 2006. Horizontal axes in **a–c** give the end year of a ten-year backward rolling window. The first valid window in the first column ends in 1969 (data from 1960) and in the second and third columns in 1980 (data from 1971), reflecting the start of the paper-level crowding series. Dotted vertical lines mark the treatment year. Because windows look back ten years, those ending between 1994 and 2003 still contain pre-entry years.

Entrants are pooled into a single treated unit: an individual country contributes only three to roughly twenty disruptive papers per ten-year window, and a country-level fit has a pre-treatment root mean squared prediction error worse than simply taking the pre-treatment mean. The synthetic control is obtained by non-negative least squares with weights constrained to be non-negative and to sum to one, placing it inside the convex hull of the donor countries rather than extrapolating beyond it. Donor weights are concentrated (Canada 0.41 for relative disruption intensity, Australia 0.59 for crowding of new ideas, the United States 0.45 for crowding of the knowledge base); **g–i** show that dropping any single donor leaves the conclusions unchanged. The donor pool comprises the 25 countries in the universe that are neither entrants nor signatories of a Europe Agreement; the latter each experienced an institutional shock between 1991 and 2001 and serve as a natural control in this paper, so they are not suitable untreated units. Including them in the donor pool raises the P value for crowding of new ideas from 0.008 to 0.105 without changing its sign. **a–c**, In-space placebo. For each synthetic control, 2,000 placebo groups are drawn, each formed by sampling 3 to 16 donor countries at random and refitting against the remaining donors; 220 are plotted for legibility. Grey lines are placebo gaps (actual minus synthetic); red lines are the entrant group. Placebo groups are random subsets rather than individual donors because the treated unit is itself a pool of 21 countries and its series is far smoother than that of any single country, making the two incomparable. Vertical axes are clipped at the 1st and 99th percentiles of the placebo gaps. **d–f**, Permutation distribution of the placebo ATT, with the observed value marked in red. Of the 2,000 placebo groups, 1,771, 1,687 and 1,803 respectively pass the pre-treatment fit filter and enter the distribution; the criterion is that a group's pre-treatment root mean squared prediction error not exceed twice the standard deviation of its own pre-treatment series. Groups that fail already fit poorly before treatment, so a large post-treatment gap in those groups is unrelated to treatment and would inflate the tails of the null distribution. The reported P is a rank statistic, not a sampling P value, and its resolution is bounded by the size of the donor pool. Previous work sets out two routes to inference: the permutation distribution of the ratio of post- to pre-treatment root mean squared prediction error, or the distribution of the mean post-treatment gap after discarding placebo groups with poor pre-treatment fit. We take the second. We do not use the ratio statistic because it divides by each unit's own pre-treatment fit, and a pooled treated unit and placebo groups of varying size are not comparable on that quantity. The criterion for poor pre-treatment fit also differs from that work: it benchmarks each placebo group against the treated unit's pre-treatment error (2, 5 or 20 times), whereas we benchmark each group against the standard deviation of its own pre-treatment series, because groups of 3 to 16 countries differ substantially in scale and normalising by each group's own dispersion is what makes them comparable. Recomputing with those three thresholds leaves the conclusions unchanged: P for relative disruption intensity lies between 0.033 and 0.055, and both crowding measures remain at or below 0.015. **g–i**, Leave-one-out. Each point in the upper row re-estimates the ATT after dropping one entrant country, and each point in the lower row after dropping one donor country; dashed lines mark the baseline ATT and solid lines zero. All 46 re-estimates share the sign of the baseline for each of

the three synthetic controls. Country-specific synthetic controls, fitted for relative disruption intensity using each country's own entry year, give a negative ATT for all four entrants with a median of at least ten disruptive papers per ten-year window (-0.87 to -0.26), three of them at $P < 0.05$; the remaining entrants have medians of only three to nine papers per window, which is why the main specification pools the 21 countries into a single treated unit. Crowding cannot be fitted country by country: those series begin in 1971, so countries entering before 1990 have fewer than twelve pre-treatment windows. Varying the assumed treatment year gives an ATT between -0.269 and -0.064 for relative disruption intensity over 1988–2002, largest in absolute value at 1994; the corresponding ranges over 1992–2002 are $+0.031$ to $+0.045$ ($P \leq 0.053$) for crowding of new ideas and $+0.026$ to $+0.060$ ($P \leq 0.014$) for crowding of the knowledge base.

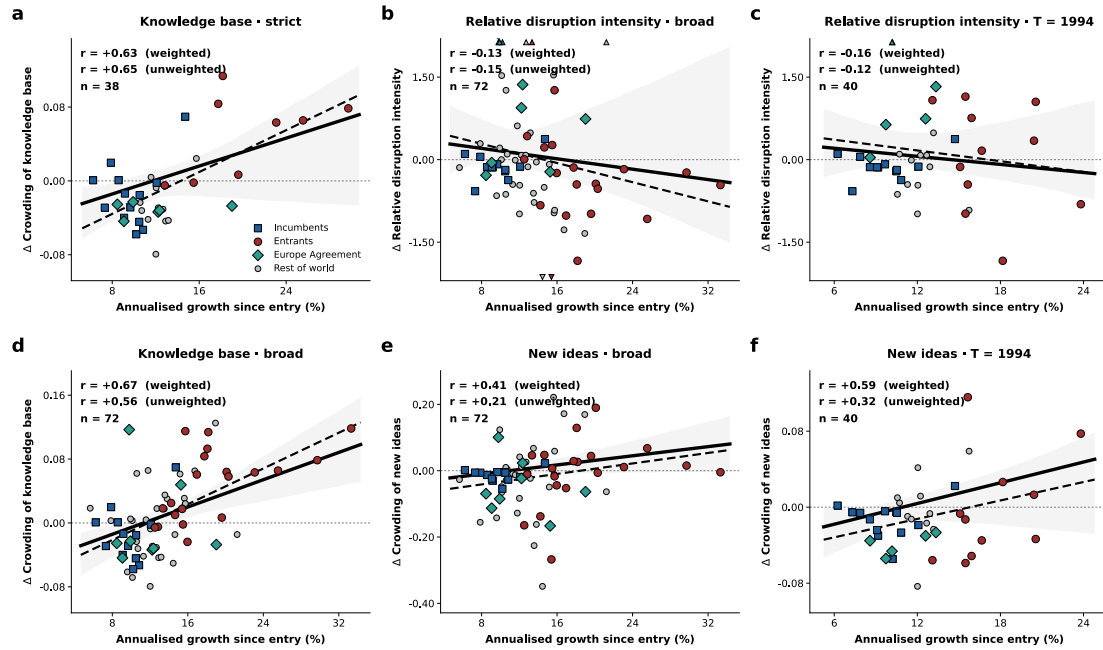


Fig. S11 Robustness of the dose–response relationship between expansion rate and RDI and global crowding. Each point is a country; the horizontal axis is the annualised growth in publication output since entry. Solid lines are fits weighted by the late-period denominator, dashed lines unweighted, and shaded bands are 2,000-draw bootstrap 95% intervals for the weighted fit. Two sample thresholds apply: the crowding measure used in that panel must have at least 40 computable values in the pre-entry window and in 2011–2015 separately, and the denominator of relative disruption intensity — high-impact papers with a computable CD5 index — must reach 80 in each period. The strict sample is filtered on coverage of new-ideas crowding (38 countries, as in the main text); the broad sample removes the coverage threshold (72 countries). All six correlations share the sign reported in the main text; their magnitudes vary with the sample and the specification. **a, d**, Crowding of the knowledge base is +0.63 on the strict sample and +0.67 on the broad sample (unweighted +0.65 and +0.56). This measure covers 99.9% of papers and is therefore almost insensitive to the sample. **b**, Relative disruption intensity falls from -0.31 in the main text to -0.13 on the broad sample (unweighted -0.22 to -0.15). Of the 34 countries added, 20 belong to the rest of the world and their denominator for relative disruption intensity has a median of 160 papers, against 1,768 for the original 38; the larger measurement error in the outcome for these countries (its standard deviation rises from 1.07 to 1.75) attenuates the correlation towards zero. **e**, Crowding of new ideas falls from +0.71 to +0.41 (unweighted +0.45 to +0.21). The measure covers only 13–15% of papers and the correlation declines monotonically as the coverage threshold is relaxed (40 papers / 38 countries, +0.71; 20 / 48, +0.56; 10 / 60, +0.49; 5 / 68, +0.45; 0 / 72, +0.41), the added countries having a median of 43 pre-entry papers with a new keyword pair and as few as 2. **c, f**, Using a common entry year of $T = 1994$ in place of each country’s own entry year gives -0.16 for relative disruption intensity (unweighted -0.12) and +0.60 for crowding of new ideas (+0.32). Both correlations weaken relative to the main text, as a common entry year mismeasures the horizontal axis. The fixed pre-entry window falls after expansion had already begun for

earlier entrants, omitting their earliest expansion; for later entrants it falls before expansion began, counting years of no expansion towards their growth rate. Measurement error in the explanatory variable of this kind attenuates the correlation towards zero. The sample changes from 38 to 40 because the fixed window alters how many papers with a computable value of new-ideas crowding each country has within it. Neither threshold drives the result. Re-estimating across the nine combinations of 40/50/60 and 80/90/100 leaves the sign of all three correlations unchanged: relative disruption intensity -0.37 to -0.30 (unweighted -0.26 to -0.21), crowding of new ideas $+0.71$ to $+0.73$ ($+0.44$ to $+0.54$) and crowding of the knowledge base $+0.67$ to $+0.67$ ($+0.56$ to $+0.58$); sample sizes range from 31 to 38 countries for new ideas and 65 to 72 for the knowledge base. Points falling outside the vertical range in **b** and **c** are marked by triangles at the axis boundary.

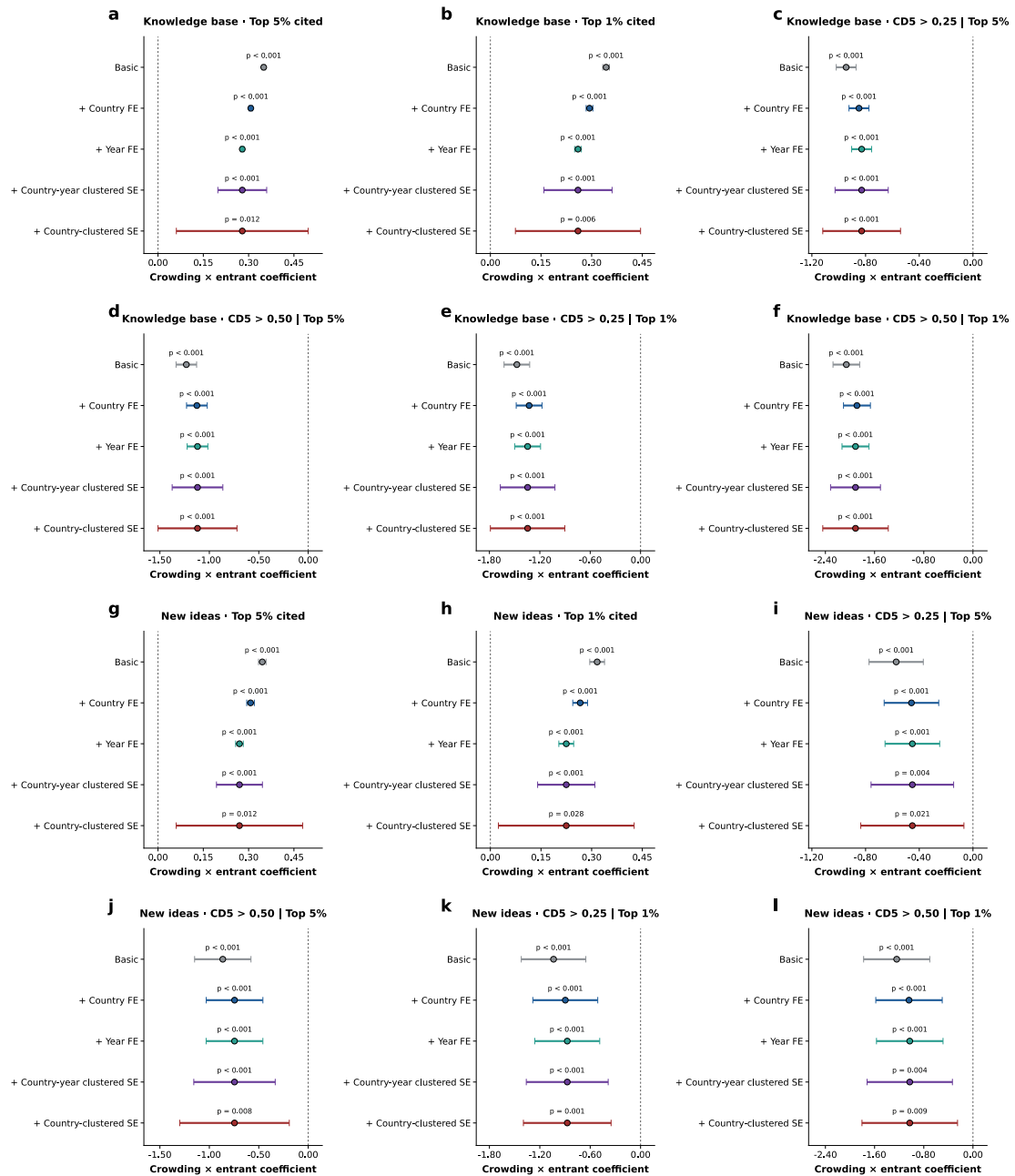


Fig. S12 Entrants gain citation return and lose disruptive advantage as crowding rises. Each panel shows the interaction term from a grouped binomial regression, $\text{logit}(\text{outcome}) = \alpha + \beta \cdot \text{crowding} + \gamma \cdot (\text{crowding} \times \text{entrant}) + \text{controls}$. Crowding is the percentile among all papers with a value of the measure in that year, divided into 100 groups of equal paper counts and rescaled to 0–1. The upper two rows use crowding of the knowledge base and the lower two crowding of new ideas. Outcomes are of two kinds: the probability that a paper is among the top 5% or top 1% by citations (**a**, **b**, **g**, **h**), and the probability that a highly cited paper reaches CD5 > 0.25 or CD5 > 0.50 (**c**–**f**, **i**–**l**), the latter restricted to papers in that citation tier for which CD5 is computable. The five specifications, from top to bottom, are: crowding, entrant and their interaction alone; adding country fixed effects, which absorb the entrant main effect so that the interaction is identified from differences in the crowding slope within countries; adding year fixed effects; and then replacing the standard errors with ones clustered by country-year and by country. The last two leave the coefficients unchanged and alter only the

standard errors, confidence intervals and P values. Points are interaction estimates and horizontal lines 95% confidence intervals. The interaction γ is the difference in the crowding slope between the two groups: incumbents have slope β and entrants $\beta + \gamma$. A positive γ means the entrant outcome rises faster with crowding and the gap narrows; a negative γ means it falls faster and the gap widens. The interaction is significantly positive for every citation outcome (approximately +0.22 to +0.35) and significantly negative for every disruption outcome (−0.44 to −1.9) under all five specifications, matching the directions of the two curves in main Fig. 4a. The disruption coefficients grow in magnitude as the definition tightens, reaching −1.9 for $CD5 > 0.50$ within the top 1%. Coefficients move only slightly from the basic specification to the one with both sets of fixed effects, indicating that the pattern is not driven by country composition or period trends; widening the standard errors to the country-year and then the country level broadens the confidence intervals, but none of the twelve interactions crosses zero.

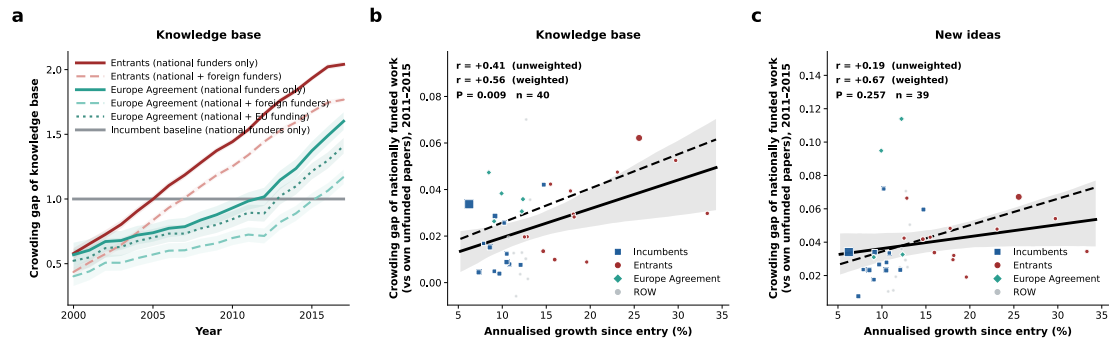


Fig. S13 The main result on the knowledge base layer, and funding moves onto crowded ground wherever expansion has been fast, in every group of countries. a, Fig. 4a reproduced on Knowledge base instead of New ideas. The ordering is preserved and the entrants rise further — $\times 2.04$ (95% CI 2.02–2.06) on national money, $\times 1.77$ (1.75–1.79) with foreign money. The Europe Agreement countries’ all-funded line reaches $\times 1.17$ (1.11–1.23). **b, c,** Each point is a country. The horizontal axis is annualised growth in top-5% output from the five years before the country’s entry year to 2011–2015; incumbents and the rest of the world have no entry event and are assigned $T = 1994$. The vertical axis is that country’s crowding gap for nationally funded work over 2011–2015, measured on Knowledge base (**b**) and New ideas (**c**). Solid lines are unweighted least squares, dashed lines are weighted by the number of nationally funded papers, and shaded bands are bootstrap 95% intervals for the unweighted fit. Marker area is proportional to the number of nationally funded papers. Countries enter the panels only if they have at least 400 papers with a measurable crowding value in both the pre-entry and the late window and at least 400 funded and 400 unfunded papers in the late window; 400 is the strictest threshold that retains all four groups (13 incumbents, 13 entrants, 5 Europe Agreement countries and 9 others on Knowledge base; 4 Europe Agreement countries on New ideas). The relationship strengthens as the threshold rises, as expected when the dependent variable is measured with error: at thresholds of 40, 100, 200, 400, 800 and 1,500 papers the unweighted correlation on Knowledge base is $+0.26, +0.27, +0.28, +0.41, +0.61$ and $+0.58$ ($n = 98, 72, 52, 40, 30, 22$), and on New ideas $+0.01, +0.13, +0.11, +0.19, +0.36$ and $+0.56$. At the threshold shown, Knowledge base gives $r = +0.41$ unweighted and $+0.56$ weighted ($P < 0.01, n = 40$) and New ideas $r = +0.19$ and $+0.67$ ($P = 0.26, n = 39$). Above 1,500 papers the Europe Agreement countries drop out entirely, so the tighter samples no longer speak to entry.

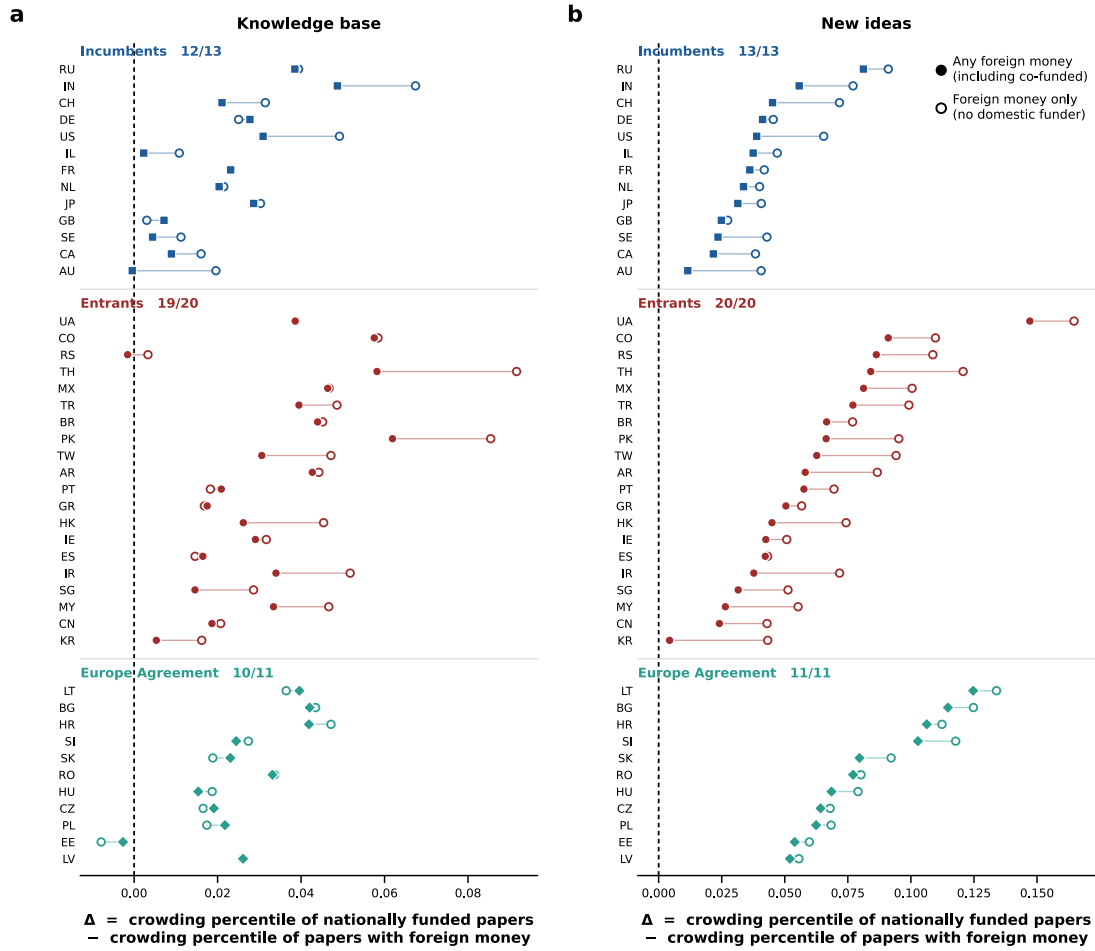


Fig. S14 Papers touching foreign money sit in less crowded positions than purely nationally funded papers, in almost every country and region. For each of the 44 countries in the three defined groups, Δ is the mean within-year crowding percentile of papers whose funders are all domestic minus that of papers acknowledging at least one foreign or supranational funder, pooled over 1991–2017, on New ideas (a) and Knowledge base (b). Filled markers use the definition applied throughout the paper, in which a paper counts as touching foreign money whether or not a domestic funder is also named; open markers tighten it to papers with no domestic funder at all. Countries are ordered within each group by Δ on New ideas, and the same order is used in both panels. Only countries with at least 200 papers in each category are shown; all 44 qualify. Δ is positive in 44 of 44 countries on New ideas under both definitions, with group medians of +0.036 for incumbents, +0.058 for entrants and +0.077 for the Europe Agreement countries. On Knowledge base it is positive in 41 of 44 under the broader definition (Australia, Serbia and Estonia are the exceptions) and 43 of 44 under the stricter one (Estonia alone), with group medians of +0.021, +0.032 and +0.025. The open markers lie to the right of the filled markers almost everywhere, so the effect is stronger the less domestic money is involved. This panel establishes the second factor in the decomposition of Table S4. A pull-back of $w \times \Delta$ requires $\Delta > 0$; that it holds almost without exception means the pull-back depends on how much foreign money a country takes in, not on whether foreign money behaves differently there. Note the sign

convention: foreign money is always less crowded than *national* money, but not always less crowded than *no* money, and that second, weaker claim does not hold universally (60/92 in our data). Romania signed a Europe Agreement in 1993 and also meets the publication-based entrant criteria (1999). It is assigned to the Europe Agreement group in the funding analysis, which therefore reports 20 entrants, and to the entrant group elsewhere.

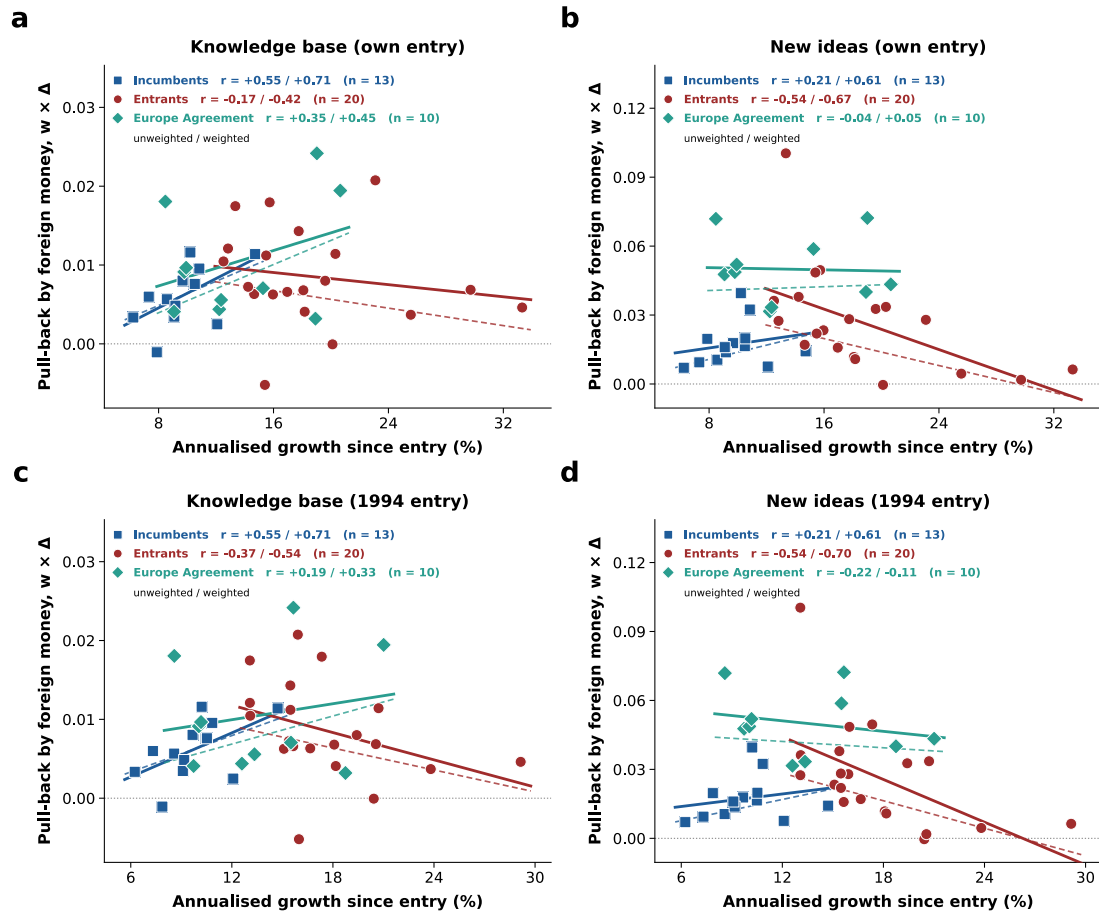


Fig. S15 The faster an entrant expanded, the less foreign money pulled it back; incumbents and Europe Agreement countries move the other way. Each point is a country. The horizontal axis is annualised growth in top-5% output from the five years before entry to 2011–2015; the vertical axis is the pull-back $w \times \Delta$ over 2011–2015, the amount by which adding foreign money moves the group’s funded output towards less crowded positions. Rows are Knowledge base (**a**, **b**) and New ideas (**c**, **d**); the left column anchors the growth rate at each country’s own entry year and the right column at $T = 1994$ for every country. Incumbents have no entry event and are assigned $T = 1994$ in both columns, so their fitted line is identical across columns. Fits are per group: solid unweighted, dashed weighted by the number of nationally funded papers. Correlations are reported as unweighted / weighted. With each country anchored at its own entry year, the correlation is negative for entrants ($-0.17 / -0.42$ on Knowledge base; $-0.54 / -0.67$ on New ideas) and positive for incumbents ($+0.55 / +0.71$; $+0.21 / +0.61$) and Europe Agreement countries ($+0.35 / +0.45$; $-0.04 / +0.05$). The sign pattern is the same under the uniform anchor. Pooling the three groups gives close to nothing, because the two directions cancel. The entrant line is firm on New ideas and softer on Knowledge base, where the unweighted correlation is -0.17 . Both components of $w \times \Delta$ contribute; the share w is the more robust of the two and is shown separately in Fig. S16.

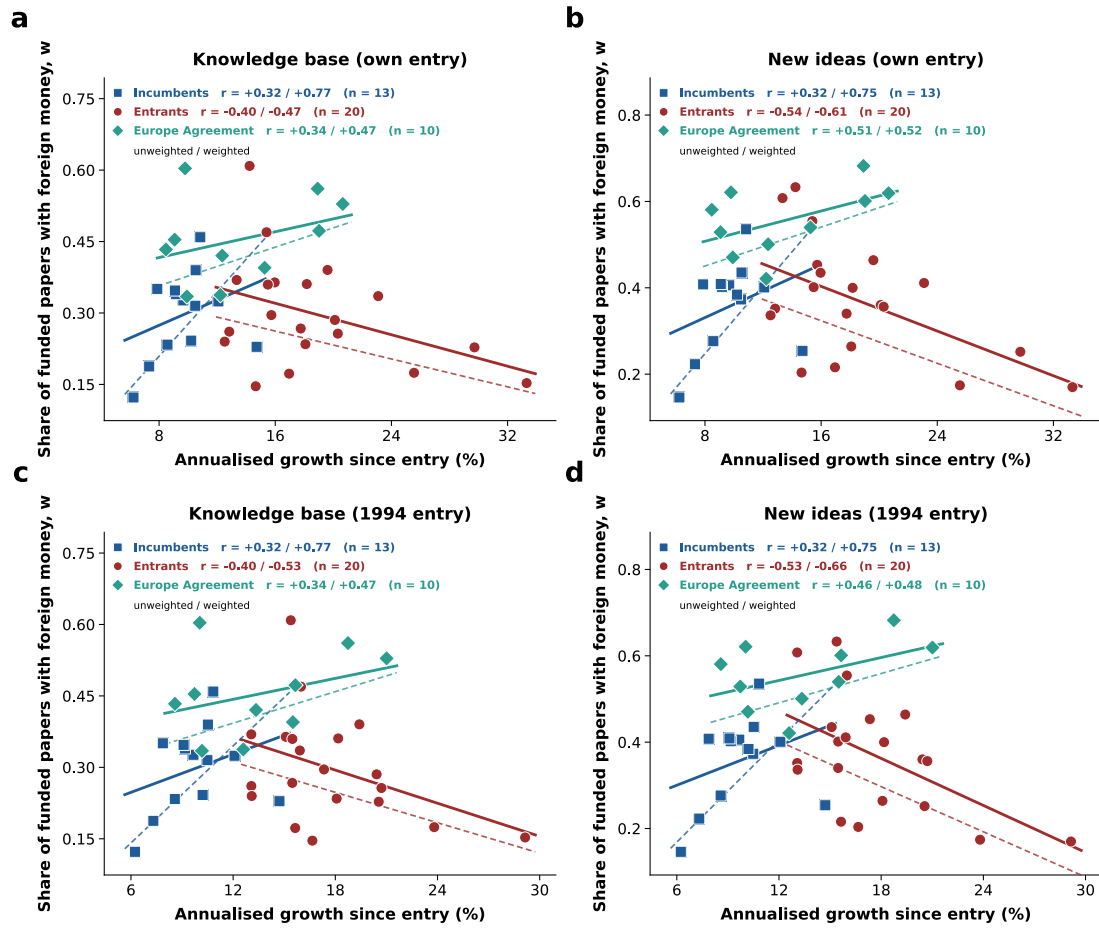


Fig. S16 Incumbents and Europe Agreement countries carried foreign funding up with them as they expanded; entrants did not. Layout, samples and entry time settings as in Fig. S15. The vertical axis is w , the share of a country's funded papers acknowledging at least one foreign or supranational funder over 2011–2015. This quantity contains no crowding measure at all; it records only how far a country's researchers appear on grants awarded elsewhere. The correlation is positive for incumbents ($+0.32 / +0.77$ on both layers) and for Europe Agreement countries ($+0.34 / +0.47$ on Knowledge base; $+0.51 / +0.52$ on New ideas), and negative for entrants ($-0.40 / -0.47$; $-0.54 / -0.61$). All four panels agree in sign, weighted and unweighted estimates agree, and the pattern survives both entry time settings. The entrants' foreign-funded output grew in absolute terms over this period; what fell was its share, because domestic funding grew faster still. The decline is therefore not a retreat from international work but a failure of institutional access to expand in proportion — and access of this kind is constrained on the other side, since a national agency abroad does not enlarge its programmes because a partner country has expanded. w measures entry into another country's funding framework rather than co-authorship: a foreign collaborator listed as an author who attaches no grant does not raise it.

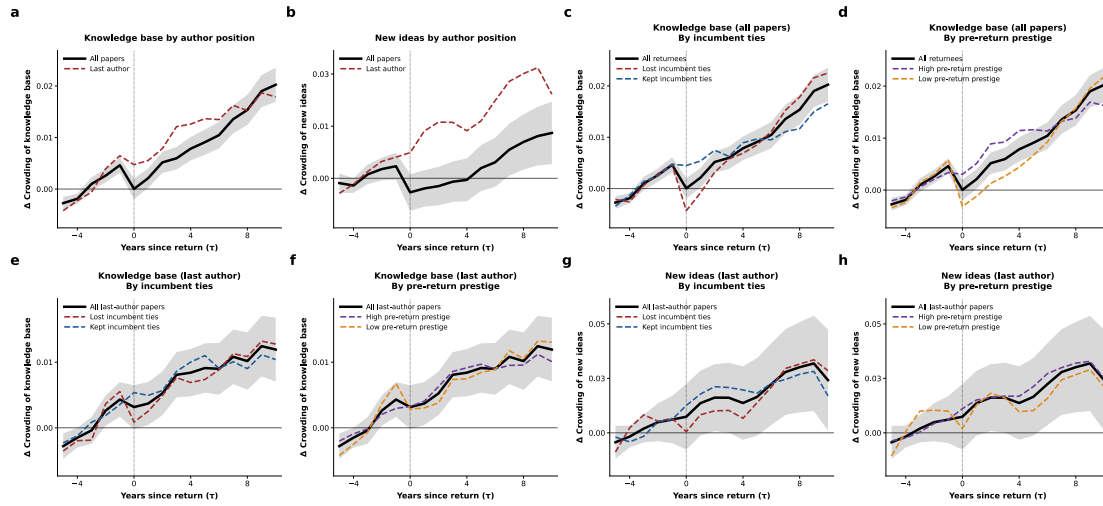


Fig. S17 Crowding after return, by author scope and subgroup. All panels use the matched event-study design described in Fig. 4. Coefficients are normalised so that the three pre-return reference years average to zero, shaded bands are 95% confidence intervals for the series drawn in black, and standard errors are clustered by author. Numerical estimates for every series are in Table S6. **a, b**, Crowding of the knowledge base and of new ideas, comparing all papers with papers on which the returnee is last author. Last author is used as a proxy for corresponding author because the corresponding-author flag in OpenAlex is populated for only 4.3% of author-years. **c, d**, Knowledge-base crowding over all papers, split by post-return incumbent ties and by pre-return prestige. **e, f**, The same two splits restricted to last-authored papers. **g, h**, The same two splits for crowding of new ideas, last-authored papers only. Knowledge-base crowding rises by +0.0146 over $\tau \in [5, 10]$ across all returnees ($P < 0.001$, $n = 81,159$ authors), and by +0.0159 on last-authored papers ($P < 0.001$). In this layer the subgroup contrasts are sharp at return and fade afterwards. At $\tau = 0$ returnees who kept incumbent ties lie +0.0088 above those who lost them ($P < 0.001$) and those with high pre-return prestige lie +0.0062 above those with low prestige ($P < 0.01$). The former gap closes by $\tau = 5$ and reverses by $\tau = 8$, the kept-ties series lying -0.0062 below the lost-ties series ($P < 0.05$). The same contrasts are not resolved in the sparser new-ideas layer or on last-authored papers alone, where every between-group difference has P above 0.19.

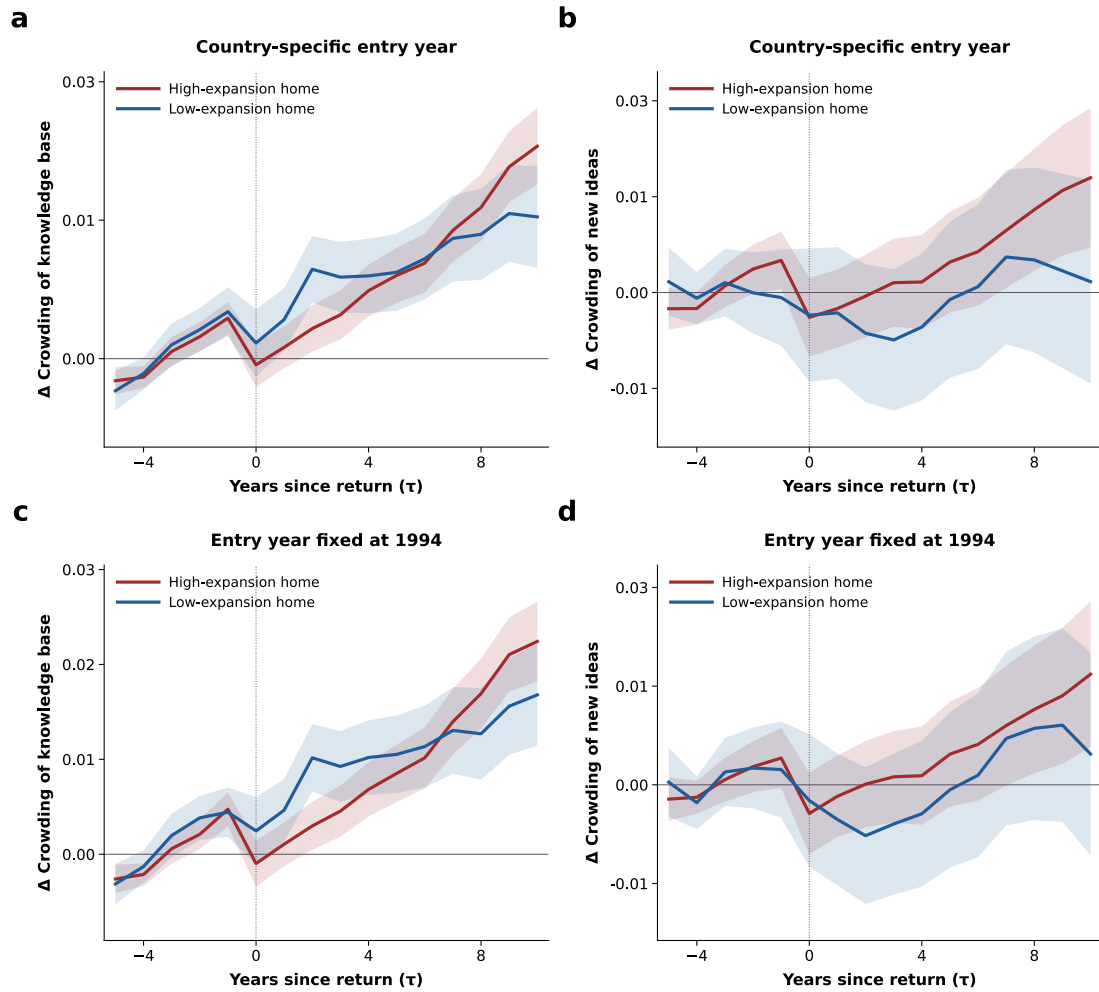


Fig. S18 Crowding by the expansion rate of the home country. The expansion rate is the annualised growth of a country's publication output from the five years before its entry year to 2011–2015, and the twenty-one entrant countries are split at the median. Design, normalisation, bands and clustering as in Fig. 4. **a, b**, Knowledge-base and new-ideas crowding using each country's own entry year (median growth 14.0% per year). **c, d**, The same two outcomes with the entry year fixed at 1994 for all countries (median 13.1% per year). All panels use all papers. Both groups show a positive late-period effect under both definitions. Returnees to high-expansion countries are significant throughout, with knowledge-base estimates of +0.0156 and +0.0155 and new-ideas estimates of +0.0113 ($P < 0.01$) and +0.0103 ($P < 0.05$). Returnees to low-expansion countries are significant in the knowledge base (+0.0130 and +0.0133) but not in new ideas (+0.0026, $P = 0.69$; +0.0050, $P = 0.42$). Between-group differences do not reach conventional significance under either definition ($P = 0.26$ to 0.95), so group-specific estimates are reported rather than the contrast. Excluding China, which accounts for 44% of returnees, leaves the high-expansion estimates unchanged or stronger in the knowledge base (+0.0178 under both definitions) and in new ideas under the country-specific definition (+0.0159, $P < 0.05$), so the pattern is not driven by the single largest entrant country.

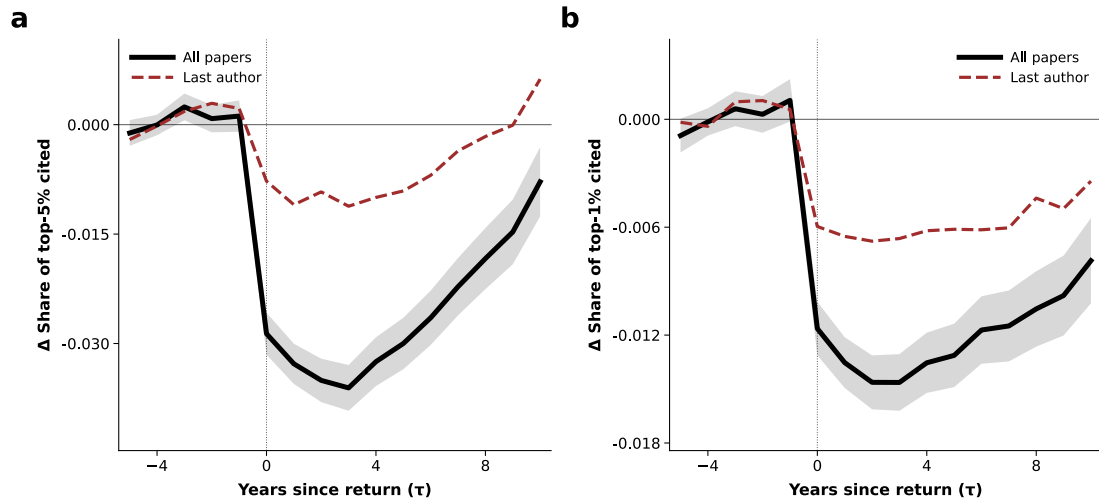


Fig. S19 Highly cited output after return. Both outcomes are shares computed over all of a returnee's papers in the rolling window. Design, normalisation, bands and clustering as in Fig. 4. Estimates are in Table S9. **a**, Share of papers in the top 5% of citations. **b**, Share in the top 1%. Each panel shows all papers in black and last-authored papers as the second series. Highly cited output falls sharply at return and recovers only in part. Over $\tau \in [0,2]$ the top-5% share is -0.0322 below the counterfactual ($P < 0.001$) and the top-1% share is -0.0133 ($P < 0.001$). By $\tau \in [5,10]$ the shortfalls narrow to -0.0199 and -0.0108 , the improvement between the two windows being $+0.0122$ and $+0.0025$ (both $P < 0.001$). The recovery is therefore partial and proportionally weaker in the top 1%, where 19% of the initial fall is made up against 38% in the top 5%. On last-authored papers the initial fall is smaller (-0.0093 and -0.0064) and the top-5% shortfall is no longer distinguishable from zero in the late window.

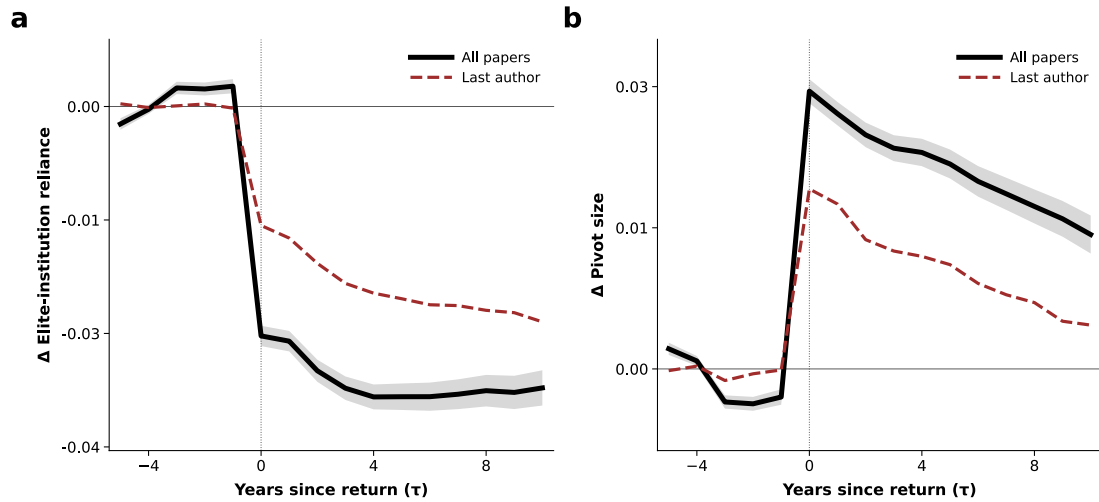


Fig. S20 Pivot size and reliance on top institutions. Pivot size is the distance between a paper's field-frequency distribution and that of its author's preceding three years, and top-institution reliance is the share of a paper's references drawn from the top 100 institutions. Neither outcome entered the matching distance, so all five pre-return coefficients are genuine out-of-sample tests. Design, normalisation, bands and clustering as in Fig. 4. Estimates are in Table S9 and pre-return diagnostics in Table S8. **a**, Pivot size. **b**, Top-institution reliance. Each panel shows all papers in black and last-authored papers as the second series. Both series move before the move itself. Over the placebo window $\tau \in [-5, -4]$ pivot size is already falling (slope -0.0013 , $P < 0.01$) and reliance on top institutions rising ($+0.0019$, $P < 0.001$), and between the two pre-return windows they shift by -0.0049 and $+0.0038$ (both $P < 0.001$), equal to 17% and 13% of the jump at $\tau = 0$. Return itself reverses both. Pivot size jumps to $+0.0272$ over $\tau \in [0, 2]$ ($P < 0.001$) as returnees carry external knowledge into a new research environment, while reliance on top institutions falls to -0.0321 ($P < 0.001$). Over $\tau \in [5, 10]$ pivot size narrows to $+0.0180$ and reliance falls further to -0.0379 , the changes between the two post-return windows being -0.0092 and -0.0058 (both $P < 0.001$). Access to leading institutions is not regained, and the search narrows instead.

Table S1 Robustness of the association between global crowding and top-tier disruption to alternative explanations.

	(1)	(2)	(3)	(4)	(5)
Global crowding	-0.0365*** (-43.9)	-0.0337*** (-40.5)	-0.0270*** (-32.6)	-0.0306*** (-34.4)	-0.0312*** (-35.0)
log(papers on edge)		-0.0516*** (-61.0)	-0.0120*** (-13.0)	-0.0119*** (-13.0)	-0.0117*** (-12.7)
log(citations, C5)			-0.1014*** (-108.0)	-0.1008*** (-107.3)	-0.1016*** (-107.9)
log(cumulative papers)				0.0101*** (11.0)	0.0027* (2.5)
log(keyword age)					0.0133*** (13.5)
Year $\times$ field FE	Yes	Yes	Yes	Yes	Yes
Within R ²	0.0013	0.0038	0.0117	0.0118	0.0119
Share of baseline retained	100%	92%	74%	84%	86%
N	1,468,933	1,468,933	1,468,933	1,468,933	1,468,933

Regression results progressively controlling for the number of papers on an edge, their mean citations, the cumulative papers accumulated on the two units of knowledge before the edge formed, and keyword age. All models include year $\times$ field fixed effects, with all variables standardized. The negative association between global crowding and top-tier disruption remains robust across all specifications: adding prior accumulation does not attenuate the coefficient but raises it, indicating that these variables suppress rather than account for the relationship. *t*-statistics are reported in parentheses. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

Table S2 Mechanisms underlying the association between global crowding and long-term impact.

	(1)	(2)	(3)	(4)
Global crowding	0.1654*** (256.0)	0.1253*** (194.5)	0.0643*** (98.2)	0.0628*** (95.7)
Team prestige		0.2066*** (320.6)	0.1203*** (177.0)	0.1256*** (179.3)
Elite-institution reliance			0.2375*** (336.7)	0.2361*** (334.1)
Prestige × Elite reliance				−0.0200*** (−31.4)
Year × field fixed effects	Yes	Yes	Yes	Yes
Within R ²	0.0274	0.0684	0.1116	0.1120
Share of effect explained	0.0%	24.2%	61.1%	62.0%
Observations	2,331,222	2,331,222	2,331,222	2,331,222

Regression results progressively adding team prestige, elite-institution reliance, and their interaction. All models include year × field fixed effects, with all variables demeaned within fixed-effect cells and scaled by their within-cell standard deviation; interaction terms are formed from mean-centred components, and the sample is held constant across columns. The positive association between global crowding and Hit 5pct attenuates substantially as these characteristics are added, indicating that most of the impact dividend is carried by the convergence of reputation and resources. “Share of effect explained” is calculated as $1 - \beta/\beta_0$, where β is the coefficient on global crowding in a given column and β_0 that in column (1); columns are not nested. *t*-statistics are reported in parentheses. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

Table S3 Mechanisms underlying the association between global crowding and top-tier disruption.

Panel A. Columns (1)–(5)

	(1)	(2)	(3)	(4)	(5)
Global crowding	−0.0361*** (−43.7)	−0.0202*** (−24.2)	−0.0109*** (−13.3)	−0.0087*** (−10.5)	−0.0052*** (−6.2)
Team knowledge distance		0.0935*** (112.2)		0.0025* (2.4)	−0.0052*** (−5.0)
Pivot size			0.1533*** (185.5)	0.1391*** (129.5)	0.1296*** (118.0)
Elite-institution reliance					−0.0558*** (−58.6)
Distance × Pivot				0.0371*** (42.6)	
Year × field fixed effects	Yes	Yes	Yes	Yes	Yes
Within R ²	0.0013	0.0098	0.0242	0.0254	0.0264
Share of effect explained	0.0%	44.1%	69.6%	75.8%	85.6%
Observations	1,468,897	1,468,897	1,468,897	1,468,897	1,468,897

Panel B. Columns (6)–(9)

	(6)	(7)	(8)	(9)
Global crowding	−0.0049*** (−5.9)	0.0000 (0.0)	−0.0036*** (−4.3)	−0.0033*** (−3.9)
Pivot size	0.1268*** (134.7)	0.0927*** (90.9)	0.1263*** (133.9)	0.1240*** (129.6)
Elite-institution reliance	−0.0553*** (−58.4)	−0.0777*** (−79.4)	−0.0552*** (−58.4)	−0.0566*** (−59.5)
Neighborhood overlap			−0.0065*** (−7.8)	−0.0082*** (−9.8)
Pivot × Elite reliance		−0.0763*** (−85.9)		
Pivot × Neighborhood overlap				−0.0119***

				(-14.3)
Year $\times$ field fixed effects	Yes	Yes	Yes	Yes
Within R ²	0.0264	0.0313	0.0265	0.0266
Share of effect explained	86.5%	100.1%	89.9%	91.0%
Observations	1,468,897	1,468,897	1,468,897	1,468,897

Regression results across nine specifications, split into two panels for typesetting; the sample, dependent variable and fixed effects are identical throughout. Panel A adds team knowledge distance, pivot size, elite-institution reliance, and the interaction between distance and pivot size; Panel B adds neighborhood overlap and the interactions of pivot size with elite-institution reliance and with neighborhood overlap. All models include year $\times$ field fixed effects, with all variables demeaned within fixed-effect cells and scaled by their within-cell standard deviation; interaction terms are formed from mean-centred components. The negative association between global crowding and top-tier disruption attenuates substantially as these characteristics are added, indicating that most of the breakthrough penalty is carried by the contraction of exploration. Regressors that enter only a subset of the specifications are shown only in the relevant panel: team knowledge distance and Distance $\times$ Pivot appear in columns (2), (4) and (5); neighborhood overlap and the Pivot interactions appear in columns (7)–(9). Empty cells indicate that the regressor is excluded from that specification. “Share of effect explained” is calculated as $1 - \beta/\beta_0$, where β is the coefficient on global crowding in a given column and β_0 that in column (1); columns are not nested. t -statistics are reported in parentheses. $*P < 0.05$; $**P < 0.01$; $***P < 0.001$.

Table S4 Expansion, foreign-funding share, and how far foreign money pulls each group back

Group	Funded papers	national only	touching foreign money	all funded papers	national + EU only	Foreign share w (EU)	Δ	Pull-back w $\times \Delta$ [95% CI]	Extra pull-back vs incumbents [95% CI]
Panel a Knowledge Base									
1991–2017									
Incumbents (13)	8,488,964	+0.0288 ($\times 1.00$)	+0.0056	+0.0238 ($\times 0.83$)	+0.0272	0.216 (0.077)	0.0232	+0.0050 [+0.0049, +0.0051]	0 (ref.)
Entrants (20)	3,237,198	+0.0453 ($\times 1.58$)	+0.0186	+0.0392 ($\times 1.36$)	+0.0428	0.230 (0.062)	0.0268	+0.0061 [+0.0060, +0.0063]	+0.0011 Z = 11.2 [+0.0009, +0.0013]
Europe Agreement (11)	370,119	+0.0295 ($\times 1.02$)	+0.0050	+0.0203 ($\times 0.71$)	+0.0251	0.374 (0.227)	0.0245	+0.0092 [+0.0084, +0.0099]	+0.0042 Z = 11.2 [+0.0034, +0.0049]
1991–2003									
Incumbents (13)	1,804,560	+0.0474 ($\times 1.00$)	+0.0288	+0.0439 ($\times 0.93$)	+0.0472	0.188 (0.050)	0.0186	+0.0035 [+0.0033, +0.0037]	0 (ref.)
Entrants (20)	237,553	+0.0364 ($\times 0.77$)	+0.0088	+0.0289 ($\times 0.61$)	+0.0340	0.272 (0.063)	0.0275	+0.0075 [+0.0068, +0.0082]	+0.0040 Z = 10.6 [+0.0032, +0.0047]
Europe Agreement (11)	47,248	+0.0314 ($\times 0.66$)	+0.0075	+0.0231 ($\times 0.49$)	+0.0289	0.349 (0.107)	0.0239	+0.0084 [+0.0065, +0.0103]	+0.0049 Z = 5.0 [+0.0030, +0.0068]
2004–2017									
Incumbents (13)	6,684,404	+0.0232 ($\times 1.00$)	-0.0000	+0.0180 ($\times 0.78$)	+0.0215	0.224 (0.085)	0.0232	+0.0052 [+0.0051, +0.0053]	0 (ref.)
Entrants (20)	2,999,645	+0.0458 ($\times 1.98$)	+0.0194	+0.0398 ($\times 1.72$)	+0.0433	0.226 (0.062)	0.0265	+0.0060 [+0.0058, +0.0062]	+0.0008 Z = 7.3 [+0.0006, +0.0010]

Group	Funded papers	national only	touching foreign money	all funded papers	national + EU only	Foreign share w (EU)	Δ	Pull-back w $\times \Delta$ [95% CI]	Extra pull-back vs incumbents [95% CI]
Europe Agreement (11)	322,871	+0.0326 ($\times 1.40$)	+0.0084	+0.0234 ($\times 1.01$)	+0.0285	0.378 (0.243)	0.0242	+0.0091 [+0.0084, +0.0099]	+0.0039 Z = 9.8 [+0.0032, +0.0047]
Panel b New Ideas									
1991–2017									
Incumbents (13)	1,196,294	+0.0347 ($\times 1.00$)	-0.0027	+0.0255 ($\times 0.73$)	+0.0286	0.246 (0.103)	0.0374	+0.0092 [+0.0089, +0.0095]	0 (ref.)
Entrants (20)	441,215	+0.0550 ($\times 1.59$)	+0.0129	+0.0440 ($\times 1.27$)	+0.0459	0.263 (0.094)	0.0422	+0.0111 [+0.0106, +0.0116]	+0.0018 Z = 6.2 [+0.0013, +0.0024]
Europe Agreement (11)	41,329	+0.0512 ($\times 1.48$)	-0.0268	+0.0155 ($\times 0.45$)	+0.0162	0.459 (0.339)	0.0780	+0.0358 [+0.0332, +0.0383]	+0.0266 Z = 20.2 [+0.0240, +0.0291]
1991–2003									
Incumbents (13)	250,246	+0.0329 ($\times 1.00$)	+0.0248	+0.0314 ($\times 0.95$)	+0.0321	0.191 (0.059)	0.0082	+0.0016 [+0.0010, +0.0021]	0 (ref.)
Entrants (20)	19,792	+0.0267 ($\times 0.81$)	+0.0083	+0.0212 ($\times 0.64$)	+0.0238	0.299 (0.091)	0.0185	+0.0055 [+0.0029, +0.0081]	+0.0040 Z = 2.9 [+0.0013, +0.0066]
Europe Agreement (11)	4,351	+0.0268 ($\times 0.81$)	-0.0033	+0.0169 ($\times 0.51$)	+0.0229	0.331 (0.124)	0.0301	+0.0099 [+0.0039, +0.0160]	+0.0084 Z = 2.7 [+0.0023, +0.0144]
2004–2017									
Incumbents (13)	946,048	+0.0350 ($\times 1.00$)	-0.0086	+0.0236 ($\times 0.67$)	+0.0273	0.261 (0.115)	0.0436	+0.0114 [+0.0110, +0.0117]	0 (ref.)
Entrants (20)	421,423	+0.0565 ($\times 1.61$)	+0.0132	+0.0452 ($\times 1.29$)	+0.0470	0.261 (0.094)	0.0432	+0.0113 [+0.0108, +0.0118]	-0.0001 Z = -0.4 [-0.0007, +0.0005]

Group	Funded papers	national only	touching foreign money	all funded papers	national + EU only	Foreign share w (EU)	Δ	Pull-back w $\times \Delta$ [95% CI]	Extra pull-back vs incumbents [95% CI]
Europe Agreement (11)	36,978	+0.0556 ($\times 1.59$)	-0.0270	+0.0165 ($\times 0.47$)	+0.0168	0.474 (0.362)	0.0826	+0.0391 [+0.0363, +0.0419]	+0.0278 Z = 19.3 [+0.0249, +0.0306]

The table gives the numbers behind Fig. 4a and separates the pull-back into its two factors. For every group, period and layer it reports where nationally funded work sits relative to that group's own unfunded work, how far foreign money moves it, and whether that movement exceeds what incumbents obtain. Three periods are reported for each layer, and crowding on both layers is measured as the within-year percentile rank at the paper level. Papers are classified into five mutually exclusive categories: domestic, where every named funder is a national body of the author country; EU, used throughout as shorthand for supranational funders, comprising the European Union's framework programmes, the European Research Council and a small number of international bodies classified by organisation type rather than by country of registration, where a paper naming any supranational funder falls in this category whatever else it names; foreign, where every named funder is a national body of another country; mixed, combining domestic and foreign national funders; and unfunded. Writing n_d , n_{EU} , n_x and n_n for the paper counts in the domestic, EU, foreign-plus-mixed and unfunded categories, and $n_f = n_d + n_{EU} + n_x$ for all funded papers, $w = (n_{EU} + n_x) / n_f$. It divides into an EU share n_{EU} / n_f and a non-EU foreign share n_x / n_f , which sum to w and are the two quantities reported in the main text. The figure in brackets in the w column is a different quantity, $n_{EU} / (n_d + n_{EU})$, the EU share among papers naming no non-EU foreign funder; its denominator excludes n_x , so it is not additive with w . Δ is the mean crowding percentile of the n_d papers minus that of the $n_{EU} + n_x$ papers, and the pull-back decomposes exactly as $w \times \Delta = (n_{EU} / n_f) \cdot \Delta_{EU} + (n_x / n_f) \cdot \Delta_x$, where Δ_{EU} and Δ_x are each the gap between the n_d papers and the n_{EU} or n_x papers respectively. The decomposition is defined in Methods. Every group is measured against its own unfunded papers. For instance, suppose a country produced 100 papers, of which 30 acknowledge any funder and 70 acknowledge none; of the 30, twenty name only domestic funders and ten name at least one foreign or supranational funder. The 'national only' column is the mean crowding percentile of the 20 minus that of the 70; the 'all funded papers' column is the mean of all 30 minus that of the 70; $w = 10/30 = 0.333$; and Δ is the mean of the 20 minus the mean of the 10. The pull-back is then $0.333 \times \Delta$. The four gap columns are positive when the funded papers sit in more crowded positions. The pull-back column has the opposite sign convention: it is positive when adding foreign money moves the group towards less crowded positions; it is a different quantity from the $\times$ multiples, which are levels. $\times$ = multiple of the incumbents' national-only gap in the same period and layer, i.e. the y-axis of Fig. 4a. w = share of a group's funded papers acknowledging at least

one foreign or supranational funder, whether or not a domestic funder is also named; the figure in brackets counts EU and supranational funders only. w therefore records entry into another country's funding framework rather than co-authorship: a foreign collaborator who is listed as an author but does not attach a grant does not raise it. The split is 2004, the first year whose ten-year rolling window lies entirely after 1994, the median entry year of the entrant countries; it is closely aligned with the year in which the entrants' nationally funded line crosses the incumbent benchmark in Fig. 4a, 2005 on New ideas and Knowledge base, but that crossing is a level crossing rather than a change in trend and carries roughly a year of sampling uncertainty, so it is reported as corroboration and not used to define the split. The full period is dominated by the second half, which holds about twenty times as many funded papers as the first, so the pooled figures track the later period closely. The one column where the periods disagree is the extra pull-back for entrants: pooled over 1991–2017 it is positive, because entrants did hold an advantage in the first period, whereas after the crossing it is indistinguishable from zero, indicating entrants entered more plugged in than incumbents and lost that position while expanding. Papers are pooled within each period. The new ideas layer cannot be taken back before 1991. It is defined only for papers that use a new combination born within the preceding five years, which covers 13–15% of papers, and dating an edge's birth requires looking back far enough into the corpus, which begins in 1971, so birth years in the 1970s and early 1980s are unreliable. Applying our sample threshold — at least 200 funded and 200 unfunded papers in a given country-window — no Europe Agreement country and no entrant country reaches it on the new ideas layer in the ten-year window ending 1980, against 6 of 11 and 20 of 20 respectively on the knowledge base layer; the new ideas layer reaches the threshold for all groups only in the window ending 1991. Both layers are reported over the same windows for comparability. 95% CI are paper-level, from $\text{Var}(\text{percentile}) = 1/12$, which is exact for the within-year marginal by construction; they assume papers are independent and so are a lower bound. Z is the extra pull-back divided by its own standard error; the corresponding two-sided P values are below 0.001 wherever $|Z|$ exceeds 3.3, and are 0.004 (entrants, 1991–2003, New ideas), 0.007 (Europe Agreement, 1991–2003, New ideas) and 0.69 (entrants, 2004–2017, New ideas).

Table S5 Covariate balance in the matching window and the placebo window

Role	Covariate	Matching window $\tau \in [-3, -1]$			Placebo window $\tau \in [-5, -4]$		
		Returnees	Stayers	Std. diff.	Returnees	Stayers	Std. diff.
<i>Matched</i>	Crowding of knowledge base	0.5703	0.5691	0.0084	0.5702	0.5606	0.0505
	Crowding slope (knowledge base)	0.0033	0.0029	0.0018	0.0035	0.0017	0.0088
	$\log(1 + \text{annual publications})$	1.2187	1.2180	0.0013	1.1233	1.1116	0.0209
	Share first-authored	0.4146	0.4103	0.0140	0.4570	0.4216	0.0975
	Share last-authored	0.1216	0.1180	0.0181	0.1188	0.1243	-0.0230
<i>Not matched</i>	Crowding of new ideas	0.5741	0.5766	-0.0106	0.5667	0.5698	-0.0122
	Pivot size	0.1109	0.1154	-0.0518	0.1121	0.1178	-0.0588
	Top-institution reliance	0.2410	0.2293	0.0967	0.2437	0.2304	0.1002
	Share of top-5% cited	0.1911	0.1986	-0.0340	0.1900	0.1919	-0.0068
	Share of top-1% cited	0.0488	0.0509	-0.0189	0.0477	0.0487	-0.0071
	Mean CD5	-0.0023	-0.0013	-0.0244	-0.0027	-0.0018	-0.0187
	Share CD5 > 0	0.1450	0.1506	-0.0250	0.1394	0.1486	-0.0354
	Share CD5 > 0.25	0.0049	0.0062	-0.0254	0.0057	0.0063	-0.0106

Covariate balance between returnees and matched stayers. Columns 3–5 cover the matching window $\tau \in [-3, -1]$ and columns 6–8 the placebo window $\tau \in [-5, -4]$. Standardised difference is the returnee mean minus the weight-adjusted stayer mean, divided by the pooled standard deviation. The placebo window is excluded from the matching distance and provides the out-of-sample check. Matching yields 36,887 returnees and 65,392 distinct stayer $\times$ pseudo-return-year units drawn from 44,304 distinct stayers, with total control weight 36,887. A returnee is a scientist who moved from an entrant country to an incumbent country and later came back to the same entrant country. Each scientist's career is divided into spells of consecutive years spent in the same country type, where a year is assigned to whichever type contributes most of that year's papers, ties go to the entrant country, and years without institutional country information are left missing without breaking the surrounding spell. A return requires a spell of at least four years in an incumbent country, followed within two years by a spell of at least five years back in the scientist's origin country, which is the entrant country appearing most often in the earliest entrant spell; where a scientist returns more than once, the first return is used. Stayers never satisfy this pattern, are still in an incumbent country in their final spell, and have published from an entrant country at some point; their pseudo-return years are placed at least four years into an incumbent spell, with at least five further years observed. Both groups must be at least five years past their first publication at the return or pseudo-return year. This yields 45,398 returnees and 2,761,323 stayer $\times$ pseudo-return-year candidates drawn from 185,602 stayers, of which 36,887 returnees have complete pre-return covariates. Each returnee is matched within an exact block defined by origin country, first publication year (± 1)

and return year (± 2), which jointly fix career age at return; the standardised differences for these two quantities are 0.0038 and 0.0006 and, being unit-level constants that do not vary across windows, they are not tabulated. Within a block the two nearest stayers are selected on a standardised Euclidean distance over the five covariates listed above, all measured in $\tau \in [-3, -1]$ and requiring at least two years with a defined crowding value, with each scale factor taken as the standard deviation of the candidate pool: the level and slope of crowding of the knowledge base fix the pre-return crowding preference, log annual output fixes the scale of production, and the shares of first- and last-authored papers fix the authorship role and career stage. Matching is 1:2 with replacement, returnees carrying weight 1.0 and each matched stayer 0.5. The median distance to the second neighbour exceeds that to the first by 19% (0.507 against 0.399), so the bias cost of the second match is small, and 1:1, 1:2 and 1:3 matching all give the same conclusion of higher late-period crowding among returnees. Crowding of new ideas is not used in the matching distance because it is defined for only 23.9% of author-years, against 95.7% for the knowledge base; requiring it to be non-missing would restrict the sample to returnees with unusually dense publication records and change the estimand. It is reported here instead as an out-of-sample check and is balanced in both windows (-0.011 and -0.012). Two further pre-return quantities are excluded by design. The share of internationally co-authored papers and the cumulative count of top-10% papers are the pre-return counterparts of the variables used later to split returnees by post-return incumbent ties and by pre-return prestige, and matching on them would align the groups in advance and remove the contrast those splits are meant to test. Their standardised differences are -0.219 and -0.047 . Within the table, every standardised difference reported is at or below 0.10, in the matched and the unmatched covariates alike, the largest being top-institution reliance in the placebo window at approximately 0.10, which is consistent with the anticipation effect documented in Table S8. Crowding is a paper-level within-year global percentile that depends only on which keywords or keyword pairs a paper uses. An author-year value is the sum of percentiles over that year's papers divided by the number of papers, and the series is smoothed with a ± 1 -year rolling window that pools papers rather than averaging annual means, is computed on a completed τ grid so that missing years is computed on a completed τ grid so that missing years neither enter the numerator nor the denominator, and does not cross $\tau = 0$, so that $\tau = -1$ uses $\{-2, -1\}$ and $\tau = 0$ uses $\{0, +1\}$. The rolling grid is keyed on author $\times$ return year rather than author, since one stayer may enter with several pseudo-return years. Observations require at least one paper in the smoothed window and are truncated at 2017 in both layers. Coefficients come from a weighted two-way fixed-effects event study,
$$y_{it} = \sum_{\tau \neq -1} \beta_{\tau} \cdot \mathbf{1}\{\text{returnee}_i, \tau_{it} = \tau\} + \alpha_i + \delta_t + \varepsilon_{it} ,$$

where α_i is an author $\times$ return-year fixed effect and δ_t a calendar-year fixed effect.

The omitted category comprises $\tau = -1$ together with all control observations; weights are 1.0 for returnees and the matching weight for stayers, and the within transformation uses the same weights. No time-varying controls are included: output and authorship shares are themselves post-return outcomes, and career age is collinear with the two

fixed effects. Standard errors are clustered by author. Results are reported to $\tau = +10$. Returns are restricted to 2013 and earlier so that the return year itself is observed. Since crowding is measured to 2017, only returns up to 2007 contribute at $\tau = +10$ and later cohorts are censored, so 38% of returnees are represented there and the number with a defined value falls to 32% of the $\tau = 0$ count in the knowledge base and 45% in the new-ideas layer. Returnees are also on average 14.9 years past their first publication when they return, so $\tau = +10$ already corresponds to the twenty-fifth year of a career. Estimation extends to $\tau = +12$ although results are reported only to $\tau = +10$, because under ± 1 smoothing the value at $\tau = +10$ pools years $\{9, 10, 11\}$: if the panel were truncated at $\tau = +10$ that observation would be built from $\{9, 10\}$ alone, so extending the window makes full use of the available years. Re-estimating the model on the shorter panel $\tau \in [-5, +10]$ gives a consistent late-period effect of +0.0142 in the knowledge base, against +0.0146 from the reported specification $\tau \in [-5, +12]$; the difference comes from the truncated endpoint, not from the reporting window, which is $\tau \in [-5, +10]$ in both cases. Coefficients are shifted so that the three pre-return reference years $\tau = -5, -4$ and -1 average to zero, so no single year is fixed at zero and the $\tau = -1$ estimate carries sampling error. The dip is the mean of $\tau \in [0, 2]$ and the late effect the mean of $\tau \in [5, 10]$, each relative to that reference; both are linear combinations of β whose standard errors are computed from the same covariance matrix, and between-group differences use disjoint-sample standard errors. The late-period increase in crowding reported in Fig. 4c,d is unchanged under a series of alternative specifications: additionally requiring the same transit incumbent country within the block; adding a sixth distance dimension given by the Euclidean distance between the two scientists' field-frequency distributions over their entire pre-return publication record, and selecting the two nearest neighbours on that augmented distance; career-age thresholds of five, six, seven or eight years; 1:1 and 1:3 rather than 1:2 matching; excluding China, which strengthens rather than weakens the estimates and shows that the pattern is not driven by the single largest entrant country; and splitting returnees by return cohort, where both cohorts move in the same direction and the later one is stronger.

Table S6 Crowding after return, by author scope and subgroup

Outcome	Author scope	Group	Authors	Observations	Early $\tau \in [0,2]$	Late $\tau \in [5,10]$	Late – early
Knowledge base	All papers	All returnees	81,159	1,105,770	+0.0024** (0.0009)	+0.0146*** (0.0013)	+0.0122*** (0.0010)
	All papers	Lost incumbent ties	43,580	564,423	−0.0007 (0.0013)	+0.0161*** (0.0017)	+0.0168*** (0.0014)
	All papers	Kept incumbent ties	44,706	541,347	+0.0058*** (0.0012)	+0.0122*** (0.0018)	+0.0065*** (0.0014)
	All papers	High prestige	43,316	534,837	+0.0057*** (0.0011)	+0.0139*** (0.0017)	+0.0082*** (0.0014)
	All papers	Low prestige	44,196	570,933	−0.0010 (0.0014)	+0.0144*** (0.0018)	+0.0154*** (0.0014)
	Last author	All returnees	65,052	605,590	+0.0060* (0.0024)	+0.0159*** (0.0030)	+0.0098*** (0.0022)
	Last author	Lost incumbent ties	34,780	300,725	+0.0041 (0.0037)	+0.0161*** (0.0044)	+0.0119*** (0.0031)
	Last author	Kept incumbent ties	35,628	304,865	+0.0080** (0.0031)	+0.0152*** (0.0040)	+0.0072* (0.0030)
	Last author	High prestige	35,385	321,665	+0.0068* (0.0029)	+0.0148*** (0.0038)	+0.0080** (0.0028)
	Last author	Low prestige	34,409	283,925	+0.0048 (0.0043)	+0.0165*** (0.0048)	+0.0116*** (0.0033)
New ideas	All papers	All returnees	65,791	577,379	−0.0031 (0.0025)	+0.0086* (0.0036)	+0.0116*** (0.0028)
	All papers	Lost incumbent ties	35,022	285,649	−0.0032 (0.0038)	+0.0103* (0.0049)	+0.0134*** (0.0037)
	All papers	Kept incumbent ties	36,502	291,730	−0.0028 (0.0034)	+0.0061 (0.0049)	+0.0089* (0.0040)
	All papers	High prestige	36,623	312,409	−0.0037 (0.0031)	+0.0071 (0.0046)	+0.0108** (0.0038)
	All papers	Low prestige	34,254	264,970	−0.0004 (0.0044)	+0.0108* (0.0054)	+0.0113** (0.0040)
	Last author	All returnees	29,520	174,821	+0.0124† (0.0075)	+0.0255* (0.0100)	+0.0131* (0.0065)
	Last author	Lost incumbent ties	15,410	83,603	+0.0063 (0.0125)	+0.0263† (0.0150)	+0.0200* (0.0090)

Last author	Kept incumbent ties	16,398	91,218	+0.0172† (0.0093)	+0.0229† (0.0128)	+0.0057 (0.0088)
Last author	High prestige	17,590	103,201	+0.0141† (0.0083)	+0.0280* (0.0115)	+0.0139† (0.0079)
Last author	Low prestige	13,951	71,620	+0.0110 (0.0164)	+0.0212 (0.0188)	+0.0102 (0.0106)

Early and late effects are means of $\tau \in [0,2]$ and $\tau \in [5,10]$ relative to the pre-return reference. The third column is the difference between the two post-return windows; the reference terms cancel, so it measures movement within the post-return period alone and is unaffected by any pre-return trend. Crowding rises further between the two windows in every subgroup of the knowledge base and in the pooled new-ideas series. Subgroups by incumbent ties are defined on post-return collaboration and are descriptive; subgroups by prestige use cumulative top-10% papers before return and are pre-determined. Last author is used as a proxy for corresponding author because the corresponding-author flag in OpenAlex is populated for only 4.3% of author-years; in most fields the last author is the corresponding author. Crowding of new ideas is observed for 23.9% of author-years against 95.7% for the knowledge base, so its standard errors are two to three times larger. Weighted two-way fixed-effects event study with author $\times$ return-year and calendar-year fixed effects; returnees carry weight 1.0 and each matched stayer 0.5. Estimates are expressed relative to the mean of three pre-return reference years ($\tau = -5, -4$ and -1). Standard errors, in parentheses, are clustered by author. Authors counts distinct scientists; observations counts author-year records entering the regression, so one scientist contributes several observations and, among stayers matched more than once, one scientist may enter under several pseudo-return years. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, † $P < 0.1$. Using $\tau \in [6,10]$ or $[7,10]$ in place of $[5,10]$ leaves every coefficient in this table within 0.005 of the value shown; the largest shift in significance is one tier and the sign is unchanged throughout.

Table S7 Disruptiveness of post-return output

Outcome	Author scope	Restriction	Authors	Observations	Early $\tau \in [0,2]$	Late $\tau \in [5,10]$	Late – early
Mean CD5	All papers	None	80,802	1,060,201	+0.0018*** (0.0003)	+0.0014*** (0.0004)	–0.0004 (0.0003)
	Last author	None	58,268	501,490	+0.0003 (0.0007)	–0.0003 (0.0009)	–0.0005 (0.0006)
	All papers	Top-5% cited	70,926	679,043	+0.0014*** (0.0004)	+0.0013** (0.0005)	–0.0000 (0.0004)
	Last author	Top-5% cited	32,818	213,392	+0.0016† (0.0009)	–0.0002 (0.0012)	–0.0018* (0.0008)
Share CD5 > 0	All papers	None	80,802	1,060,201	+0.0309*** (0.0016)	+0.0303*** (0.0021)	–0.0006 (0.0018)
	Last author	None	58,268	501,490	+0.0096* (0.0041)	+0.0155** (0.0051)	+0.0059 (0.0037)
	All papers	Top-5% cited	70,926	679,043	+0.0190*** (0.0025)	+0.0222*** (0.0035)	+0.0032 (0.0029)
	Last author	Top-5% cited	32,818	213,392	+0.0077 (0.0069)	+0.0076 (0.0090)	–0.0001 (0.0063)
Share CD5 > 0.25	All papers	None	80,802	1,060,201	+0.0004 (0.0003)	–0.0006 (0.0005)	–0.0010* (0.0004)
	Last author	None	58,268	501,490	–0.0009 (0.0010)	–0.0024* (0.0012)	–0.0015† (0.0008)
	All papers	Top-5% cited	70,926	679,043	+0.0001 (0.0005)	+0.0002 (0.0007)	+0.0001 (0.0006)
	Last author	Top-5% cited	32,818	213,392	+0.0005 (0.0014)	–0.0019 (0.0018)	–0.0024* (0.0012)

The jump in pivot size at return raises mean CD5 and the share of papers with CD5 > 0 in the early window, consistent with returnees carrying external knowledge into their home system. The upper tail does not follow: the share with CD5 > 0.25 is flat in the early window. As pivot size narrows over the late window neither mean CD5 nor the share above zero rises further, their late-minus-early differences being indistinguishable from zero, while the share above 0.25 declines. Rows restricted to the top 5% of citations require an author-year whose three-year window contains a paper that is both highly cited and has a defined CD5; the restriction is itself an outcome of the period studied, since highly cited output becomes scarcer after return (Fig. S19; Table S9), and those rows are reported for reference rather than as primary evidence. Effects on the share above 0.25 lie in the fourth decimal and are reported rather than plotted.

Weighted two-way fixed-effects event study with author $\times$ return-year and calendar-year fixed effects; returnees carry weight 1.0 and each matched stayer 0.5. Estimates are expressed relative to the mean of three pre-return reference years ($\tau = -5, -4$ and -1). Standard errors, in parentheses, are clustered by author. Authors counts distinct scientists; observations counts author-year records entering the regression, so one scientist contributes several observations and, among stayers matched more than once, one scientist may enter under several pseudo-return years. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, † $P < 0.1$. Using $\tau \in [6,10]$ or $[7,10]$ in place of $[5,10]$ leaves every coefficient in this table within 0.005 of the value shown; the largest shift in significance is one tier and the sign is unchanged throughout.

Table S8 Pre-return diagnostics and anticipation effect of returnees

Outcome	Authors	Observations	Placebo slope $\tau \in [-5, -4]$	Pre-return shift	Matching dimension
Knowledge base	81,159	1,105,770	+0.0008 (0.0008)	+0.0050*** (0.0009)	Yes
New ideas	65,791	577,379	-0.0007 (0.0016)	+0.0042† (0.0023)	No
Pivot size	80,804	1,066,409	-0.0013** (0.0004)	-0.0049*** (0.0005)	No
Top-institution reliance	80,934	1,070,753	+0.0019*** (0.0005)	+0.0038*** (0.0006)	No
Share of top-5% cited	81,159	1,105,578	+0.0011 (0.0012)	+0.0021† (0.0012)	No
Share of top-1% cited	81,159	1,105,578	+0.0008 (0.0006)	+0.0012† (0.0007)	No
Mean CD5	80,802	1,060,201	-0.0000 (0.0003)	-0.0001 (0.0003)	No
Share CD5 > 0	80,802	1,060,201	-0.0001 (0.0013)	-0.0022 (0.0014)	No
Share CD5 > 0.25	80,802	1,060,201	-0.0001 (0.0003)	-0.0006† (0.0003)	No

This table reports two quantities and refers to a third test. The placebo slope is the OLS slope between $\tau = -5$ and $\tau = -4$, both of which lie inside the placebo window $\tau \in [-5, -4]$, a window excluded from the matching distance; it tests whether the gap between the two groups is still changing within that window. The pre-return shift is the mean of $\tau \in [-3, -1]$ minus the mean of $\tau \in [-5, -4]$, and measures how far the gap has already moved by the time the matching window begins. The joint pre-trend test (Methods) takes $\tau = -1$ as its baseline and imposes $\beta_{-5} = \beta_{-4} = 0$ simultaneously; because $\tau = -1$ falls inside the matching window, that test likewise compares the level difference between the placebo window and the matching window, so it is closely related to the pre-return shift and differs mainly in its degrees of freedom. Variables outside the

matching distance provide direct evidence of anticipation. Pivot size and top-institution reliance move significantly in both windows, with placebo slopes of -0.0013 ($P = 0.003$) and $+0.0019$ ($P = 0.0002$) and pre-return shifts of -0.0049 and $+0.0038$ (both $P < 10^{-10}$), equal to 17% and 13% of the jump at $\tau = 0$. Neither entered the matching distance, so their movement in both windows is movement in the raw data: returnees begin to narrow their research pivots and to cite the work of top institutions more heavily two to three years before they move, and because the placebo window also shows a significant difference, this adjustment appears to begin even before $\tau = -3$. Highly cited output and disruptiveness move in opposite directions over the same window. The shares of papers in the top 5% and top 1% of citations both rise (pre-return shift $P = 0.096$ and 0.086), while the share of highly disruptive papers ($CD5 > 0.25$) falls ($P = 0.059$); mean $CD5$ and the share with $CD5 > 0$ show no significant change. Both point to the same adjustment. Selection and evaluation in entrant countries reward impact rather than disruption, and in the years before moving returnees shift effort towards work more likely to be highly cited. Together these constitute a third body of evidence for anticipation. Crowding behaves consistently with the evidence of anticipation above. Placebo slopes are indistinguishable from zero in both crowding layers (knowledge base $+0.0008$, $P = 0.31$; new ideas -0.0007 , $P = 0.66$), indicating no differential movement within the uncontaminated placebo window $\tau \in [-5, -4]$. The joint pre-trend test is rejected for knowledge-base crowding but passes for new-ideas crowding ($P = 0.134$). This difference reflects the fact that the joint test compares the early placebo periods ($\tau = -5$ and $\tau = -4$) with the reference period $\tau = -1$, which lies inside the matching window where crowding has already shifted before the return event. In the knowledge-base layer, the pre-return shift over $\tau \in [-3, -1]$ is $+0.0050$ ($P < 0.001$), consistent with the adjustment patterns documented for the other unmatched outcomes. Because the level and slope of knowledge-base crowding over $\tau \in [-3, -1]$ both entered the matching distance, a test anchored on $\tau = -1$ is not an independent placebo test for this outcome; the uncontaminated placebo diagnostic is the within-placebo slope, which remains statistically indistinguishable from zero. Return is therefore a process rather than a date, and scientists adjust before they move, so we treat $\tau \in [-3, -1]$ as partially exposed. Normalising on $\tau \in [-5, -4]$ alone would be cleaner, since it anchors on the uncontaminated window and counts the pre-return adjustment as part of the effect rather than netting it out; we report $\tau = -5$, -4 and -1 throughout, the more conservative choice, under which all conclusions are consistent. The pre-return adjustment admits two readings that this design cannot separate. Under the first, anticipation, scientists have already decided to return and adapt their research to the criteria of the receiving system. Under the second, selection, receiving institutions recruit on impact, so those hired already display this pattern before they move. The two are observationally identical in the pre-return window, and we do not observe unsuccessful applicants, so they cannot be distinguished. This uncertainty bears on how the pre-return window is read, not on the post-return effects themselves. The difference between the late effect ($\tau \in [5, 10]$) and the early effect ($\tau \in [0, 2]$) is measured entirely after return, the reference terms cancel, and it is therefore unaffected by any pre-return trend or selection. The conclusions in the main text concerning the late-

period rise in crowding rest on that within-period difference. Weighted two-way fixed-effects event study with author $\times$ return-year and calendar-year fixed effects; returnees carry weight 1.0 and each matched stayer 0.5; 1:2 matching with replacement. Standard errors, in parentheses, are clustered by author. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, † $P < 0.1$.

Table S9 The post-return adjustment path

Outcome	Author scope	Authors	Observations	Early $\tau \in [0,2]$	Late $\tau \in [5,10]$	Late – early
Share of top-1% cited	All papers	81,159	1,105,578	−0.0133*** (0.0007)	−0.0108*** (0.0010)	+0.0025*** (0.0007)
	Last author	64,998	604,397	−0.0064*** (0.0016)	−0.0052* (0.0021)	+0.0012 (0.0014)
Share of top-5% cited	All papers	81,159	1,105,578	−0.0322*** (0.0013)	−0.0199*** (0.0019)	+0.0122*** (0.0014)
	Last author	64,998	604,397	−0.0093** (0.0031)	−0.0025 (0.0041)	+0.0068* (0.0028)
Top-institution reliance	All papers	80,934	1,070,753	−0.0321*** (0.0007)	−0.0379*** (0.0010)	−0.0058*** (0.0007)
	Last author	59,771	521,933	−0.0180*** (0.0015)	−0.0268*** (0.0019)	−0.0088*** (0.0013)
Pivot size	All papers	80,804	1,066,409	+0.0272*** (0.0006)	+0.0180*** (0.0008)	−0.0092*** (0.0007)
	Last author	59,610	520,337	+0.0168*** (0.0015)	+0.0075*** (0.0019)	−0.0093*** (0.0013)

Return severs access to leading institutions and depresses highly cited output immediately. Reliance on top institutions and the shares of top-5% and top-1% cited papers all fall in the early window, while pivot size jumps at the same moment as returnees carry external knowledge into a new research environment. That jump is not without value. Mean CD5 and the share of papers with CD5 above zero both rise in the early window, so the imported knowledge does translate into modestly more disruptive work, although the upper tail does not follow and the share above 0.25 is flat from the outset (Table S7). The late window shows a different adjustment. Access to leading institutions is not regained but lost further, reliance lying below the counterfactual by more in the late window than in the early one. With that channel closed, the recovery of impact comes from narrowing the search instead. Pivot size contracts relative to the early window and crowding rises across the same interval in both layers, by +0.0122 in the knowledge base and +0.0116 in new ideas (Table S6). Highly cited output recovers in part, the shares of top-5% and top-1% cited papers both moving back towards the counterfactual. The corresponding cost of this shift falls on disruptiveness. Once pivot size contracts, neither mean CD5 nor the share above zero rises any further, their late-minus-early differences being indistinguishable from zero, while the share above 0.25 declines further (Table S7). Returnees regain part of their impact by working on more crowded and less exploratory questions, and the modest disruptive advantage created by the initial pivot does not survive that shift. All movements between the early and late windows are measured within the post-return period, so the reference terms cancel and they are unaffected by any pre-return trend. Weighted two-way fixed-

effects event study with author $\times$ return-year and calendar-year fixed effects; returnees carry weight 1.0 and each matched stayer 0.5. Estimates are expressed relative to the mean of three pre-return reference years ($\tau = -5, -4$ and -1). Standard errors, in parentheses, are clustered by author. Authors counts distinct scientists; observations counts author-year records entering the regression, so one scientist contributes several observations and, among stayers matched more than once, one scientist may enter under several pseudo-return years. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, † $P < 0.1$. Using $\tau \in [6,10]$ or $[7,10]$ in place of $[5,10]$ leaves every coefficient in this table within 0.005 of the value shown; the largest shift in significance is one tier and the sign is unchanged throughout.